

* PERCENTAGES (%) *

★ DEFINITION :

Percentage is a way of expressing a number as a fraction of 100.

Symbol : %

Example : 25% means 25 out of 100 $\rightarrow \frac{25}{100}$

1) CONVERSIONS :

FRACTION $\leftrightarrow$ PERCENTAGE

$\frac{1}{2}$	= 50%
$\frac{1}{3}$	= 33.33....%
$\frac{1}{4}$	= 25%
$\frac{1}{5}$	= 20%
$\frac{1}{8}$	= 12.5%
$\frac{1}{10}$	= 10%

DECIMAL $\leftrightarrow$ PERCENTAGE

0.01	= 1%
0.05	= 5%
0.1	= 10%
0.25	= 25%
0.5	= 50%
0.75	= 75%

4) SHORTCUTS :

- 10% of a number = $\frac{\text{Number}}{10}$
- 20% = $\frac{1}{5}$ of the number
- 25% = $\frac{1}{4}$ of the number
- 50% = $\frac{1}{2}$ of the number
- 75% = $\frac{3}{4}$ of the number
- To find x%, multiply by x and divide by 100.

5) PROFIT & LOSS PERCENTAGE BASICS :

- Profit % = $\left(\frac{\text{Profit}}{\text{Cost Price}} \right) \times 100$
- Loss % = $\left(\frac{\text{Loss}}{\text{Cost Price}} \right) \times 100$
- Selling Price = Cost Price + Profit
- Selling Price = Cost Price - Loss

6) COMMON MISTAKES :

- > Confusing % increase and % decrease.
- > Wrongly taking percentage on wrong value (e.g., on new value instead of old).
- > Forgetting to divide by 100.
- > In successive change formula, forgetting $\left(\frac{x \times y}{100} \right)$ part.

2) IMPORTANT FORMULAS :

- Percentage of a number = $x\% \text{ of } y = \frac{x}{100} \times y$
- To find what % one number is of another : $\% = \left(\frac{\text{Part}}{\text{Whole}} \right) \times 100$
- Percentage Increase = $\left(\frac{\text{New} - \text{Old}}{\text{Old}} \right) \times 100$
- Percentage Decrease = $\left(\frac{\text{Old} - \text{New}}{\text{Old}} \right) \times 100$

3) SUCCESSIVE PERCENTAGE CHANGE :

If a quantity is increased by x% and then decreased by y%, then

$$\text{Net change \%} = x - y - \left(\frac{x \times y}{100} \right)$$

If increased by x% and then increased by y%,

$$\text{Net change \%} = x + y + \left(\frac{x \times y}{100} \right)$$

7) TCS SHORTCUT TIPS :

- ★ Always check options and estimate.
- ★ For 50%, 25%, 75% use quick mental math.
- ★ In increase/decrease problems, write base value as 100, then calculate.
- ★ In successive % problems, use shortcut formula to save time.

8) PRACTICE QUESTIONS (TCS LEVEL) :

- 35 is what percent of 70?
a) 40% b) 50% c) 60% d) 70%
- 20% of 150 is equal to?
a) 25 b) 30 c) 35 d) 40
- A number is increased by 15%. If the new number is 115, what is the original number?
a) 90 b) 95 c) 100 d) 110
- A number is decreased by 20%. If the new number is 80, what is the original number?
a) 90 b) 95 c) 100 d) 110
- What is the net % change when a number is first increased by 20% and then decreased by 10%?
a) 8% b) 9% c) 10% d) 12%
- 12.5% of a number is 25. What is 50% of that number?
a) 50 b) 75 c) 100 d) 150
- A shirt is sold for ₹960 after giving 20% discount. What is the marked price?
a) ₹1100 b) ₹1150 c) ₹1200 d) ₹1250
- A shopkeeper gains 25% by selling a product for ₹250. What is the cost price?
a) ₹180 b) ₹200 c) ₹210 d) ₹220
- The population of a town increased from 80,000 to 96,000. What is the percentage increase?
a) 15% b) 18% c) 20% d) 25%
- What is 60% of 250 + 20% of 150?
a) 120 b) 130 c) 140 d) 150

LOGICAL REASONING

NUMBER SERIES

★ DEFINITION :

A number series is a sequence of numbers that follows a certain pattern or rule. We have to find the missing number(s) in that sequence.

1) TYPES OF NUMBER SERIES :

① Arithmetic Series (A.P.)	Constant difference between terms.
② Geometric Series (G.P.)	Constant ratio between terms.
③ Square Series	Numbers are perfect squares.
④ Cube Series	Numbers are perfect cubes.
⑤ Prime Series	All prime numbers in order.
⑥ Alternate Series	Two patterns alternate one by one.

3) EXAMPLES :

★ A.P. : 2, 5, 8, 11, 14, ?

$$d = 3 \quad \text{Next term} = 14 + 3 = \boxed{17}$$

★ G.P. : 3, 6, 12, 24, ?

$$r = 2 \quad \text{Next term} = 24 \times 2 = \boxed{48}$$

★ Square Series : 1, 4, 9, 16, 25, ?

$$\text{Next term} = 6^2 = \boxed{36}$$

★ Cube Series : 1, 8, 27, 64, ?

$$\text{Next term} = 4^3 = \boxed{64}$$

★ Prime Series : 2, 3, 5, 7, 11, ?

$$\text{Next term} = \boxed{13}$$

★ Alternate Series : 1, 2, 4, 7, 11, 16, ?

$$\begin{array}{cccccc} & +1 & +2 & +3 & +4 & +5 \\ \text{Next term} & = & 16 & + & 6 & = \boxed{22} \end{array}$$

2) SHORTCUT METHODS :

• For A.P. → Find common difference (d)

$$d = T_2 - T_1 = T_3 - T_2 = T_4 - T_3 = \dots$$

$$T_n = a + (n-1)d \quad (a = \text{first term})$$

• For G.P. → Find common ratio (r)

$$r = \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3} = \dots$$

$$T_n = a \times r^{(n-1)} \quad (a = \text{first term})$$

• For Missing Middle Term (A.P.) →

$$\text{Missing term} = \left(\frac{\text{Before} + \text{After}}{2} \right)$$

• For Missing Middle Term (G.P.) →

$$\text{Missing term} = \sqrt{\text{Before} \times \text{After}}$$

4) COMMON PATTERNS TO IDENTIFY :

✓ Check differences between terms (A.P.)

✓ Check ratios between terms (G.P.)

✓ Check squares → 1, 4, 9, 16, 25, 36, ...

✓ Check cubes → 1, 8, 27, 64, 125, ...

✓ Check prime numbers → 2, 3, 5, 7, 11, ...

✓ Check alternating patterns (add/subtract increasing/decreasing numbers)

5) TCS TIPS :

★ Always observe at least 2-3 steps before finalizing rule.

★ Try multiple patterns, don't stick to one.

★ Eliminate options using last term.

★ Don't waste time, if stuck move to next and come back.

6) PRACTICE QUESTIONS (TCS LEVEL) :

1. 2, 5, 8, 11, 14, ?

- a) 16 b) 17 c) 18 d) 20

2. 3, 6, 12, 24, 48, ?

- a) 72 b) 84 c) 96 d) 108

3. 4, 9, 16, 25, 36, ?

- a) 45 b) 49 c) 64 d) 81

4. 1, 8, 27, 64, ?, 216

- a) 81 b) 100 c) 125 d) 135

5. 2, 3, 5, 7, 11, 13, ?

- a) 15 b) 17 c) 19 d) 21

6. 1, 3, 6, 10, 15, ?

- a) 18 b) 20 c) 21 d) 25

7. 5, 10, 20, 35, 55, ?

- a) 80 b) 85 c) 90 d) 95

8. 100, 90, 80, 70, ?

- a) 50 b) 55 c) 60 d) 65

9. 2, 6, 12, 20, 30, ?

- a) 40 b) 42 c) 44 d) 45

10. 1, 4, 7, 11, 16, ?

- a) 20 b) 21 c) 22 d) 23

7) ANSWER KEY : 1-b 2-a 3-b 4-c 5-b 6-c 7-d 8-c 9-a 10-c

Practice more... Identify pattern faster... Solve smarter! 😊

⇒ SUBJECT - VERB AGREEMENT ⇐

★ DEFINITION :

Subject - Verb Agreement means the verb must agree with its subject in number (Singular/Plural).

1) BASIC RULES :

RULE	EXAMPLE
• Singular subject takes singular verb (add -s/-es).	He plays cricket. She writes a letter.
• Plural subject takes plural verb (no -s/-es).	They play cricket. We write letters.
• Subjects joined by AND take plural verb.	Ravi and Sita are friends.
• Subjects joined by OR / NOR take verb of the nearest subject.	Either you or I am right. Neither Rahul nor his friends are coming.
• 'Each' and 'Every' take singular verb.	Each student is responsible. Every book has a value.
• 'Along with', 'As well as', 'Together with' → verb agrees with first subject.	Rohan along with his brothers is going. Priya as well as her parents is coming.
• Collective nouns take singular verb.	The team is winning. The crowd is excited.
• A number + of + noun takes plural verb.	A bunch of flowers are beautiful. A lot of students are here.

2) SPECIAL CASES & EXCEPTIONS :

- ★ 'Some', 'Most', 'Many', 'Few' → plural verb (when used with countable nouns).
- ★ 'None', 'All', 'More than one' → plural verb.
- ★ 'Somebody', 'Someone', 'Nobody', 'Anyone' → singular verb.
- ★ 'Everybody', 'Everyone', 'Everything' → singular verb.
- ★ Titles like 'The rich', 'The poor' → plural verb.

3) COMMON MISTAKES :

- Using plural verb with singular subject.
(He go → He goes)
- Using singular verb with plural subject.
(They is → They are)
- Ignoring words between subject and verb.
(The list of items is long.)
- Confusing with collective nouns.
(The family are → The family is)

4) PRACTICE QUESTIONS (TCS LEVEL) :

- The boy _____ playing in the ground.
a) is b) are c) am d) be
- Ravi and his friends _____ going to the movie.
a) is b) are c) was d) were
- Neither Radha nor her sisters _____ coming.
a) is b) are c) am d) be
- Each of the books _____ interesting.
a) is b) are c) was d) were
- The list of items _____ on the table.
a) is b) are c) has d) have
- Some of the chocolates _____ missing.
a) is b) are c) was d) were
- Everyone in the class _____ present today.
a) is b) are c) was d) were
- Either you or I _____ going to be late.
a) is b) am c) are d) be
- A number of students _____ absent today.
a) is b) are c) was d) were
- The team _____ practicing hard for the match.
a) is b) are c) was d) were
- My mother along with my sisters _____ coming.
a) is b) are c) was d) were
- The rich _____ getting richer day by day.
a) is b) are c) was d) were
- Each of the players _____ given a medal.
a) is b) are c) was d) were
- All of the water in the tank _____ spilled.
a) is b) are c) was d) were
- Somebody _____ left their bag here.
a) has b) have c) is d) are
- Nobody _____ perfect in this world.
a) is b) are c) was d) were
- The crowd _____ cheering for the team.
a) is b) are c) was d) were
- Plenty of time _____ available to finish the work.
a) is b) are c) was d) were
- One of my pens _____ missing.
a) is b) are c) was d) were
- The couple _____ visiting India next month.
a) is b) are c) was d) were

5) ANSWER KEY : 1-b 2-b 3-a 4-a 5-a 7-a 8-b 9-b 10-b

11-b 12-b 13-a 14-b 15-a 17-b 17-b 19-a 20-b

TCS TIP : Identify the real subject, ignore the extra words in between, and match number with verb. 😊

VOCABULARY

SYNONYMS & ANTONYMS

1) WHY VOCABULARY IS IMPORTANT ?

- » Helps in Reading Comprehension.
- » Improves communication.
- » Important for Verbal Ability in placements.
- » Good vocabulary = Good Score.

2) MEMORY TRICKS :

- ★ Read daily 10 new words.
- ★ Learn with meanings, synonyms & antonyms.
- ★ Use words in sentences.
- ★ Revise regularly (Spaced Repetition).
- ★ Make short notes / flashcards.

3) HIGH FREQUENCY SYNONYMS :

WORD	MEANING	SYNONYMS
① Abundant	Plentiful	Ample, Plenty, Copious
② Accurate	Correct	Exact, Precise, True
③ Acquire	Obtain	Gain, Obtain, Procure
④ Adverse	Unfavorable	Unpleasant, Harmful, Detrimental
⑤ Amend	Improve	Improve, Modify, Correct
⑥ Benevolent	Kind	Kind, Generous, Compassionate
⑦ Commence	Begin	Start, Initiate, Launch
⑧ Diligent	Hardworking	Industrious, Dedicated, Assiduous
⑨ Eloquent	Well spoken	Fluent, Expressive, Articulate
⑩ Genuine	Real	Authentic, True, Sincere

4) HIGH FREQUENCY ANTONYMS :

WORD	MEANING	ANTONYMS
① Abandon	Give up	Retain, Keep, Continue
② Absurd	Ridiculous	Sensible, Logical, Reasonable
③ Ancient	Very old	Modern, New, Contemporary
④ Brave	Courageous	Cowardly, Fearful, Timid
⑤ Complex	Complicated	Simple, Easy, Clear
⑥ Create	Bring into existence	Destroy, Ruin, Demolish
⑦ Efficient	Effective	Inefficient, Ineffective
⑧ Generous	Giving	Mean, Stingy, Selfish
⑨ Happy	Joyful	Sad, Unhappy, Miserable
⑩ Important	Significant	Unimportant, Insignificant

5) TCS VOCABULARY LIST (Must Know Words):

• Analyze - Examine in detail	• Consider - Think about	• Exhibit - Display	• Prevent - Stop
• Assume - Take for granted	• Construct - Build	• Factor - Element	• Prove - Demonstrate
• Assist - Help	• Decline - Decrease	• Focus - Concentrate	• Provide - Supply
• Benefit - Advantage	• Demand - Request	• Ignore - Disregard	• Receive - Get
• Challenge - Difficulty	• Discipline - Control	• Indicate - Show	• Remove - Eliminate
• Clarify - Make clear	• Ensure - Make sure	• Interpret - Explain	• Support - Back
• Collect - Gather	• Estimate - Calculate	• Motivate - Encourage	• Utilize - Use
• Compare - Examine similarities	• Evaluate - Assess	• Perform - Carry out	• Verify - Check

6) PRACTICE QUESTIONS :

A) Choose the correct SYNONYM.

- The manager is very _____ in his work.
a) lazy b) careless c) diligent d) idle
- Please _____ the form with accurate details.
a) ignore b) fill c) destroy d) avoid
- She gave a very _____ speech.
a) unclear b) eloquent c) dull d) weak
- We must _____ every opportunity.
a) reject b) acquire c) waste d) ignore
- The weather was _____ to their plan.
a) helpful b) supportive c) adverse d) friendly

B) Choose the correct ANTONYM.

- He showed great _____ by facing the problem.
a) bravery b) fear c) courage d) strength
- This is a very _____ solution to the problem.
a) complex b) difficult c) simple d) tough
- She decided to _____ the old structure.
a) create b) build c) remove d) design
- He felt very _____ after losing the match.
a) happy b) joyful c) sad d) excited
- The data is not _____ yet.
a) important b) relevant c) accurate d) verified

7) ANSWER KEY :

Synonyms → 1-c 2-b 3-b 4-b 5-c

Antonyms → 1-c 2-c 3-c 4-c 5-d

★ TCS TIP : Read the sentence carefully.

Identify the correct meaning of the word in the context before choosing answer.

Context is the key!



TECHNICAL BASICS

PYTHON FUNDAMENTALS

1) WHAT IS PYTHON ?

- Python is a high level, interpreted, general purpose programming language.
- Created by Guido van Rossum and released in 1991.
- It is simple, readable and beginner friendly.
- Widely used in Web Development, Data Science, AI/ML, Automation, etc.

2) FEATURES OF PYTHON :

- ★ Easy to learn and use
- ★ Readable and clean syntax
- ★ Interpreted - no compilation needed
- ★ Cross-platform
- ★ Large standard library
- ★ Supports multiple programming paradigms
- ★ High demand in IT industry

3) BASIC SYNTAX RULES :

- Python is case sensitive.
- Indentation is very important (use 4 spaces).
- Every statement ends with a new line.
- Use # for single line comments.
- Use ''' or """ for multi-line comments.

4) VARIABLES IN PYTHON :

- A variable is a name given to a memory location to store data.
- No need to declare variable type.
- Variable name rules :
 - ✓ Use letters (a-z, A-Z), digits (0-9) and underscore (_).
 - ✓ Must start with a letter or underscore.
 - ✓ Case sensitive (age, Age and AGE are different).
 - ✓ No spaces or special characters.

EXAMPLE :

```
name = "Santosh"    # string
age = 20             # integer
cgpa = 9.18          # float
is_student = True    # boolean
```

5) DATA TYPES :

Data Type	Description	Example
int	Integer (whole numbers)	10, -5, 0
float	Decimal numbers	3.14, -0.5
str	String (text)	"Hello", 'Python'
bool	Boolean (True / False)	True, False
list	Ordered, changeable collection	[1, 2, 3]
tuple	Ordered, unchangeable collection	(1, 2, 3)
set	Unordered, unique elements	{1, 2, 3}
dict	Key-Value pairs	{'a': 1, 'b': 2}

6) BASIC INPUT/OUTPUT :

- Use input() to take input from user.
name = input("Enter your name: ")
- Use print() to display output.
print("Hello, " + name)

OUTPUT :

```
Enter your name: Santosh
Hello, Santosh
```

7) BASIC OPERATORS :

Type	Operators	Example	Result
Arithmetic	+ - * / // % **	5 + 2	7
Comparison	== != > < >= <=	5 > 2	True
Assignment	= += -= *= /= //= % =	x = 10	10
Logical	and or not	5 > 2 and 3 > 1	True
Membership	in not in	'a' in 'apple'	True
Identity	is is not	x is y	False

8) COMMON SHORTCUTS :

- ★ print() → Used to display output.
- ★ type() → Used to check data type.
- ★ len() → Used to find length of string/list.
- ★ int(), float(), str() → Type casting functions.
- ★ range(n) → Generates numbers from 0 to n-1.
- ★ # → Single line comment.
- ★ '''...''' or """...""" → Multi-line comment.

EXAMPLE :

```
x = 5
print(type(x))
print(len("python"))
print(int(3.7))
OUTPUT :
<class 'int'>
6
3
```

9) PRACTICE QUESTIONS (PYTHON FUNDAMENTALS)

- Which of the following is a valid variable name ?
a) 2name b) _name c) name\$ d) name-1
- What will be the output of print(2 + 3 * 4) ?
a) 20 b) 14 c) 10 d) 24
- What will be the output of type(3.14) ?
a) <class 'int'> b) <class 'float'>
c) <class 'str'> d) <class 'bool'>
- What is the output of print(10 // 3) ?
a) 3.33 b) 4 c) 3 d) 3.0
- Which operator is used for exponentiation ?
a) ^ b) ** c) %% d) ^^

- What will be the output of print("Hello" * 3) ?
a) Hello3 b) Hello Hello Hello c) Hello*3 d) Error
- Which of the following is a mutable data type ?
a) tuple b) list c) str d) int
- What is the output of print(5 > 3 and 2 < 1) ?
a) True b) False c) Error d) None
- What will be the output of len([1, 2, 3, 4, 5]) ?
a) 4 b) 5 c) 6 d) Error
- Which keyword is used to create a function in Python ?
a) func b) def c) define d) function

10) ANSWER KEY :

- 1-b 2-b 3-b 4-c 5-b
6-b 7-b 8-b 9-b 10-b

TCS TIP : Be clear with basics. Most coding questions in placements are from fundamentals. Practice daily and build strong logic. Consistency = Success! 😊

★ DEFINITION :

Profit is the gain when Selling Price (SP) is more than Cost Price (CP).

Loss is the **loss** when Selling Price (SP) is less than Cost Price (CP).

2) BASIC FORMULAS :

Profit (P)	$P = SP - CP$
Loss (L)	$L = CP - SP$
Profit %	$\text{Profit \%} = \left(\frac{P}{CP} \right) \times 100$
Loss %	$\text{Loss \%} = \left(\frac{L}{CP} \right) \times 100$
Selling Price	$SP = CP \left(1 + \frac{\text{Profit \%}}{100} \right)$ $SP = CP \left(1 - \frac{\text{Loss \%}}{100} \right)$
Cost Price	$CP = SP \times \frac{100}{(100 + \text{Profit \%})}$ $CP = SP \times \frac{100}{(100 - \text{Loss \%})}$

3) SHORT TRICKS :

- ① $\text{Profit \%} = \left(\frac{\text{Profit}}{CP} \right) \times 100$
- ② $\text{Loss \%} = \left(\frac{\text{Loss}}{CP} \right) \times 100$
- ③ If SP is given, find CP quickly using ratios.
- ④ For equal CP :
 - Gain % : SP ratio
 - Loss % : CP ratio
- ⑤ If $P\% = L\% \rightarrow$ S.P. is same in both cases.
- ⑥ For small % $\rightarrow$ Do in your head.

7) PRACTICE QUESTIONS (TCS LEVEL):

1. A pen is bought for ₹40 and sold for ₹50. Find the profit %.
a) 20% b) 25% c) 15% d) 10%.
2. A book is sold at a loss of 10%. If CP is ₹200, find SP.
a) ₹180 b) ₹190 c) ₹210 d) ₹220
3. If SP is ₹360 and profit is 20%, find CP.
a) ₹250 b) ₹280 c) ₹300 d) ₹320
4. A shirt is bought for ₹500 and sold for ₹450. Find the loss %.
a) 5% b) 8% c) 10% d) 15%
5. A watch is sold for ₹1200 with a profit of 25%. Find its cost price.
a) ₹900 b) ₹960 c) ₹1000 d) ₹800

4) IMPORTANT POINTS :

- ★ Always identify CP and SP correctly.
- ★ Profit is always calculated on CP.
- ★ In loss, percent is calculated on CP.
- ★ Convert % to decimal while calculating.
- ★ Check answer using estimation.

5) COMMON MISTAKES :

- Mixing up CP and SP.
- Applying % on wrong amount.
- Forgetting + or - sign.
- Not reading the question carefully.
- Calculation mistakes in percentage.

6) TCS TIPS :

- ★ Read the question twice.
- ★ Underline key numbers.
- ★ Decide first $\rightarrow$ Profit or Loss.
- ★ Use shortcut formulas.
- ★ Don't calculate big numbers directly, simplify first.

6. If CP is ₹750 and loss% is 20%, find SP.
a) ₹600 b) ₹550 c) ₹650 d) ₹700
7. A man sold his bike for ₹28,000 at a loss of 30%. What was the cost price?
a) ₹40,000 b) ₹38,000 c) ₹36,000 d) ₹35,000
8. If a trader gains 15% by selling an item for ₹690, what is the cost price?
a) ₹550 b) ₹600 c) ₹630 d) ₹650
9. An article is sold for ₹276 at a loss of 25%. Find the cost price.
a) ₹350 b) ₹360 c) ₹368 d) ₹400
10. By selling a bag for ₹840, a shopkeeper gains 20%. Find the cost price.
a) ₹600 b) ₹650 c) ₹700 d) ₹720

8) ANSWER KEY : 1-b 2-a 3-c 4-c 5-b 6-a 7-c 8-b 9-c 10-b

Remember :

Small mistakes in %

create big loss in marks!



1) DEFINITION :

Coding-Decoding is the process of converting information (words / letters / numbers / symbols) into a secret code and vice-versa using a given rule or pattern.

2) TYPES OF CODING-DECODING :

TYPE	DESCRIPTION
① Letter Coding	Letters are coded as other letters based on a rule.
② Number Coding	Letters or words are coded using numbers.
③ Word Coding	Words are coded as other words based on position / rule.
④ Symbol Coding	Letters are coded as symbols or symbols as letters.
⑤ Mixed Coding	Combination of letters, numbers and symbols.

4) EXAMPLES :

① If in a certain code, CAT is written as DBU.

What is BAT written as?

Solution : $C \rightarrow D (+1)$, $A \rightarrow B (+1)$, $T \rightarrow U (+1)$

$B \rightarrow C$, $A \rightarrow B$, $T \rightarrow U \Rightarrow$ **CBU**

② In a code, "HELLO" is written as "KHOOR".

How is "WORLD" written?

Solution : $H \rightarrow K (+3)$, $E \rightarrow H (+3)$, $L \rightarrow O (+3)$, $L \rightarrow O (+3)$, $O \rightarrow R (+3)$

$W \rightarrow Z$, $O \rightarrow R$, $R \rightarrow U$, $L \rightarrow O$, $D \rightarrow G \Rightarrow$ **ZRUOG**

③ If "SILENT" is written as "TJMFUO", what is "LISTEN" written as?

Solution : Each letter is +1

$L \rightarrow M$, $I \rightarrow J$, $S \rightarrow T$, $T \rightarrow U$, $E \rightarrow F$, $N \rightarrow O \Rightarrow$ **MJTUFO**

7) PRACTICE QUESTIONS (TCS LEVEL):

1. If in a code, BOOK is written as CPPL, how is READ written?

a) SFBE b) SFDC c) SFDB d) SEDC

2. In a certain code, 198 is written as 746.

What is 275 written as?

a) 523 b) 514 c) 514 d) 413

3. If "PLANE" is written as "SODQH", how is "TRAIN" written?

a) WUDLK b) UQZHM c) USZGK d) UQZGK

4. In a code, "APPLE" is written as "@#\$\$%".

What is "MANGO" written as?

a) &%@#@ b) &%@#@ c) @&%@#@ d) %&@#@

5. If in a code, ZEBRA is written as AYDQZ.

How is TIGER written?

a) SHFDQ b) SHFEQ c) SHFER d) SJFER

3) COMMON CODING PATTERNS :

- ★ +n Pattern : Each letter is replaced by the letter n steps forward in the alphabet.
- ★ -n Pattern : Each letter is replaced by the letter which is n steps backward in the alphabet.
- ★ Alternate Pattern : +n for odd positions, -n for vice-versa.
- ★ Reverse Order : Letters are written in reverse.
- ★ Position Based : Letters are coded based on their position in the word.
- ★ Number / Symbol Replacement : Letters are replaced by numbers or symbols.

5) TRICK BOX :

- ★ Convert letters to their positions ($A=1, B=2, \dots, Z=26$).
- ★ After adding/subtracting, if the value > 26 , subtract 26.
- ★ If the value ≤ 0 , add 26.
- ★ For alternate pattern, check odd-even positions.
- ★ Always verify the pattern using the given example.

6) TCS TIPS :

- ★ Read the question carefully and identify the pattern.
- ★ Try with first 2-3 letters before applying on the whole word.
- ★ In number/symbol coding, remember the mapping.
- ★ Don't assume - confirm the rule from examples.
- ★ Manage time: Don't spend too much on one question.

8) ANSWER KEY : 1-b 2-b 3-a 4-a 5-b

6-b 7-a 8-a 9-a 10-b

6. In a certain code, JUMP is written as LUNQ.

How is GAME written?

a) HBNF b) HCNF c) HBMF d) HBNG

7. If in a code, 3 5 7 is written as 9 11 13.

How is 4 6 8 written in that code?

a) 10 12 14 b) 9 11 13 c) 8 10 12 d) 10 11 13

8. In a code, "DOG" is written as "FQH".

What is "CAT" written as?

a) EBU b) DBU c) EBT d) DCS

9. If "HEAD" is written as "IFBE" in a code.

How is "READ" written?

a) SFBE b) SFDB c) SFBC d) SFCF

10. In a code, "KING" is written as "LQPI".

How is "POINT" written?

a) RQJOU b) QJOUN c) RQJNT d) ROJNT

★ TCS TIP : Identify the pattern $\rightarrow$ apply carefully $\rightarrow$ verify the answer. Practice more to become faster! 😊

1) DEFINITION :

Tenses show the time of an action or event.

There are three main tenses :

Present, Past and Future.

2) TYPES OF TENSES :

	TENSE	STRUCTURE	EXAMPLE
PRESENT TENSE	Simple Present	S + V ₁ (s/es) + O	He plays cricket.
	Present Continuous	S + is/am/are + V-ing + O	He is playing.
	Present Perfect	S + has/have + V ₃ + O	He has played.
	Present Perfect Continuous	S + has/have + been + V-ing + O	He has been playing.
PAST TENSE	Simple Past	S + V ₂ + O	He played.
	Past Continuous	S + was/were + V-ing + O	He was playing.
	Past Perfect	S + had + V ₃ + O	He had played.
	Past Perfect Continuous	S + had + been + V-ing + O	He had been playing.
FUTURE TENSE	Simple Future	S + will/shall + V ₁ + O	He will play.
	Future Continuous	S + will/shall + be + V-ing + O	He will be playing.
	Future Perfect	S + will/shall + have + V ₃ + O	He will have played.
	Future Perfect Continuous	S + will/shall + have + been + V-ing + O	He will have been playing.

5) SIGNAL WORDS :

TENSE	SIGNAL WORDS	EXAMPLE
Present Simple	always, usually, often, sometimes, never, every day	She always wakes up early.
Present Continuous	now, at the moment, currently, look!, listen!	I am reading a book now.
Past Simple	yesterday, last week, ago, in 2010	We visited the museum yesterday.
Past Continuous	while, when, at that time	She was cooking when I arrived.
Future Simple	tomorrow, next week, soon, later	I will go to the park tomorrow.
Future Continuous	this time tomorrow, next week at this time	I will be studying at 8 pm.

7) PRACTICE QUESTIONS (TCS LEVEL) :

- He _____ to school every day.
a) go b) goes c) going d) gone
- I _____ a movie yesterday.
a) watch b) watched c) watching d) watches
- They _____ dinner when I called them.
a) eat b) ate c) were eating d) have eaten
- She _____ this book for two hours.
a) reads b) read c) is reading d) has read
- We _____ to Mumbai next week.
a) go b) went c) will go d) going

- By next year, I _____ my degree.
a) complete b) will complete c) completed d) completing
- He _____ in this company since 2020.
a) works b) worked c) has been working d) is working
- I _____ my homework already.
a) do b) did c) have done d) doing
- She _____ very tired last night.
a) is b) was c) are d) were
- Tomorrow at this time, we _____ in the train.
a) shall travel b) will be travelling c) travel d) travelled

- 8) ANSWER KEY : 1-b 2-b 3-c 4-c 5-c
6-b 7-c 8-c 9-b 10-b

TCS TIP : Practice all tenses daily. Understand the time, identify the helping verbs and choose the correct tense. 😊

3) KEY POINTS :

- ★ Use Present Simple for habits and general truths.
(I go to school every day.)
- ★ Use Present Continuous for actions happening now.
(I am studying now.)
- ★ Use Past Simple for completed actions in the past.
(I visited Delhi last year.)
- ★ Use Past Continuous for actions in progress in the past.
(I was watching TV at 8 pm.)
- ★ Use Future Simple for decisions, promises or predictions.
(I will call you tomorrow.)

4) COMMON MISTAKES :

- He go to school every day. X
He goes to school every day. ✓
- She don't like coffee. X
She doesn't like coffee. ✓
- I am went to market. X
I went to market. ✓
- They was playing. X
They were playing. ✓
- He have finished his work. X
He has finished his work. ✓

6) TCS TIPS :

- ★ Identify the time of action.
- ★ Focus on helping verbs
(is/am/are/was/were/has/have/will).
- ★ Check subject-verb agreement.
- ★ Look for signal words in the sentence.
- ★ Eliminate options by using process of elimination.
- ★ Practice is the key to perfection! 😊

⇒ ONE-WORD SUBSTITUTIONS ⇒

1) WHAT ARE ONE-WORD SUBSTITUTIONS ?

They are words or phrases that can be replaced by a single word without changing the meaning.

Example :

- ✓ A person who loves books → Bibliophile
- ✓ Fear of heights → Acrophobia

2) WHY ARE THEY IMPORTANT ?

- ★ Improve vocabulary and expression.
- ★ Frequently asked in TCS NQT & other exams.
- ★ Save time in reading and writing.
- ★ Helps in scoring well in Verbal Ability section.

3) IMPORTANT ONE-WORD SUBSTITUTIONS :

MEANING	ONE-WORD	MEANING	ONE-WORD
① A person who loves books	Bibliophile	⑪ Someone who helps the poor	Philanthropist
② One who speaks the truth	Sincere	⑫ Excessive self love	Egoism
③ A person who never gives up	Perseverant	⑬ Great fear	Terror
④ Fear of heights	Acrophobia	⑭ A person who eats a lot	Glutton
⑤ Strong dislike or hatred	Aversion	⑮ One who avoids others	Introvert
⑥ One who is always happy	Optimist	⑯ Difficult to understand	Obscure
⑦ One who tells lies	Liar	⑰ A person with evil plans	Schemer
⑧ A person who is always busy	Workaholic	⑱ One who is brave	Courageous
⑨ A person who is greedy	Miser	⑲ Having many knowledge	Learned
⑩ Careless in spending money	Spendthrift	⑳ One who sings songs	Lyricist

4) CATEGORIES :

People	Bibliophile, Philanthropist, Workaholic, Introvert
Qualities	Sincere, Brave, Greedy, Courageous, Learned
Emotions	Aversion, Terror, Egoism, Love, Hatred
Actions	Admit, Appreciate, Initiate, Evaluate, Conclude
Conditions	Obscure, Ancient, Modern, Intact, Definite

5) TIPS TO REMEMBER :

- ★ Break the meaning and relate with the word.
- ★ Read daily and note new words.
- ★ Use in sentences.
- ★ Revise regularly.
- ★ Make flashcards / short notes.

6) COMMONLY CONFUSED WORDS :

WORDS	MEANING	ONE-WORD
Very glad	Extremely happy	Elated
Very angry	Extremely angry	Furious
Very tired	Extremely tired	Exhausted
Very afraid	Extremely afraid	Terrified
Very kind	Extremely kind	Benevolent

7) PRACTICE QUESTIONS (TCS LEVEL) :

- A person who writes books.
a) Author b) Poet c) Novelist d) Editor
- Fear of darkness.
a) Claustrophobia b) Nyctophobia c) Aerophobia d) Aquaphobia
- One who is indifferent to pain.
a) Stoic b) Epicure c) Optimist d) Pessimist
- A person who cannot be trusted.
a) Honest b) Sincere c) Reliable d) Unreliable
- One who is very punctual.
a) Latecomer b) Timekeeper c) Punctual d) Scheduler
- A person who talks too much.
a) Taciturn b) Talkative c) Chatty d) Silent
- One who is broad-minded.
a) Narrow-minded b) Open-minded c) Close-minded d) Selfish
- One who studies stars.
a) Geologist b) Astronomer c) Biologist d) Botanist
- A person who is very ambitious.
a) Indifferent b) Ambitious c) Lazy d) Humble
- One who is fond of art.
a) Artist b) Pianist c) Connoisseur d) Engineer

- A person who steals.
a) Robber b) Thief c) Burglar d) Smuggler
- One who is quick to learn.
a) Dull b) Intelligent c) Timid d) Quick-witted
- Excessive pride in oneself.
a) Humility b) Arrogance c) Modesty d) Patience
- One who is not easily moved by emotion.
a) Emotional b) Sensitive c) Stoic d) Nervous
- One who breaks the law.
a) Citizen b) Loyalist c) Criminal d) Patriot
- One who is skilled in music.
a) Musician b) Vocalist c) Composer d) Singer
- A person who believes in equality.
a) Egalitarian b) Individualist c) Capitalist d) Socialist
- A person who talks in a clever way.
a) Foolish b) Articulate c) Dumb d) Shy
- One who saves money regularly.
a) Spender b) Investor c) Miser d) Saver
- A person who hates mankind.
a) Philanthropist b) Misogynist c) Misanthrope d) Optimist

- 8) ANSWER KEY : 1-c 2-b 3-a 4-d 5-c 6-b 8-b 9-b
11-b 12-d 13-b 14-c 15-c 16-a 17-a 18-b 19-d 20-c

TCS TIP : Learn a few words every day, use them in sentences and revise regularly.

Consistency = Better Vocabulary = Better Score! 😊

1) WHAT IS OOP ?

- ★ OOP stands for Object Oriented Programming.
- ★ It is a programming paradigm based on the concept of "Objects".
- ★ It helps in writing modular, reusable and maintainable code.

2) KEY OOP CONCEPTS :

CONCEPT	DESCRIPTION
★ Class	A blueprint or template for creating objects.
★ Object	An instance of a class. It has state (data) and behavior (methods).
★ Encapsulation	Binding data (variables) and methods together in a class and restricting direct access to some of its components.
★ Abstraction	Hiding unnecessary details and showing only essential features.
★ Inheritance	Deriving a new class from an existing class. It promotes code reusability.
★ Polymorphism	The ability of an object to take many forms. Same method can behave differently.

5) REAL-LIFE EXAMPLES :

- Class → Mobile Phone
- Object → iPhone, Samsung, OnePlus
- Encapsulation → Password in ATM
- Abstraction → Driving a car (don't know engine details)
- Inheritance → Dog is an Animal
- Polymorphism → Same function "play()" in Dog, Cat behaves differently.

7) PRACTICE QUESTIONS (TCS LEVEL) :

- What is OOP?
 - Object Oriented Programming
 - Ordered Oriented Programming
 - Object Operating Program
 - None of these
- Which is not a pillar of OOP?
 - Encapsulation
 - Abstraction
 - Compilation
 - Polymorphism
- Which concept allows code reusability?
 - Inheritance
 - Polymorphism
 - Abstraction
 - Class
- What is a class in OOP?
 - An object
 - A blueprint or template
 - A method
 - A variable
- Which keyword is used for inheritance in Python?
 - extends
 - inherit
 - super
 - implements

3) OOP PILLARS :

- ★ Encapsulation → Security
- ★ Abstraction → Simplicity
- ★ Inheritance → Reusability
- ★ Polymorphism → Flexibility

4) EXAMPLE :

```
class Car:
    def __init__(self, brand, color):
        self.brand = brand    # attributes
        self.color = color

    def display(self):
        print(f"Car: {self.brand}, Color: {self.color}")

# Creating objects
c1 = Car("Toyota", "Red")
c2 = Car("Honda", "Black")

c1.display()    # Output: Car: Toyota, Color: Red
c2.display()    # Output: Car: Honda, Color: Black
```

6) COMMON OOP TERMS :

- ★ Attribute → Variable inside a class
- ★ Method → Function inside a class
- ★ Constructor → Special method __init__()
- ★ Instance → Individual object
- ★ Super() → Refers to parent class
- ★ Self → Refers to current object

- Which method is called automatically when an object is created?
 - __init__()
 - display()
 - constructor()
 - start()
- What does self refer to?
 - Parent class
 - Current object
 - Child class
 - Global variable
- Which concept is used to hide unnecessary details?
 - Encapsulation
 - Abstraction
 - Inheritance
 - Polymorphism
- Same method name but different behavior in different class is an example of _____.
 - Polymorphism
 - Inheritance
 - Abstraction
 - Encapsulation
- Which is an example of abstraction?
 - Hiding password in login system
 - Multiple inheritance
 - Overloading methods
 - Using loops in program

8) ANSWER KEY :

1-a 2-c 3-a 4-b 5-c 6-a 7-b 8-a 9-a 10-a

TCS TIP : Understand concepts → Practice code →

Solve questions. Strong fundamentals = Strong placement! 😊

1) DEFINITION :

Ratio is the comparison of two quantities of the same kind by division.

Proportion is an equation stating that two ratios are equal.

2) BASIC FORMULAS :

CONCEPT	FORMULA
Ratio	$a : b = \frac{a}{b}$
Simplest Form of Ratio	Divide both terms by HCF
Proportion	$a : b = c : d \Rightarrow \frac{a}{b} = \frac{c}{d}$
Product of Extremes	$a \times d = b \times c$
Product of Means	$a \times c = b \times d$
Continued Proportion	a, b, c are in proportion if $b^2 = a \times c$
Dividing in a Ratio	If A is divided in the ratio $a : b$, First part = $A \times \frac{a}{a+b}$ Second part = $A \times \frac{b}{a+b}$

3) EXAMPLES :

- The ratio of 12 to 18 is $12 : 18 = \boxed{2 : 3}$ ($\div 6$)
- If $3 : 5 = 6 : 10$
 $3 \times 10 = 5 \times 6 = 30$ ✓ Proportion
- Find fourth proportion to $2 : 3 :: 4 : ?$
 $\frac{2}{3} = \frac{4}{x} \Rightarrow x = \frac{3 \times 4}{2} = \boxed{6}$
- Divide ₹720 in the ratio $2 : 3 : 5$
Sum of parts = $2 + 3 + 5 = 10$
Parts $\rightarrow 2, 3, 5$
Value of one part = $\frac{720}{10} = 72$
Required parts $\rightarrow 2 \times 72 = 144, 3 \times 72 = 216, 5 \times 72 = 360$
Answer $\rightarrow ₹144, ₹216, ₹360$

7) PRACTICE QUESTIONS (TCS LEVEL) :

- The ratio of 15 to 25 is _____.
a) 2 : 3 b) 3 : 5 c) 3 : 4 d) 4 : 5
- If $4 : 6 = 8 : x$, then $x =$ _____.
a) 8 b) 10 c) 12 d) 14
- The third proportional to 2 and 5 is _____.
a) 2.5 b) 6 c) 10 d) 12.5
- If $a : b = 7 : 9$ and $a + b = 64$, then $a =$ _____.
a) 24 b) 27 c) 28 d) 32
- Divide 540 in the ratio $3 : 6 : 9$.
a) 90, 180, 270 b) 60, 120, 180
c) 100, 200, 200 d) 80, 160, 300

4) IMPORTANT POINTS :

- ★ Ratios should be in the simplest form.
- ★ Quantities must be in the same unit.
- ★ In proportion, product of extremes = product of means.
- ★ Check your answer by substituting in the given ratio.
- ★ Practice is the key to speed and accuracy.

5) COMMON MISTAKES :

- Not simplifying the ratio.
- Using different units in comparison.
- Confusing means and extremes.
- Wrong cross multiplication.
- Not checking the answer.

6) TCS TIPS :

- ★ Simplify first, solve fast.
- ★ In proportion problems, use cross multiplication.
- ★ Use cancelation method for faster calculation.
- ★ In divided ratio problems, find value of one part first.
- ★ Practice previous year questions. 😊

8) ANSWER KEY :

1-a 2-c 3-c 4-b 5-a 6-b 7-c 8-a 9-b 10-b

- If $5 : 8 = x : 40$, then $x =$ _____.
a) 20 b) 25 c) 30 d) 32
- The mean proportional between 3 and 12 is _____.
a) 5 b) 6 c) 9 d) 10
- If $2a : 3b = 4 : 9$ and $b = 15$, then $a =$ _____.
a) 10 b) 12 c) 16 d) 18
- If $6 : 7 = 12 : x$, then $x =$ _____.
a) 14 b) 16 c) 18 d) 20
- The ratio of boys to girls in a class is $5 : 7$.
If there are 30 boys, how many girls are there?
a) 35 b) 40 c) 42 d) 45

★ TCS TIP : Understand the concept $\rightarrow$ Apply formula $\rightarrow$ Simplify $\rightarrow$ Solve $\rightarrow$ Verify your answer. 😊

1) DEFINITION :

Blood Relations questions are based on the relationships within a family.

We have to find the relation between two persons in a family.

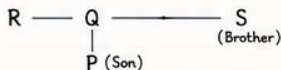
2) COMMON RELATIONSHIPS :

RELATION	SYMBOL	DESCRIPTION
Male	□	Square
Female	○	Circle
Marriage	—	Horizontal line
Siblings	—	Horizontal line (from same parents)
Parent - Child		Vertical line
Twins	(from a point)	Identical / fraternal twins
Difference of Generation	(no direct line)	• Uncle, Aunt • Cousins, etc.

5) EXAMPLES :

- ① P is the son of Q. Q is the wife of R. R is the brother of S. How is S related to P?

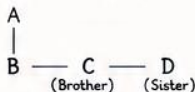
Solution :



Answer → P is the **nephew** of S.

- ② A is the father of B. B is the brother of C. C is the sister of D. How is D related to A?

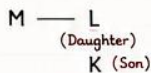
Solution :



Answer → D is the **daughter** of A.

- ③ K is the son of L. L is the daughter of M. How is M related to K?

Solution :



Answer → M is the **grandmother** of K.

7) PRACTICE QUESTIONS (TCS LEVEL) :

- A is the son of B. B is the wife of C. C is the brother of D. How is D related to A?
a) Uncle b) Aunt c) Brother d) Cousin
- P is the father of Q. Q is the sister of R. R is the brother of S. How is P related to S?
a) Father b) Uncle c) Grandfather d) Brother
- M is the daughter of N. N is the husband of O. O is the mother of P. How is M related to P?
a) Sister b) Cousin c) Aunt d) Mother
- X is the brother of Y. Y is the father of Z. Z is the sister of W. How is X related to W?
a) Brother b) Uncle c) Cousin d) Nephew
- A is the husband of B. B is the daughter of C. C is the father of D. How is A related to D?
a) Father b) Uncle c) Grandfather d) Cousin

3) IMPORTANT RULES :

- ★ A generation is the set of people born around the same time.
- ★ People in the same generation are represented on the same horizontal line.
- ★ A person is related to himself as "Self".
- ★ Siblings share the same parents.
- ★ A father of your father is your Grandfather.
- ★ A mother of your mother is your Grandmother.
- ★ Father's brother → Paternal Uncle.
- ★ Mother's sister → Maternal Aunt.
- ★ Sibling's child → Nephew / Niece.
- ★ First cousin → Children of your uncle / aunt.

4) TRICK BOX :

- ★ Draw a neat family tree.
- ★ Start from the given person and move step by step.
- ★ Use M's Brother → MB, F's Sister → FS, etc.
- ★ Count generations carefully.
- ★ Practice more diagrams to become fast.
- ★ Eliminate wrong options by using logic.

6) TYPES OF QUESTIONS ASKED :

- ★ How is A related to B?
- ★ How is A's sister / brother related to B?
- ★ How is A's father / mother related to B?
- ★ Who is the parent of A?
- ★ Who is the child of A?
- ★ Who is the wife / husband of A?
- ★ Based on family tree diagram, find the relation.

- 8) ANSWER KEY : 1-b 2-b 3-c 4-b 5-b
6-b 7-d 8-b 9-d 10-c

- ★ TCS TIP : Draw diagram → Identify relation →
Count generations → Choose the correct answer. 😊

- There are three persons P, Q and R. Q is the mother of P. R is the brother of Q. How is R related to P?
a) Father b) Uncle c) Brother d) Cousin
- S is the son of T. T is the brother of U. U is the mother of V. How is S related to V?
a) Brother b) Uncle c) Cousin d) Nephew
- If A is the sister of B and B is the father of C, how is A related to C?
a) Sister b) Aunt c) Mother d) Cousin
- P has a son Q. Q has a sister R. R is married to S. How is S related to P?
a) Son b) Father c) Uncle d) Father-in-law
- G is the daughter of H. H is the sister of I. I is the father of J. How is G related to J?
a) Sister b) Aunt c) Cousin d) Grandmother

⇒ VERBAL ABILITY ⇐

⇒ ARTICLES & PREPOSITIONS ⇐

1) ARTICLES :

Articles are small words used before nouns.

There are three articles in English : a, an, the

ARTICLE	USAGE	EXAMPLE
a	Used before singular countable nouns starting with a consonant sound.	- a book - a university
an	Used before singular countable nouns starting with a vowel sound (a, e, i, o, u).	- an apple - an hour
the	Used for specific nouns or nouns already known to the listener/reader.	- the sun - the Taj Mahal

2) RULES TO USE ARTICLES :

- ★ Use "a" before singular nouns starting with consonant sound.
- ★ Use "an" before singular nouns starting with vowel sound.
- ★ Use "the" before specific nouns.
- ★ Use "the" before superlatives. (the best)
- ★ Use "the" before ordinal numbers. (the first)
- ★ Use "the" before unique things. (the earth, the sky)
- ★ Do not use any article before plural or uncountable nouns. (e.g., I like books. / Water is essential.)

3) PREPOSITIONS :

Prepositions are words used before a noun, pronoun or noun phrase to show the relation to another word in the sentence.

4) COMMON PREPOSITIONS AND USAGE :

PREPOSITION	USAGE	EXAMPLE
in	Inside a place/area	• She is in the room.
on	On a surface	• The book is on the table.
at	At a specific point	• He arrived at 5 o'clock.
by	Near / beside / by means of	• He is sitting by the window.
with	Together with	• I went to market with my friend.
for	For a purpose / duration	• This gift is for you.
to	Towards	• She is going to school.
of	Belonging to	• The cover of the book.
from	Starting point	• He came from Delhi.
into	Movement to inside	• He jumped into the pool.

8) PRACTICE QUESTIONS (TCS LEVEL) :

- She is _____ honest girl.
a) a b) an c) the d) no article
- I saw _____ elephant in the zoo.
a) a b) an c) the d) no article
- He is _____ best player in the team.
a) a b) an c) the d) no article
- We should respect _____ teachers.
a) a b) an c) the d) no article
- He is sitting _____ the chair.
a) in b) on c) at d) by

5) COMMON MISTAKES :

- Using 'a' instead of 'an'. (an apple X / ✓)
- Using 'the' for general things. (I saw the dog. X)
- Using any article with plurals. (the books X)
- Wrong preposition usage.
(He is good in mathematics. X)
- Omitting preposition. (He is interested _____ music. X)

6) EXAMPLES :

- ★ She is an honest girl.
- ★ I saw a movie yesterday.
- ★ The Ganges is a holy river.
- ★ He is taller than his brother.
- ★ We should be thankful to our parents.
- ★ He is good at solving puzzles.
- ★ This is the pen I was looking for.

7) TIPS TO REMEMBER :

- ★ Think about the noun and its number.
- ★ Identify vowel/consonant sound, not just letter.
- ★ For "the", think: Is it specific or already known?
- ★ For prepositions, think of the relation in the sentence.
- ★ Read more to improve natural usage. 😊

- She reached _____ the station early.
a) in b) on c) at d) by
- This pen is _____ my brother.
a) in b) on c) of d) with
- Please give this book _____ him.
a) to b) for c) by d) at
- They are going _____ Mumbai tomorrow.
a) on b) to c) in d) into
- He came _____ the office late.
a) in b) at c) on d) from

ANSWER KEY : 1-b 2-b 3-c 4-c 5-b
6-c 7-c 8-a 9-b 10-d

TCS TIP : Learn the rules → Practice daily →
Analyze mistakes → You will master it! 😊

VOCABULARY

IDIOMS & PHRASES

1) WHAT ARE IDIOMS ?

Idioms are phrases whose meaning is different from the literal meaning of the words.

Example :

- ✓ Break the ice → Start a conversation
- ✓ Hit the nail on the head → Exactly correct

2) COMMONLY USED IDIOMS :

IDIOM	MEANING	EXAMPLE
Break the ice	Start a conversation	He told a joke to break the ice.
Hit the nail on the head	Exactly correct	You hit the nail on the head.
Spill the beans	Reveal a secret	Don't spill the beans about the party.
Once in a blue moon	Very rarely	I visit my village once in a blue moon.
Bite the bullet	Face a difficult situation bravely	He bit the bullet and applied again.
Under the weather	Feeling ill	I'm a bit under the weather.
Piece of cake	Very easy	The test was a piece of cake.
Burn the midnight oil	Work late into the night	She burned the midnight oil to finish the project.

5) TIPS TO REMEMBER :

- ★ Read idioms and phrases daily.
- ★ Learn with examples - it helps in remembering.
- ★ Try to use them in your sentences.
- ★ Revise regularly and note new ones.
- ★ Practice is the key to improve vocabulary! 😊

7) PRACTICE QUESTIONS (TCS LEVEL) :

- He tried to break the ice by telling a funny story.
a) speak clearly b) start a conversation c) make noise d) stop talking
- You hit the nail on the head with your explanation.
a) missed the point b) spoke correctly c) spoke loudly d) avoided question
- Don't spill the beans about the surprise party.
a) eat something b) reveal a secret c) break a jar d) cook food
- I visit my grandparents once in a blue moon.
a) very often b) rarely c) every day d) never
- He had to bite the bullet and accept the truth.
a) accept bravely b) eat something c) sleep well d) run away
- She felt under the weather and didn't go to college.
a) very excited b) not feeling well c) very happy d) very angry
- The assignment was a piece of cake for him.
a) very hard b) very easy c) very long d) very boring
- He had to burn the midnight oil to finish his work.
a) sleep early b) work late c) take a break d) go for walk
- He was welcomed with open arms by the family.
a) with anger b) with warm welcome c) with fear d) with silence
- He succeeded in spite of many problems.
a) because of b) with the help of c) although d) due to

3) WHAT ARE PHRASES ?

Phrases are group of words that work together but do not contain a subject and a verb.

Example :

- ✓ in the morning
- ✓ with great effort
- ✓ because of you

4) COMMON PHRASES & THEIR MEANINGS :

PHRASE	MEANING
As soon as possible	At the earliest time
At the end of the day	Ultimately
In a hurry	With speed
On the other hand	In contrast
Out of the question	Not possible
With open arms	With a warm welcome
In the long run	In the future
At any rate	Anyway / In any case
By and large	Generally
In spite of	Despite

6) COMMON MISTAKES :

- Using idioms in the wrong situation.
- Understanding idioms literally.
- Forgetting prepositions in phrases.
- Spelling mistakes.
- Not using idioms and phrases in practice.

- 8) ANSWER KEY : 1-b 2-b 3-b 4-b 5-a 6-b 7-b 9-b 10-c
11-b 12-b 13-b 14-a 15-b 16-b 17-b 18-a 19-a 20-b

- I will call you as soon as possible.
a) immediately b) at the earliest time c) after some time d) never
- At the end of the day, honesty is important.
a) in the morning b) finally c) at noon d) at night
- I am in a hurry, so I can't talk now.
a) moving slowly b) with speed c) feeling tired d) feeling happy
- He is intelligent. On the other hand, he is lazy.
a) in contrast b) in addition c) similarly d) as a result
- This plan is out of the question.
a) possible b) not possible c) very good d) important
- They received the guest with open arms.
a) with anger b) with warm welcome c) with fear d) with doubt
- In the long run, hard work always pays off.
a) in the past b) in the future c) right now d) yesterday
- At any rate, we should try our best.
a) anyway b) as a result c) for example d) because
- By and large, the meeting was successful.
a) generally b) usually c) rarely d) never
- She succeeded in spite of facing many failures.
a) because of b) despite c) due to d) with the help of

TCS TIP : Learn 2-3 idioms & phrases every day →
Use in sentences → Build strong vocabulary! 😊

1) WHAT ARE DATA STRUCTURES ?

Data Structures are ways to organize and store data in computer memory so that operations can be performed efficiently.

Goal : To manage data in an efficient way.

2) TYPES OF DATA STRUCTURES :

TYPE	DESCRIPTION	EXAMPLES
Primitive (Built-in)	Basic data types provided by the language.	int, char, float, double, boolean
Non-Primitive (User Defined)	Derived from primitive types. Used to store collections of data.	Array, Linked List, Stack, Queue, Tree, Graph
Linear	Elements are arranged in a sequence.	Array, Linked List, Stack, Queue
Non-Linear	Elements are not arranged sequentially.	Tree, Graph

3) ARRAY (Linear DS) :

- ★ Stores elements of same data type in contiguous memory locations.
- ★ Access time is $O(1)$.

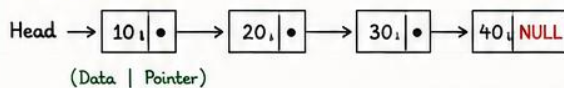
Example : `int arr[5] = {10, 20, 30, 40, 50};`

Index :	0	1	2	3	4
Value :	[10]	[20]	[30]	[40]	[50]

4) LINKED LIST (Linear DS) :

- ★ A collection of nodes where each node contains data and a pointer to the next node.
- ★ Dynamic size.

Singly Linked List :

**9) PRACTICE QUESTIONS (TCS LEVEL) :**

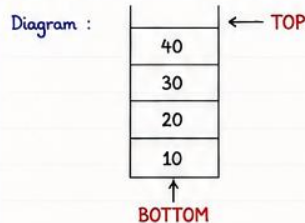
- Which data structure uses LIFO principle?
a) Queue b) Stack c) Array d) Linked List
- In a queue, insertion takes place at _____.
a) Front b) Rear c) Middle d) Any position
- Which data structure allows random access to elements?
a) Array b) Linked List c) Stack d) Queue
- In a stack, deletion is performed from _____.
a) Bottom b) Top c) Middle d) Rear
- Which of the following is a non-linear data structure?
a) Array b) Stack c) Tree d) Queue

10) **ANSWER KEY :** 1-b 2-b 3-a 4-b 5-c
6-b 7-c 8-b 9-c 10-b

5) STACK (Linear DS) :

- ★ Follows LIFO (Last In First Out) principle.
- ★ Insertion and deletion at one end called TOP.

Operations : Push (insert), Pop (delete), Peek (top element)



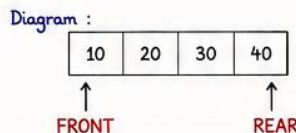
Example :

Push 10
Push 20
Push 30
Push 40
Pop → 40
Pop → 30
Top → 20

6) QUEUE (Linear DS) :

- ★ Follows FIFO (First In First Out) principle.
- ★ Insertion at REAR and deletion at FRONT.

Operations : Enqueue (insert), Dequeue (delete), Front, Rear



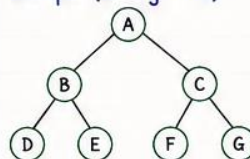
Example :

Enqueue 10, 20, 30, 40
Dequeue → 10
Dequeue → 20
Front → 30
Rear → 40

7) TREE (Non-Linear DS) :

- ★ Hierarchical structure of nodes.
- ★ One root node and subtrees.

Example (Binary Tree) :



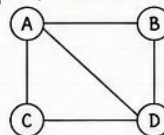
Terms :

- A is Root
- B, C are children of A
- D, E are children of B
- Leaf Nodes : D, E, F, G

8) GRAPH (Non-Linear DS) :

- ★ A collection of vertices (nodes) connected by edges.
- ★ Can be directed / undirected.

Example (Undirected Graph) :



Applications :

- Social networks
- Maps and routes
- Web page linking

- The starting node of a linked list is called _____.
a) Tail b) Head c) Root d) Parent
- Which operation is used to insert an element in a queue?
a) Push b) Pop c) Enqueue d) Dequeue
- In a binary tree, the maximum number of children a node can have is _____.
a) 1 b) 2 c) 3 d) Any number
- Which data structure is used in recursion?
a) Queue b) Array c) Stack d) Linked List
- In graphs, edges connect _____.
a) Data items b) Nodes c) Arrays d) All of these

★ **TCS TIP :** Understand basic operations of each data structure → Practice diagrams → Solve more problems → Build strong concepts! 😊

1) DEFINITION :

Average is a value that is calculated by dividing the sum of all observations by the total number of observations.

2) BASIC FORMULAS :

TYPE	FORMULA
Simple Average	$\text{Average} = \frac{\text{Sum of all values}}{\text{Number of values}}$
Average with Frequencies	$\text{Average} = \frac{\text{Sum of values} \times \text{Frequency}}{\text{Sum of frequencies}}$
Combined Average	$\text{Average} = \frac{(\text{Sum}_1 + \text{Sum}_2 + \dots)}{(N_1 + N_2 + \dots)}$
Weighted Average	$\text{Average} = \frac{(a_1w_1 + a_2w_2 + \dots + a_nw_n)}{(w_1 + w_2 + \dots + w_n)}$

3) IMPORTANT POINTS :

- ★ The unit of average is the same as the unit of the given observations.
- ★ If each observation is increased/decreased by a number k , the average also becomes increased/decreased by k .
- ★ If each observation is multiplied/divided by a number k , the average is also multiplied/divided by the same number k .
- ★ Average is affected by extreme values.
- ★ Always add all values first, then divide.

5) PRACTICE QUESTIONS (TCS LEVEL) :

- The average of 5 numbers is 20. If one number 30 is added to the group, the new average is:
a) 22 b) 24 c) 23 d) 25
- The average of 8 numbers is 16. The sum of the numbers is:
a) 112 b) 118 c) 120 d) 128
- The average of A and B is 25. If A is 10 more than B, then A is:
a) 30 b) 35 c) 25 d) 40
- The average of first 6 natural numbers is:
a) 3.5 b) 3 c) 4 d) 3.6
- The average of 3 numbers is 40. If two of them are 30 and 50, the third number is:
a) 30 b) 35 c) 40 d) 45

4) EXAMPLES :

- ① Find the average of 12, 18, 24, 30, 36.

Solution : Sum = 12 + 18 + 24 + 30 + 36 = 120

Number of values = 5

$$\text{Average} = \frac{120}{5} = 24$$

- ② Find the average of numbers 10, 20, 30 with frequencies 2, 3, 5.

Solution : Sum = $(10 \times 2) + (20 \times 3) + (30 \times 5)$

$$= 20 + 60 + 150 = 230$$

Sum of frequencies = 2 + 3 + 5 = 10

$$\text{Average} = \frac{230}{10} = 23$$

- ③ The average of 6 numbers is 15. Find the sum of the numbers.

Solution : Average = $\frac{\text{Sum}}{\text{Number of values}}$

$$15 = \frac{\text{Sum}}{6} \Rightarrow \text{Sum} = 15 \times 6 = 90$$

- ④ The average of boys' marks is 72 and the average of girls' marks is 68. The average of all 50 boys and 50 girls is?

Solution : Total = $(72 \times 50) + (68 \times 50)$

$$= 3600 + 3400 = 7000$$

$$\text{Average} = \frac{7000}{50 + 50} = \frac{7000}{100} = 70$$

- The average weight of 30 students is 45 kg. If 5 more students with average weight 50 kg join the group, the new average weight is:
a) 46 b) 46.5 c) 47 d) 48
- The average of 4 consecutive even numbers is 11. The numbers are:
a) 8, 10, 12, 14 b) 9, 11, 13, 15
c) 7, 9, 11, 13 d) 10, 12, 14, 16
- The average of 3 numbers is 18. Two of the numbers are 15 and 21. The third number is:
a) 16 b) 18 c) 20 d) 21
- The average of 10 numbers is 24. If each number is increased by 6, the new average is:
a) 24 b) 28 c) 30 d) 32
- The average of 5 numbers is 13. If each number is multiplied by 3, the new average is:
a) 13 b) 23 c) 26 d) 39

- 8) ANSWER KEY : 1-c 2-c 3-a 4-a 5-b
6-b 7-a 8-b 9-b 10-c

★ TCS TIP : Understand the formula → Practice daily
→ Do mental math → Solve previous year questions. 😊

DIRECTION SENSE

1) DEFINITION :

Direction Sense questions test your ability to find directions and positions. You have to imagine directions in all the four main directions : North, South, East and West.

2) BASIC DIRECTIONS :

DIRECTION	SYMBOL	OPPOSITE
North	↑	South
South	↓	North
East	→	West
West	←	East

3) INTERMEDIATE DIRECTIONS :

DIRECTION	SYMBOL	REPRESENTATION
North-East (NE)	↗	
South-East (SE)	↘	
South-West (SW)	↙	
North-West (NW)	↖	

4) IMPORTANT RULES :

- ★ Always face North unless otherwise mentioned.
- ★ Turn to your Right → Clockwise.
- ★ Turn to your Left → Anticlockwise.
- ★ Right turn = 90° Left turn = 90°
- ★ Two right turns = 180°
- ★ Three right turns = 270° (or one left turn)
- ★ Four right turns = 360° (back to original direction)
- ★ Use the diagram or rough sketch to avoid mistakes.

5) COMMON MISTAKES :

- Confusing left and right.
- Not maintaining the reference direction.
- Forgetting the number of turns.
- Mistakes in opposite directions.
- Not drawing rough diagram.
- Overlooking units in distance based questions.

6) TIPS TO REMEMBER :

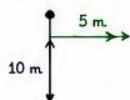
- ★ Draw a rough diagram.
- ★ Take North as reference.
- ★ Mark each step clearly.
- ★ Keep track of turns.
- ★ Practice more questions.



7) EXAMPLES :

- ① Starting from a point, A walks 10 m towards North, then turns right and walks 5 m. In which direction is he moving now?

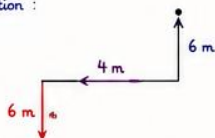
Solution :



Answer : East

- ② Starting from a point, B walks 6 m South, then 4 m West, then 6 m North. Where is he now?

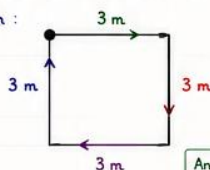
Solution :



Answer : West

- ③ Starting from a point, C goes 3 m East, 3 m South, 3 m West and 3 m North. Where is he now?

Solution :



Answer : At starting point

7) PRACTICE QUESTIONS (TCS LEVEL) :

- If North is to the left, then South will be to the _____.
a) Left b) Right
c) Front d) Back
- If you face East and turn right, you will face _____.
a) North b) South
c) West d) East
- If A walks 10 m North, then 5 m East, then turns left and walks 5 m, he is facing _____.
a) North b) East
c) South d) West
- From a point, P goes 7 m South, then 4 m West. In which direction is P from the starting point?
a) North-West b) South-West
c) North-East d) South-East
- If you are facing West and turn 180° to your left, which direction will you face?
a) North b) South
c) East d) West
- Starting from a point, R walks 5 m East, 5 m North, 5 m West and 5 m South. Where is R now?
a) 5 m North b) 5 m South
c) At starting point d) 5 m East
- If you face North and turn right 3 times, you will face _____.
a) North b) South
c) East d) West
- A goes 8 m North, then 6 m West. In which direction is A from the starting point?
a) North-West b) South-West
c) South-East d) North-East
- If East is to your right, then North is to your _____.
a) Front b) Left
c) Back d) Right
- Starting from a point, M walks 12 m South, then turns left and walks 5 m. In which direction is M moving now?
a) East b) West
c) North d) South

- 9) ANSWER KEY : 1-b 2-a 3-b 4-b 5-b
6-c 7-c 8-a 9-d 10-c

★ TCS TIP : Draw diagram → Take reference direction →

Move step by step → Keep track → Reach the answer! 😊

VERBAL ABILITY

ERROR SPOTTING

1) WHAT IS ERROR SPOTTING ?

Error Spotting is a type of question in which a sentence is divided into parts. One part contains an error. You have to find the part that has the error.

2) TYPES OF ERRORS :

TYPE	DESCRIPTION
Subject - Verb Agreement	Subject and verb do not agree in number.
Tense	Wrong use of tense.
Pronoun	Wrong use of pronouns.
Articles	Missing or wrong use of articles (a, an, the).
Prepositions	Wrong or missing preposition.
Adverbs	Wrong use or wrong position of adverbs.
Conjunctions	Wrong use of conjunctions.
Word Order	Incorrect arrangement of words.
Spelling	Spelling mistakes.
Redundancy	Repetition of words / unnecessary words.

5) TIPS TO SOLVE :

- ★ Read the whole sentence once.
- ★ Identify the subject and the verb.
- ★ Check grammar rules one by one.
- ★ If everything is correct, choose 'No Error'.
- ★ Practice more to increase accuracy and speed. 😊

6) PRACTICE QUESTIONS (TCS LEVEL) :

- My sister / have gone / to the temple / yesterday.
A B C D
a) A b) B c) C d) D e) No Error
- The teacher / explained the lesson / clearly and / patient.
A B C D
a) A b) B c) C d) D e) No Error
- He can / speaks English / very well / in the class.
A B C D
a) A b) B c) C d) D e) No Error
- She is / one of the best / dancer / in our school.
A B C D
a) A b) B c) C d) D e) No Error
- They were / playing cricket / since morning / now.
A B C D
a) A b) B c) C d) D e) No Error

3) IMPORTANT RULES :

- ★ Check the sentence part by part.
- ★ Find the subject and the verb, check their agreement.
- ★ Check the tense of the verb.
- ★ Look for correct use of articles and prepositions.
- ★ Check the correct use of pronouns.
- ★ Check the correct placement of adverbs.
- ★ Look for spelling mistakes.
- ★ Find the part that makes the sentence grammatically incorrect or meaningless.
- ★ If the sentence is correct, mark 'No Error'.

4) EXAMPLES :

- The students / are going / to the market / yesterday.
A B C D
Error → D (yesterday is used with past tense. Correct: were going)
- She don't / like to eat / apples / in the morning.
A B C D
Error → A (Subject 'She' is third person singular. Correct: She doesn't)
- He is / the honest boy / in the class / of all.
A B C D
Error → D ('of' should be followed by 'them' or remove it. Correct: of them / (No Error))

- I have / a less money / than my brother / .
A B C D
a) A b) B c) C d) D e) No Error
- Neither of the boys / were present / in the meeting / yesterday.
A B C C D
a) A b) B c) C d) D e) No Error
- She enjoy / reading books / every day / after school.
A B C D
a) A b) B c) C d) D e) No Error
- Everyone in the class / are excited / for the tour / next week.
A B C C D
a) A b) B c) C d) D e) No Error
- The baby / is crying / because he is / hungry.
A B C D
a) A b) B c) C d) D e) No Error

- ### 7) ANSWER KEY :
- 1-b 2-d 3-a 4-c 5-d
6-b 7-b 8-a 9-a 10-c

- ### ★ TCS TIP :
- Practice daily → Analyse mistakes →
Improve grammar rules → You will ace this section! 😊

⇒ FREQUENTLY CONFUSED WORDS ⇐

1) WHAT ARE FREQUENTLY CONFUSED WORDS ?

These are words that look similar, sound similar, or have similar meanings but are used in different ways.

Learning the difference helps in better communication and fewer mistakes.

2) COMMONLY CONFUSED WORDS :

WORD 1	WORD 2	DIFFERENCE
Accept	Except	Accept = to receive Except = other than
Affect	Effect	Affect = to influence Effect = result
Advice	Advise	Advice = a suggestion (noun) Advise = to give suggestion (verb)
Allusion	Illusion	Allusion = a reference Illusion = a false image
Complement	Compliment	Complement = complete/supplement Compliment = praise
Discrete	Discreet	Discrete = separate, distinct Discreet = careful, cautious
Principal	Principle	Principal = head of an institution Principle = rule or belief
Stationary	Stationery	Stationary = not moving Stationery = writing materials
Than	Then	Than = used in comparison Then = used for time/order
Lose	Loose	Lose = to misplace Loose = not tight

3) MORE CONFUSED WORDS :

WORD 1	WORD 2	DIFFERENCE
Lay	Lie	Lay = to put something down Lie = to rest
Sit	Set	Sit = to be seated Set = to place
Bring	Take	Bring = to come with Take = to go with
Farther	Further	Farther = physical distance Further = more/ additionally
Hear	Here	Hear = to listen Here = in this place
Its	It's	Its = belonging to it It's = it is
Your	You're	Your = belonging to you You're = you are

4) TIPS TO REMEMBER :

- ★ Read the sentence carefully.
- ★ Understand the meaning of the whole sentence.
- ★ Identify the part of speech (noun, verb, adjective, etc.).
- ★ Think of the correct usage.
- ★ Practice daily to improve vocabulary. 😊

5) PRACTICE QUESTIONS (TCS LEVEL) :

- I will accept your invitation _____ you are busy.
a) accept b) except c) effect d) affect
- Climate change will _____ the lives of future generations.
a) effect b) affect c) reflect d) infect
- She gave me good _____ on how to improve my skills.
a) advice b) advise c) device d) advise
- The magician created an amazing _____ on the stage.
a) allusion b) illusion c) delusion d) confusion
- Your dress perfectly _____ your shoes.
a) complement b) compliment c) complete d) compliment
- He was very _____ while handling the secret information.
a) discrete b) discreet c) discreate d) discreatee
- The _____ of the school addressed the students.
a) principal b) principle c) principle d) principals
- Please keep the files in the _____ drawer.
a) stationary b) stationery c) stationairy d) stationeries
- He is taller _____ his brother.
a) than b) then c) that d) there
- Don't _____ your keys. I saw them on the table.
a) loose b) lose c) looss d) looz

- 6) ANSWER KEY : 1-b 2-b 3-a 4-b 5-a
6-b 7-a 8-b 9-a 10-b

★ TCS TIP : Learn new words daily → Use in sentences →
Read books, editorials → Build strong vocabulary! 😊

1) WHAT IS DBMS ?

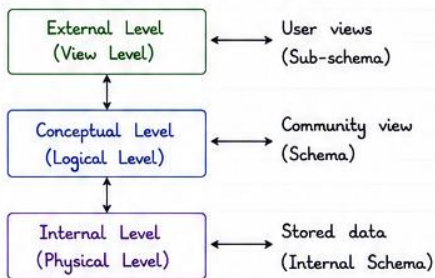
DBMS (Database Management System) is software that helps us to define, create, maintain and control access to a database.

It acts as an **interface** between the user and the database.

2) KEY FEATURES :

- ★ Data Independence (reduces dependence between data and programs)
- ★ Data Security (protects data from unauthorized access)
- ★ Data Integrity (maintains accuracy and consistency)
- ★ Concurrency Control (handles multiple users together)
- ★ Backup & Recovery (prevents data loss)
- ★ Reduced Data Redundancy
- ★ Easy Data Access

3) DBMS ARCHITECTURE (3-LEVEL) :



4) DATA MODELS :

MODEL	DESCRIPTION
Hierarchical Model	Data is organized in a tree structure.
Network Model	Data is organized in a graph structure.
Relational Model	Data is organized in tables (rows & columns).
Entity-Relationship (ER) Model	Uses entities, relationships and attributes.

7) PRACTICE QUESTIONS (TCS LEVEL) :

- DBMS stands for _____.
a) Data Base Management System b) Data Backup Management System
c) Database Management System d) Data Bank Management System
- Which of the following is NOT a key feature of DBMS?
a) Data Security b) Data Redundancy c) Concurrency Control d) Backup & Recovery
- In a relational database, data is stored in the form of _____.
a) Trees b) Tables c) Graphs d) Networks
- Which SQL command is used to retrieve data from a table?
a) GET b) SELECT c) FETCH d) RETRIEVE
- Which of the following is used to remove a table from the database?
a) REMOVE b) DELETE c) DROP d) ERASE

5) RELATIONAL DATABASE BASICS :

- ★ Data is stored in the form of tables (relations).
- ★ Each table has rows (tuples) and columns (attributes).
- ★ A table has a Primary Key to uniquely identify each row.
- ★ Tables are related using Foreign Keys.

Example Table - STUDENT

Roll_No (PK)	Name	Dept	Age
101	Rahul	CSE	20
102	Priya	IT	21
103	Karan	ECE	20
104	Sneha	CSE	22

PK - Primary Key (unique for each record)

6) SQL BASICS :

COMMAND	USE
SELECT	To retrieve data from a table
INSERT	To insert new records
UPDATE	To modify existing records
DELETE	To delete records
CREATE	To create database/table
DROP	To delete database/table

Example SQL Queries :

- Select all records : **SELECT** * FROM Student;
- Select specific columns : **SELECT** Name, Dept FROM Student;
- Insert a record : **INSERT** INTO Student **VALUES** (105, 'Amit', 'IT', 21);
- Update a record : **UPDATE** Student **SET** Age = 22 **WHERE** Roll_No = 102;
- Delete a record : **DELETE** FROM Student **WHERE** Roll_No = 103;

- ### 8) ANSWER KEY :
- 1-c 2-b 3-b 4-b 5-c
6-b 7-b 8-b 9-c 10-b

★ **TCS TIP** : Understand DBMS concepts → Practice SQL queries → Solve more problems → You will master it! 😊

⇒ QUANTITATIVE APTITUDE ⇐

⇒ TIME, SPEED & DISTANCE ⇐

1) DEFINITION :

Speed is the distance traveled by an object in unit time.

It is measured in km/h, m/s, etc.

2) BASIC FORMULAS :

CONCEPT	FORMULA	NOTES
Speed	$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$	$S = D/T$
Distance	$\text{Distance} = \text{Speed} \times \text{Time}$	$D = S \times T$
Time	$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$	$T = D/S$
Unit Conversion	$1 \text{ km/h} = \frac{5}{18} \text{ m/s}$ $1 \text{ m/s} = \frac{18}{5} \text{ km/h}$	

3) IMPORTANT POINTS :

- ★ Speed, Distance and Time are directly proportional ($S \propto D$, $S \propto 1/T$).
- ★ If speed is increased, time decreases for the same distance.
- ★ If speed is decreased, time increases for the same distance.
- ★ Always keep units same before calculation.
- ★ Average speed = Total distance / Total time

5) PRACTICE QUESTIONS (TCS LEVEL) :

- A car covers 300 km in 5 hours. What is its speed?
a) 50 km/h b) 60 km/h c) 70 km/h d) 80 km/h
- A train travels at 90 km/h. How far will it travel in 6 hours?
a) 450 km b) 500 km c) 540 km d) 600 km
- A man walks 12 km in 3 hours. What is his speed?
a) 3 km/h b) 4 km/h c) 5 km/h d) 6 km/h
- A car covers a distance of 180 km at a speed of 60 km/h. How long will it take?
a) 2 hours b) 2.5 hours c) 3 hours d) 3.5 hours
- A bus covers 150 km in 3 hours. Find the distance covered in 5 hours at the same speed.
a) 240 km b) 250 km c) 260 km d) 270 km

4) EXAMPLES :

- ① A car covers a distance of 240 km in 4 hours. Find its speed.

$$\begin{aligned} \text{Solution : Speed} &= \frac{\text{Distance}}{\text{Time}} \\ &= \frac{240}{4} = \boxed{60 \text{ km/h}} \end{aligned}$$

- ② A train runs at a speed of 72 km/h. How much distance will it cover in 5 hours?

$$\begin{aligned} \text{Solution : Distance} &= \text{Speed} \times \text{Time} \\ &= 72 \times 5 = \boxed{360 \text{ km}} \end{aligned}$$

- ③ A cyclist covers 90 km at a speed of 15 km/h. Find the time taken.

$$\begin{aligned} \text{Solution : Time} &= \frac{\text{Distance}}{\text{Speed}} \\ &= \frac{90}{15} = \boxed{6 \text{ hours}} \end{aligned}$$

- ④ A man walks 8 km in 2 hours. What is his speed in m/s?

$$\begin{aligned} \text{Solution : Speed} &= \frac{8}{2} = 4 \text{ km/h} \\ &= \frac{4 \times 5}{18} = \boxed{\frac{10}{9} \text{ m/s}} \end{aligned}$$

- A cyclist rides at a speed of 18 km/h. How much time will he take to cover 72 km?
a) 3 hours b) 4 hours c) 5 hours d) 6 hours
- A train covers 360 km in 4 hours. What is its speed in m/s?
a) 20 m/s b) 25 m/s c) 30 m/s d) 36 m/s
- A car covers half the distance at 40 km/h and the remaining half at 60 km/h. What is its average speed?
a) 45 km/h b) 48 km/h c) 50 km/h d) 55 km/h
- A man runs at a speed of 10 m/s. Find the distance he covers in 15 minutes.
a) 7.5 km b) 8 km c) 9 km d) 10 km
- A boat moves downstream at 15 km/h and upstream at 10 km/h. What is the average speed for a trip downstream and back upstream covering equal distances?
a) 11.25 km/h b) 12 km/h c) 12.5 km/h d) 13 km/h

- 6) ANSWER KEY :** 1-b 2-b 3-b 4-c 5-d
6-b 7-b 8-c 9-b 10-a

★ **TCS TIP :** Understand the formulas → Practice daily questions → Keep units consistent → You will master it! 😊

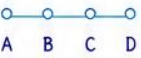
LOGICAL REASONING

SEATING ARRANGEMENT

1) INTRODUCTION :

Seating Arrangement questions test your ability to arrange people or objects in a particular order or position based on given conditions.

2) TYPES OF SEATING ARRANGEMENTS :

TYPE	DESCRIPTION	EXAMPLE
Linear (Row)	People sit in a straight line, facing the same or opposite direction.	
Circular (Facing Center)	People sit around a circle, facing the center.	
Circular (Facing Outside)	People sit around a circle, facing outside.	
Rectangular (Table)	People sit around a rectangle or square table.	

3) IMPORTANT POINTS :

- ★ Read all the conditions carefully before solving.
- ★ Identify the type of arrangement.
- ★ Fix one position and place others accordingly.
- ★ Use diagrams to avoid confusion.
- ★ Apply elimination method to reduce options.
- ★ Check all conditions after placing each person.
- ★ Practice regularly to improve speed and accuracy. 😊

6) PRACTICE QUESTIONS (TCS LEVEL) :

- Five friends A, B, C, D and E sit in a straight line facing North. B sits to the immediate right of A. C sits to the immediate left of E. D sits between B and C. Who sits in the middle?
a) A b) B c) C d) D
- Six people P, Q, R, S, T and U sit in a row facing South. P is second to the right of Q. R sits third to the left of U. T sits between P and R. Who is second from the left?
a) Q b) P c) T d) R
- Four persons A, B, C and D sit around a circular table facing center. A is to the immediate right of D. B is opposite D. Who is to the immediate right of B?
a) A b) D c) C d) Cannot be determined
- Seven persons are sitting in a row facing North. M is at one end. N is at the other end. O is third to the left of M. P is right in the middle. Q is second to the right of N. Who is fourth from the left?
a) O b) P c) Q d) N
- Five persons A, B, C, D and E are sitting around a circle facing outside. A is to the immediate left of B. C is opposite B. D is second to the right of C. Who is to the far right of A?
a) B b) C c) D d) E

4) COMMON CLUES AND MEANINGS :

CLUE / PHRASE	MEANING
Immediate right / left	Next to the right / left
Second to the right / left	One who is two places away on the right / left
Opposite	Face to face (in linear) 180° apart (in circular)
Between A and B	A person sits between them
Next to / Adjacent to	Immediately beside
Facing	Look towards
Same as	In the same direction
Different from	In the opposite direction
More than / Less than	At least one / At least two

5) TIPS TO REMEMBER :

- ★ Start with the most specific clue.
- ★ Use symbols or short forms while solving.
- ★ Draw rough diagrams quickly.
- ★ Keep checking all conditions.
- ★ If stuck, try reversing the arrangement.
- ★ Practice previous year questions. 😊

- 7) **ANSWER KEY :** 1-c 2-b 3-a 4-b 5-c
 6-d 7-b 8-a 9-c 10-a

★ **TCS TIP :** Understand the type → Identify key clues → Draw diagram → Place logically → Check all conditions → You will solve any seating arrangement easily! 😊

⇒ SENTENCE IMPROVEMENT ⇐

1) WHAT IS SENTENCE IMPROVEMENT ?

In Sentence Improvement questions, a sentence is given with an underlined part. You have to choose the option that improves the sentence grammatically and makes it meaningful.

2) COMMON GRAMMATICAL CORRECTIONS :

ERROR TYPE	INCORRECT	CORRECT	RULE / TIP
Subject - Verb Agreement	He go to school.	He goes to school.	Singular subject takes singular verb.
Tense	She will go to market yesterday.	She went to market yesterday.	Use correct tense according to time.
Articles	I saw a elephant in the zoo.	I saw an elephant in the zoo.	Use a, an, the properly.
Prepositions	He is absent in the class.	He is absent from the class.	Use correct preposition.
Pronouns	Me and he went to the park.	He and I went to the park.	Use correct subject and object forms.
Adverbs	She sings good.	She sings well.	Use correct adverb form.
Word Order	Never I have seen such a view.	I have never seen such a view.	Follow correct word order.
Redundancy	He is a unique person.	He is a unique person.	Remove extra or repeated words.
Parallelism	She likes singing, dancing and to play.	She likes singing, dancing and playing.	Use same form of words.
Spelling	He recieved the award.	He received the award.	Check spelling carefully.

3) TIPS TO IMPROVE SENTENCES :

- ★ Read the whole sentence carefully.
- ★ Identify the underlined part and its error.
- ★ Check subject, verb, tense and agreement.
- ★ Use correct articles and prepositions.
- ★ Avoid unnecessary repetition.
- ★ Choose the option that makes the sentence clear, correct and meaningful. 😊

6) PRACTICE QUESTIONS (TCS LEVEL) :

- He don't know the answer.
a) doesn't knows b) doesn't know c) don't knows d) don't know
- The boys are playing cricket in the ground.
a) on b) at c) in the d) on the
- Me and my friend went to the mall.
a) I b) Myself c) Me d) Mine
- She have been working here since 2019.
a) has b) having c) had d) have
- He is interested for sports.
a) in b) at c) to d) for

7) ANSWER KEY : 1-b 2-d 3-a 4-a 5-a
6-a 7-a 8-c 9-a 10-b

4) EXAMPLES :

- She don't like to eat spicy food.
a) dosen't likes b) doesn't likes c) doesn't like d) don't likes
Solution : Correct option → (c) doesn't like
Explanation : 'She' is third person singular, so use 'doesn't like'.
- They was playing football in the ground.
a) is b) are c) were d) be
Solution : Correct option → (c) were
Explanation : 'They' is plural, so use 'were' with past continuous.
- I have a pen. Give it to me.
a) I b) myself c) mine d) me
Solution : Correct option → (a) I
Explanation : 'Give it to I' is incorrect. Use 'I' (subject form).
- He is good in mathematics.
a) at b) on c) for d) at
Solution : Correct option → (c) for
Explanation : 'Good at' is the correct phrase.
- She sings good and dances well.
a) better b) well c) nicely d) best
Solution : Correct option → (b) well
Explanation : 'Sings well' is correct adverb usage.

5) KEY POINTS TO REMEMBER :

- ★ Subject and verb must agree.
- ★ Tense should match the time.
- ★ Use correct articles 'a', 'an', 'the'.
- ★ Prepositions show relationships — use them correctly.
- ★ Pronouns must be in correct form (I, me, he, him, etc.).
- ★ Choose the option that improves the meaning and grammar. 😊

- I have never seen such a beautiful place.
a) never b) always c) usually d) sometimes
- The teacher said, "Do your homework."
a) does b) did c) doing d) done
- She can speaks English fluntly.
a) spoke b) speaking c) speak d) spoken
- He is taller then his brother.
a) than b) them c) that d) those
- The project was completed by they.
a) them b) they c) their d) themselves

★ TCS TIP : Read more English daily → Identify errors
→ Practice regularly → Improve accuracy → You will ace it! 😊

VOCABULARY

PHRASAL VERBS

1) WHAT ARE PHRASAL VERBS ?

Phrasal verbs are combinations of a verb and a preposition or adverb. They have a meaning different from the individual words.
They are widely used in spoken and written English.

2) COMMON PHRASAL VERBS :

PHRASAL VERB	MEANING	EXAMPLE
Give up	To stop trying	Don't give up. You can do it!
Look after	To take care of	She looks after her younger sister.
Get up	To rise from bed	I get up at 6 o'clock every day.
Turn on	To switch on (a machine)	Please turn on the fan.
Turn off	To switch off (a machine)	Don't forget to turn off the lights.
Find out	To discover information	I will find out the time of the meeting.
Bring up	To raise a child	He was brought up in a small village.
Look forward to	To anticipate with pleasure	I look forward to meeting you.
Run out of	To have none left	We ran out of milk.
Put off	To postpone	They put off the meeting until next week.

3) TIPS TO REMEMBER :

- ★ Learn phrasal verbs in groups.
- ★ Read sentences to understand usage.
- ★ Practice daily in conversations and writing.
- ★ Make your own sentences for each phrasal verb.
- ★ Use flashcards to revise regularly.



6) PRACTICE QUESTIONS (TCS LEVEL) :

- Don't _____ your dreams. Keep working hard.
a) give away b) give off c) give up d) give in
- She always _____ her younger brother after school.
a) looks at b) look for c) looks after d) looked on
- Please _____ the lights before leaving the room.
a) turn up b) turn on c) turn off d) turn down
- I will _____ the answer and let you know.
a) find in b) find on c) find out d) find up
- We have _____ all the biscuits.
a) run over b) run into c) run out of d) run through

4) EXAMPLES IN SENTENCES :

- Give up
 - Meaning : To stop trying
 - Example : He never gives up easily.
- Look after
 - Meaning : To take care of
 - Example : Please look after my cat while I am away.
- Find out
 - Meaning : To discover information
 - Example : I will find out the result tomorrow.
- Look forward to
 - Meaning : To anticipate with pleasure
 - Example : We are looking forward to your visit.
- Run out of
 - Meaning : To have none left
 - Example : We ran out of water during the trip.
- Put off
 - Meaning : To postpone
 - Example : The match was put off due to rain.

5) PRACTICE TIP :

- ★ Do not translate phrasal verbs word for word.
- ★ Understand the context of the sentence.
- ★ Practice using phrasal verbs in daily life.
- ★ Improve vocabulary by reading newspapers, stories and articles.



- 7) ANSWER KEY : 1-c 2-c 3-c 4-c 5-c
6-c 7-c 8-a 9-c 10-b

- ★ TCS TIP : Learn phrasal verbs in context → Use them in sentences → Practice daily → Get fluent! 😊

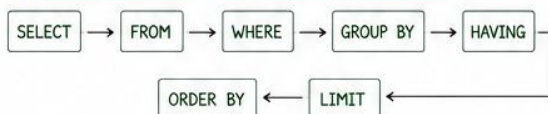
1) WHAT IS SQL ?

SQL (Structured Query Language) is a standard language used to store, retrieve, update and manage data in a Relational Database Management System (RDBMS).

2) SQL CATEGORIES :

TYPE	FULL FORM	PURPOSE
DDL	Data Definition Language	Define or modify database objects
DML	Data Manipulation Language	Manage data in tables
DCL	Data Control Language	Control access and permissions
TCL	Transaction Control Language	Manage transactions in the database

3) SQL CLAUSES (SELECT STATEMENT ORDER)



4) SQL COMMANDS OVERVIEW :

CATEGORY	COMMANDS	DESCRIPTION
DDL	CREATE, ALTER, DROP, TRUNCATE	Define, modify or delete database objects
DML	SELECT, INSERT, UPDATE, DELETE	Manipulate data in tables
DCL	GRANT, REVOKE	Control user permissions
TCL	COMMIT, ROLLBACK, SAVEPOINT	Manage transactions

7) PRACTICE QUESTIONS (TCS LEVEL) :

- Which SQL command is used to create a new table?
a) CREATE b) INSERT c) UPDATE d) SELECT
- Which clause is used to filter records?
a) FROM b) WHERE c) GROUP BY d) ORDER BY
- Which command is used to retrieve all columns from a table?
a) SELECT ALL FROM table;
c) GET * FROM table;
- Which command is used to add new records into a table?
a) ADD b) INSERT c) UPDATE d) APPEND
- Which command is used to remove a table from the database?
a) DELETE TABLE b) DROP TABLE
c) REMOVE TABLE d) ERASE TABLE

8) **ANSWER KEY :** 1-a 2-b 3-b 4-b 5-b
 6-a 7-b 8-a 9-b 10-c

5) COMMON SQL COMMANDS WITH EXAMPLES :

COMMAND	SYNTAX	EXAMPLE	DESCRIPTION
CREATE DATABASE	CREATE DATABASE db_name;	CREATE DATABASE College;	Creates a new database
CREATE TABLE	CREATE TABLE table_name (column1 datatype, column2 datatype, ...);	CREATE TABLE Student (RollNo INT, Name VARCHAR(50), Age INT);	Creates a new table
INSERT	INSERT INTO table_name (columns) VALUES (values);	INSERT INTO Student (RollNo, Name, Age) VALUES (101, 'Amit', 20);	Inserts data into table
SELECT	SELECT columns FROM table_name [WHERE condition];	SELECT * FROM Student; SELECT Name, Age FROM Student WHERE Age > 18;	Retrieves data from table
UPDATE	UPDATE table_name SET column = value WHERE condition;	UPDATE Student SET Age = 21 WHERE RollNo = 101;	Updates existing data
DELETE	DELETE FROM table_name WHERE condition;	DELETE FROM Student WHERE RollNo = 101;	Deletes data from table
DROP TABLE	DROP TABLE table_name;	DROP TABLE Student;	Deletes the table structure

6) SAMPLE TABLE : STUDENT

RollNo	Name	Dept	Age
101	Amit	CSE	20
102	Priya	IT	21
103	Karan	ECE	19
104	Sneha	CSE	22

KEY POINTS :

- ★ SQL is not case sensitive.
- ★ Each statement ends with a semicolon (;).
- ★ Use meaningful names for databases and tables.
- ★ Practice queries regularly to improve speed.



- Which SQL command is used to modify existing records?
a) UPDATE b) MODIFY c) CHANGE d) ALTER
- Which command is used to delete records from a table based on condition?
a) REMOVE b) DELETE c) DROP d) ERASE
- Which clause is used to sort the result set?
a) ORDER BY b) SORT BY c) ARRANGE BY d) GROUP BY
- Which command is used to give user permissions?
a) PERMIT b) GRANT c) ALLOW d) AUTHORIZE
- Which of the following is a DDL command?
a) INSERT b) UPDATE c) CREATE d) SELECT

★ **TCS TIP :** Learn basic SQL queries → Practice daily
→ Solve more problems → You will master SQL! 😊

1) DEFINITION :

Time and Work problems are based on the concept of work done by a person or a group in a given time. Work is either done completely or in parts. It is measured in terms of time.

2) BASIC FORMULAS :

CONCEPT	FORMULA	NOTES
Work Done by a Person	Work = Rate × Time ($W = R \times T$)	R = work done per unit time
Time Taken	Time = $\frac{\text{Work}}{\text{Rate}}$ ($T = W/R$)	If work and rate are known, time can be found.
Rate of Work	Rate = $\frac{\text{Work}}{\text{Time}}$ ($R = W/T$)	If work and time are known, rate can be found.
Total Work	Total Work = 1 Unit (or LCM of times)	We usually consider total work as 1 unit.
Efficiency	Efficiency = $\frac{1}{\text{Time taken}}$	More efficiency means less time.
Combined Work	Total Rate = $R_1 + R_2 + R_3 + \dots$	For persons working together.
Work Left	Work Left = (1 - Work Done)	If total work = 1 unit

3) IMPORTANT POINTS :

- ★ If a person can do a work in 'n' days, his 1 day work = $1/n$.
- ★ If a person's efficiency increases 'k' times, new time = original time / k.
- ★ If a person's efficiency decreases 'k' times, new time = original time × k.
- ★ If A can do a work in x days and B in y days, A and B together can do it in = $\frac{xy}{x+y}$ days.
- ★ Always convert times to same unit before solving.

5) PRACTICE QUESTIONS (TCS LEVEL) :

- A can do a work in 18 days and B in 27 days. In how many days can they do it together?
a) 8 b) 9 c) 10 d) 12
- If 12 men can complete a work in 15 days, in how many days can 8 men complete the same work?
a) 20 b) 22.5 c) 24 d) 26
- A and B can do a work in 10 days. A alone can do it in 15 days. In how many days can B do it alone?
a) 20 b) 25 c) 30 d) 35
- A can do a work in 24 days. He works for 6 days and then B completes the remaining work in 12 days. In how many days can B do the whole work alone?
a) 30 b) 28 c) 36 d) 32
- 20 men can build a wall in 18 days. How many men are required to build the same wall in 12 days?
a) 24 b) 26 c) 28 d) 30

4) EXAMPLES :

- ① A can do a work in 10 days and B can do the same work in 15 days. In how many days can they do it together?

Solution :

$$\text{A's 1 day work} = \frac{1}{10}$$

$$\text{B's 1 day work} = \frac{1}{15}$$

$$\text{Total 1 day work} = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$$

So, together they can do the work in **6 days.**

- ② A and B can do a work in 12 days together. A works for 4 days and B works for 6 days. Find the fraction of work done.

Solution :

$$\text{A's 1 day work} = \frac{1}{12} \times \frac{\text{B's days}}{\text{Total days}} = \frac{1}{12} \times \frac{6}{12} = \frac{1}{24}$$

$$\text{A's work in 4 days} = \frac{1}{24} \times 4 = \frac{1}{6}$$

$$\text{B's 1 day work} = \frac{1}{12} \times \frac{\text{A's days}}{\text{Total days}} = \frac{1}{12} \times \frac{4}{12} = \frac{1}{36}$$

$$\text{B's work in 6 days} = \frac{1}{36} \times 6 = \frac{1}{6}$$

$$\text{Total work done} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

- ③ A can do a work in 20 days. He works alone for 8 days and then B joins him. Together they finish the remaining work in 6 days. In how many days can B do the whole work alone?

Solution :

$$\text{A's 1 day work} = \frac{1}{20}$$

$$\text{Work done by A in 8 days} = \frac{1}{20} \times 8 = \frac{2}{5}$$

$$\text{Work left} = 1 - \frac{2}{5} = \frac{3}{5}$$

$$\text{Remaining work done in 6 days} = \frac{3}{5}$$

$$\text{B's 1 day work} = \frac{3}{5} \div 6 = \frac{3}{30} = \frac{1}{10}$$

So, B alone can do the whole work in **10 days.**

6. A alone can do a work in 30 days and B alone in 45 days. They work together for 5 days. How much work is left?

$$\text{a) } \frac{1}{3} \quad \text{b) } \frac{1}{4} \quad \text{c) } \frac{1}{5} \quad \text{d) } \frac{1}{6}$$

7. A, B and C can do a work in 12 days. A and B together can do it in 18 days. B and C together can do it in 20 days. In how many days can C alone do it?

$$\text{a) 30} \quad \text{b) 36} \quad \text{c) 40} \quad \text{d) 45}$$

8. A can do 40% of a work in 6 days. In how many days can he do the whole work?

$$\text{a) 12} \quad \text{b) 15} \quad \text{c) 18} \quad \text{d) 20}$$

9. A can do a work in 16 days and B in 24 days. They work alternately for one day each starting with A. In how many days will the work be completed?

$$\text{a) 18} \quad \text{b) 20} \quad \text{c) 24} \quad \text{d) 28}$$

10. If 5 machines can produce 500 units in 8 hours, how many machines are required to produce 750 units in 10 hours?

$$\text{a) 6} \quad \text{b) 7} \quad \text{c) 8} \quad \text{d) 9}$$

- 6) ANSWER KEY : 1-b 2-b 3-c 4-d 5-b
6-b 7-c 8-a 9-c 10-b

★ TCS TIP : Understand the concepts → Practice regularly
→ Speed comes with practice → You will ace it! 😊

⇒ SYLLOGISMS ⇐

1) INTRODUCTION :

Syllogisms are statements in which a conclusion is drawn from two given statements called premises. They help in testing logical reasoning and deductive ability.

2) STANDARD FORM :

PART	DESCRIPTION
Major Premise	The first statement. It contains the major term.
Minor Premise	The second statement. It contains the minor term.
Conclusion	The final statement. It gives the result.
Terms	Major Term (M) - present in major premise and conclusion. Minor Term (m) - present in minor premise and conclusion. Middle Term (S) - present in both premises but not in conclusion.

3) FIGURES (POSITION OF MIDDLE TERM) :

FIGURE	DIAGRAM	DESCRIPTION
Figure - 1		Middle term (S) is the subject in both premises.
Figure - 2		Middle term (S) is the predicate in both premises.
Figure - 3		Middle term (S) is the subject in minor and predicate in major premise.
Figure - 4		Middle term (S) is the predicate in minor and subject in major premise.

4) IMPORTANT RULES :

- ★ There must be exactly three terms - M, m, S.
- ★ The middle term must be distributed at least once.
- ★ No term should be distributed in the conclusion.
- ★ If a premise is negative, the conclusion must be negative.
- ★ If a premise is particular, the conclusion must be particular.
- ★ From two negative premises, no conclusion follows.
- ★ From two particular premises, no conclusion follows.
- ★ If one premise is negative, the other must be affirmative.
- ★ If one premise is particular, the other must be universal. 😊

5) STANDARD STATEMENTS :

TYPE	SYMBOL	EXAMPLE	DIAGRAM
Universal Affirmative	A (All S are P)	All cats are animals.	
Universal Negative	E (No S are P)	No birds are animals.	
Particular Affirmative	I (Some S are P)	Some students are athletes.	
Particular Negative	O (Some S are not P)	Some dogs are not pets.	

6) TIPS TO SOLVE :

- ★ Identify the three terms - M, m, S.
- ★ Determine the figure.
- ★ Check the mood (A, E, I, O) of premises.
- ★ Apply the rules of syllogism.
- ★ Use Venn diagrams if required.
- ★ Practice regularly for speed and accuracy. 😊

7) PRACTICE QUESTIONS (TCS LEVEL) :

- All roses are flowers. All flowers are plants. Therefore, all roses are plants.
a) A b) E c) I d) O
- No pen is a book. All books are papers. Therefore, no pen is a paper.
a) A b) E c) I d) O
- Some chairs are tables. All tables are furniture. Therefore, some chairs are furniture.
a) A b) E c) I d) O
- All doctors are humans. Some humans are artists. Therefore, some doctors are artists.
a) A b) E c) I d) O
- No cow is a lion. Some lions are animals. Therefore, some animals are not cows.
a) A b) E c) I d) O
- Some mobiles are gadgets. No gadgets are toys. Therefore, some mobiles are not toys.
a) A b) E c) I d) O

8) ANSWER KEY : 1-a 2-b 3-c 4-c 5-c

★ TCS TIP : Understand the rules → Practice more questions
→ Draw diagrams → You will master syllogisms! 😊



1) WHAT IS FILL IN THE BLANKS ?

In this type of question, a sentence is given with a blank. You have to choose the most appropriate word from the options that fits the meaning and grammar of the sentence.

2) TYPES OF BLANKS :

TYPE	DESCRIPTION	EXAMPLE
Noun	A person, place, thing or idea.	She has a strong desire to become a <u>doctor</u> .
Verb	An action or state of being.	We should always <u>tell</u> the truth.
Adjective	A word that describes a noun.	He is a <u>talented</u> musician.
Adverb	A word that describes a verb, adjective or another adverb.	She sings <u>beautifully</u> .
Preposition	Shows relationship between a noun/pronoun and other words.	The book is <u>on</u> the table.
Conjunction	Joins words, phrases or clauses.	I stayed home <u>because</u> it was raining.
Article	'a', 'an', 'the' are articles.	He is <u>a</u> honest boy.

3) GENERAL TIPS :

- ★ Read the whole sentence carefully.
- ★ Understand the meaning of the sentence.
- ★ Identify the part of speech of the blank.
- ★ Check grammar rules (tense, subject-verb agreement).
- ★ Consider the context of the sentence.
- ★ Eliminate the wrong options.
- ★ Choose the word that fits best.



6) PRACTICE QUESTIONS (TCS LEVEL) :

- He has a great passion for _____ music.
a) to b) for c) at d) of
- The students listened to the teacher _____.
a) careful b) carefully c) care d) careless
- She is one of the _____ players in the team.
a) good b) better c) best d) well
- We must protect our _____.
a) environment b) environmental c) environments d) environmentally
- He is _____ honest that everyone trusts him.
a) so b) such c) very d) too

- 7) **ANSWER KEY :** 1-d 2-b 3-c 4-a 5-b
6-a 7-a 8-c 9-a 10-c

4) EXAMPLES :

- ① She has a strong desire to become a _____.
a) artist b) singing c) doctor d) quickly

Solution : Correct option → (c) doctor

Explanation : 'Doctor' fits the meaning and is a noun.

- ② We should always _____ the truth.
a) tell b) telling c) told d) tells

Solution : Correct option → (a) tell

Explanation : 'Tell' is the base form and fits the modal 'should'.

- ③ He is an _____ boy in the class.
a) honest b) honestly c) honesty d) more honest

Solution : Correct option → (a) honest

Explanation : 'Honest' is an adjective and describes 'boy'.

- ④ She sings _____.
a) beautiful b) beautifully c) more beautiful d) beauty

Solution : Correct option → (b) beautifully

Explanation : 'Beautifully' is an adverb describing the verb 'sings'.

- ⑤ The cat is sleeping _____ the chair.
a) on b) in c) at d) by

Solution : Correct option → (a) on

Explanation : 'On' shows the position on the surface.

5) POINTS TO REMEMBER :

- ★ Always match the word with the blank's part of speech.
- ★ Check the meaning and context of the sentence.
- ★ Look for clues around the blank.
- ★ Practice daily to improve vocabulary and grammar.
- ★ Confidence comes with consistent practice.



- I will call you as soon as I _____ home.
a) reach b) reached c) reaching d) reaches
- They worked _____ to complete the project on time.
a) hard b) hardly c) harder d) hardworking
- The movie was so _____ that we watched it again.
a) boring b) bored c) exciting d) excited
- She has been studying _____ last night.
a) since b) for c) from d) at
- He gave me a book _____ I was interested in.
a) who b) what c) which d) whose

★ **TCS TIP :** Read more → Learn new words → Practice
Fill in the Blanks questions daily → Improve accuracy! 😊

1) WHAT ARE COLLOCATIONS ?

Collocations are words that are often used together. They sound natural to native speakers. Using the right collocation improves your fluency and makes your English more professional.

2) COMMON COLLOCATIONS (HIGH FREQUENCY)

WORD	COLLOCATIONS
Make	a decision, a mistake, a plan, progress, sense, an effort, money
Do	homework, a favour, research, exercise, damage, business, your best
Take	a break, a decision, a risk, notes, a chance, care, responsibility
Have	a conversation, a look, a good time, an impact, patience, a bath
Give	advice, a presentation, a chance, a reason, support, a response
Get	a job, experience, permission, tired, better, ready
Go	for a walk, on a trip, to bed, on a diet, abroad, shopping
Break	a record, a promise, the law, a habit, the news

3) TIPS TO REMEMBER :

- ★ Learn collocations in groups, not as single words.
- ★ Read more - books, articles, newspapers.
- ★ Note down new collocations and use them in sentences.
- ★ Practice daily to make them a part of your vocabulary.
- ★ Do not translate word by word; think in English. 😊

6) PRACTICE QUESTIONS (TCS LEVEL) :

- She made _____ decision to change her career.
a) a b) the c) an d) no article
- You should take _____ every hour while studying.
a) a break b) a rest c) rest d) the break
- Do your _____ and you will succeed.
a) good b) better c) best d) well
- We had _____ time at the beach last weekend.
a) a good b) good c) the good d) good a
- He gave me some useful _____ about the project.
a) suggestion b) advise c) advice d) advising

4) EXAMPLES IN SENTENCES :

- Make a decision
• She made a quick decision to join the course.
- Take a break
• You should take a break after one hour of study.
- Do your best
• Just do your best and don't worry about results.
- Have a good time
• We had a good time at the concert last night.
- Give advice
• He gave me some useful advice for the interview.
- Get permission
• Get permission from your parents before you go.
- Go for a walk
• I go for a walk every morning to stay fit.
- Break a record
• The athlete broke the world record yesterday.

5) COMMONLY CONFUSED COLLOCATIONS :

INCORRECT ✗	CORRECT ✓	MEANING
Do a mistake	Make a mistake	To commit an error
Take a rest	Take a break	To rest for some time
Have a discussion	Have a conversation	To talk with someone
Strong desire	A strong desire	A powerful wish
Do a decision	Make a decision	To decide

- Please get _____ from your teacher before submitting.
a) a permission b) permission c) an permission d) the permission
- I usually go _____ in the evening.
a) for a walk b) for walk c) on a walk d) to walk
- The team broke _____ in the final match.
a) a record b) the record c) record d) no article
- Let's have _____ about your plans for the future.
a) a discussion b) a conversation c) discussion d) talk
- She is trying to make _____ in her life.
a) progress b) progressing c) progressed d) progression

- 7) ANSWER KEY : 1-a 2-a 3-c 4-a 5-c
 6-b 7-a 8-b 9-b 10-a

★ TCS TIP : Learn collocations → Use in sentences
→ Read daily → Speak & write regularly → You will sound natural and confident! 😊

TECHNICAL BASICS

OPERATING SYSTEMS

1) INTRODUCTION :

An Operating System (OS) is system software that acts as an interface between the user and the computer hardware.

It manages all the resources and provides services for computer programs.

2) FUNCTIONS OF OPERATING SYSTEM :

- ★ **Process Management** : Creates, schedules, and terminates processes.
- ★ **Memory Management** : Allocates and deallocates memory to processes.
- ★ **File Management** : Organizes files and manages file operations.
- ★ **I/O Management** : Handles input/output devices and data.
- ★ **Security** : Protects data and controls access to resources.
- ★ **User Interface** : Provides interface for user interaction (CLI / GUI).

3) TYPES OF OPERATING SYSTEMS :

TYPE	DESCRIPTION
Batch OS	Jobs are collected and executed in batches without user interaction.
Time Sharing OS	Allows multiple users to use the system simultaneously by sharing CPU time.
Multiprogramming OS	Several programs are kept in memory and CPU switches between them.
Multitasking OS	Allows a single user to run multiple tasks concurrently.
Real Time OS	Responds to input immediately within a specific time limit.
Distributed OS	Manages resources of multiple connected computers and appears as a single system.
Network OS	Provides services and shares resources over a network.

7) PRACTICE QUESTIONS (TCS LEVEL) :

- Which of the following is the main function of an Operating System?
 - Compiling programs
 - Managing hardware resources
 - Designing circuits
 - Writing documents
- Which component of OS is responsible for allocating memory to processes?
 - CPU Scheduler
 - Memory Manager
 - File Manager
 - Device Driver
- Which OS allows multiple users to use the system simultaneously?
 - Batch OS
 - Time Sharing OS
 - Real Time OS
 - Network OS
- Which of the following is NOT a function of Operating System?
 - Process Management
 - Virus Protection
 - File Management
 - I/O Management
- Which type of OS responds to input immediately within a specific time limit?
 - Batch OS
 - Real Time OS
 - Distributed OS
 - Time Sharing OS

8) **ANSWER KEY** : 1-b 2-b 3-b 4-b 5-b
6-c 7-c 8-b 9-c 10-b

4) FEATURES OF GOOD OPERATING SYSTEM :

FEATURE	DESCRIPTION
★ Reliability	Should be reliable and available always.
★ Efficiency	Should utilize resources efficiently.
★ Scalability	Should handle growth in users and resources.
★ Security	Should protect data and prevent unauthorized access.
★ User Friendly	Should be easy to learn and use.
★ Flexibility	Should be flexible to support new features.
★ Portability	Should run on different types of hardware.

5) COMMON OPERATING SYSTEMS :

OS NAME	TYPE	USED IN
Windows	Multiuser, Multitasking	PCs, Laptops, Servers
Linux	Multiuser	Servers, Desktops, Embedded Systems
macOS	Multiuser, Multitasking	Apple Computers
Android	Mobile OS	Smartphones, Tablets
iOS	Mobile OS	iPhones, iPads
Unix	Multiuser	Servers, Workstations

6) KEY POINTS TO REMEMBER :

- ★ OS is the manager of computer resources.
- ★ OS acts as an interface between user and hardware.
- ★ Main functions: Process, Memory, File, I/O, Security and UI management.
- ★ Different OS types are designed for different needs.
- ★ Examples: Windows, Linux, macOS, Android, iOS. 😊

- Which OS is mainly used in smartphones?
 - Windows
 - Linux
 - Android
 - Unix
- Which of the following is an example of a Mobile OS?
 - Windows
 - macOS
 - Android
 - Linux
- Which OS manages resources of multiple connected computers and appears as a single system?
 - Network OS
 - Distributed OS
 - Time Sharing OS
 - Batch OS
- Which feature of a good OS ensures protection of data and prevents unauthorized access?
 - Reliability
 - Efficiency
 - Security
 - Flexibility
- Which OS is developed by Apple Inc. for its computers?
 - Windows
 - macOS
 - Linux
 - Android

★ **TCS TIP** : Understand the role of OS in real life → Learn functions and types → Practice MCQs regularly → You will score full marks in this topic! 😊

1) INTRODUCTION :

Interest is the amount paid by a borrower to the lender for the use of money.

It is of two types :

- Simple Interest (S.I.) - calculated only on the principal.
- Compound Interest (C.I.) - calculated on the principal as well as on the accumulated interest.

2) BASIC FORMULAS :

TYPE	FORMULA	WHERE
Simple Interest (S.I.)	$S.I. = \frac{P \times R \times T}{100}$ $\text{Amount (A)} = P + S.I.$	P = Principal R = Rate (% per annum) T = Time (in years)
Compound Interest (C.I.)	$\text{Amount (A)} = P \left(1 + \frac{R}{100}\right)^T$ $C.I. = A - P$	P = Principal R = Rate (% per annum) T = Time (in years) A = Amount
General Relations	• Interest = Amount - Principal • Principal = Amount - Interest • Rate = $\frac{\text{Interest} \times 100}{P \times T}$ • Time = $\frac{\text{Interest} \times 100}{P \times R}$	Use these when time, rate or principal is unknown.
Half Yearly / Quarterly Compounding	$R' = \frac{R}{2}, \quad T' = 2T$ (Use in the formula of C.I.)	When interest is compounded half yearly.

3) IMPORTANT POINTS :

- ★ In Simple Interest, interest is calculated only on the principal.
- ★ In Compound Interest, interest is calculated on principal + accumulated interest.
- ★ For C.I., time (T) must be in years.
- ★ Amount in C.I. is always greater than in S.I.
- ★ When rate or time is given in months, convert into years.
- ★ 1 year = 12 months, 6 months = $\frac{1}{2}$ year, 3 months = $\frac{1}{4}$ year.
- ★ Read the question carefully to identify S.I. or C.I. 😊

5) PRACTICE QUESTIONS (TCS LEVEL) :

- Find the Simple Interest on ₹ 7,500 for 3 years at 12% p.a.
a) 2400 b) 2700 c) 3000 d) 3200
- Find the Amount on ₹ 6,000 for 2 years at 5% p.a. Simple Interest.
a) 6200 b) 6300 c) 6400 d) 6500
- Find the Compound Interest on ₹ 8,000 for 2 years at 10% p.a.
a) 1520 b) 1600 c) 1680 d) 1760
- Find the Amount on ₹ 10,000 for 3 years at 5% p.a. Compound Interest.
a) 11550 b) 11576 c) 11630 d) 11850
- If the Simple Interest on ₹ 12,000 for 3 years is ₹ 1800, find the rate of interest.
a) 4% b) 5% c) 6% d) 7%

4) EXAMPLES :

- ① Find the Simple Interest on ₹ 8,000 for 2 years at 10% p.a.

Solution :

$$S.I. = \frac{P \times R \times T}{100} = \frac{8000 \times 10 \times 2}{100} = 1600$$

- ② Find the Amount on ₹ 5,000 for 3 years at 8% p.a. Simple Interest.

Solution :

$$S.I. = \frac{5000 \times 8 \times 3}{100} = 1200$$

$$\text{Amount} = P + S.I. = 5000 + 1200 = 6200$$

- ③ Find the Compound Interest on ₹ 10,000 for 2 years at 10% p.a. compounded annually.

Solution :

$$A = P \left(1 + \frac{R}{100}\right)^T$$

$$= 10000 \left(1 + \frac{10}{100}\right)^2 = 10000 (1.1)^2$$

$$= 10000 \times 1.21 = 12100$$

$$C.I. = A - P = 12100 - 10000 = 2100$$

- ④ Find the Amount on ₹ 12,000 for 3 years at 5% p.a. Compound Interest.

Solution :

$$A = 12000 \left(1 + \frac{5}{100}\right)^3 = 12000 (1.05)^3$$

$$= 12000 \times 1.157625 = 13891.50$$

- ⑤ Find the Rate of Interest if the Simple Interest on ₹ 9,000 for 2 years is ₹ 1800.

Solution :

$$\text{Using } S.I. = \frac{P \times R \times T}{100}$$

$$1800 = \frac{9000 \times R \times 2}{100}$$

$$R = \frac{1800 \times 100}{9000 \times 2} = 10\%$$

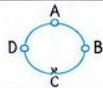
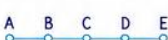
- ### 6) ANSWER KEY :
- 1-c 2-b 3-b 4-b 5-b
6-b 7-b 8-b 9-c 10-c

★ TCS TIP : Understand the formulas → Practice daily → Speed comes with practice → Accuracy brings success! 😊

1) INTRODUCTION :

Logical Puzzles test your ability to arrange, match and organize information based on given clues. They involve seating arrangements, ranking, ordering, grouping, etc.

2) TYPES OF LOGICAL PUZZLES :

TYPE	DESCRIPTION	EXAMPLE
Seating Arrangement	Arranging people around a table, in a row, etc.	
Linear Arrangement	Arranging people/things in a straight line.	
Ranking Puzzles	Finding positions of people in a race, exam, height, marks, etc.	1st > 2nd > 3rd > 4th
Categorization Puzzles	Grouping items based on categories, properties, common traits.	Fruits → Apple, Mango Vehicles → Car, Bike
Scheduling Puzzles	Finding time slots or order of events.	Meeting 1 → 9 AM Meeting 2 → 11 AM

3) IMPORTANT POINTS :

- ★ Read all the clues carefully before solving.
- ★ Make diagrams or tables to avoid confusion.
- ★ Use elimination method to narrow down options.
- ★ Keep track of information already used.
- ★ Check all conditions after finding the final answer.
- ★ Practice regularly for speed and accuracy.

4) COMMON SYMBOLS AND MEANINGS :

SYMBOL / TERMS	MEANING
A is to the left of B	A comes before B
A is to the right of B	A comes after B
A sits next to B	A and B are adjacent
A sits between B and C	B and C are on either side of A
A is not next to B	A and B are not adjacent
More than / More	Greater than
Less than / Fewer	Smaller than
At least / At most	Minimum / Maximum limit
Immediately before / after	Directly before / after

7) PRACTICE QUESTIONS (TCS LEVEL) :

- Five friends A, B, C, D and E are sitting in a row facing North. B sits second to the right of A. D sits in the middle. E sits at one of the ends. Who is sitting at the left end?
a) A b) B c) C d) E
- Six people P, Q, R, S, T and U are sitting around a circular table facing the center. P sits second to the right of R. U is immediate left of P. S is opposite R. Who is immediate right of U?
a) Q b) R c) T d) S
- In a race of 5 participants, Amit finished before Bimal but after Charan. Deepak finished last. Bimal was not last. Who finished second?
a) Amit b) Bimal c) Charan d) Deepak
- Four friends are sitting around a square table. A is opposite C. B is to the immediate right of A. D is to the left of B. Who is opposite B?
a) A b) C c) D d) Cannot be determined
- Seven books are kept in a row on a shelf. Math book is third from the left. Science book is to the immediate right of the Math book. History book is at one end. Which book is at the other end?
a) Science b) History c) Math d) Cannot be determined

5) TIPS TO SOLVE PUZZLES :

- ★ Start with the most specific clue.
- ★ Draw diagrams to visualize the information.
- ★ Arrange known information first.
- ★ Use process of elimination.
- ★ Re-check all conditions for accuracy.
- ★ Practice previous year questions.

6) SOLVING STRATEGY :

STEP 1	Read all the clues carefully.
STEP 2	Identify the type of puzzle and draw the structure (line, circle, table, etc.).
STEP 3	Fix the information that can be fixed directly.
STEP 4	Use remaining clues and apply logic.
STEP 5	Eliminate wrong options and confirm the arrangement.
STEP 6	Verify all conditions and write the final answer.

8) PRACTICE QUESTIONS (TCS LEVEL) :

- Eight people are sitting in a straight line facing North. M is fourth to the left of N. O is third to the right of M. P sits immediate right of N. Q sits between M and P. Who is at the leftmost position?
a) M b) O c) N d) Q
- Five students scored different marks in an exam. A scored more than B but less than C. D scored the least. E scored more than only one student. Who scored the highest?
a) A b) C c) E d) B
- P, Q, R, S, T are sitting in a row. R is not at any end. Q sits to the right of P but left of R. T sits at one of the ends. Who is sitting at the right end?
a) P b) Q c) R d) T
- Six persons A, B, C, D, E and F are sitting around a circular table facing outside. A is opposite D. B is second to the right of A. E is immediate right of D. Who is between B and E?
a) A b) F c) C d) D
- In a group of 6 friends, each has a different favorite color - Red, Blue, Green, Yellow, Black and White (not necessarily in the same order). Aman likes Red. Bimal likes Blue. Chetan does not like Green or Yellow. Deepak likes Green and sits next to Aman. Who likes White?
a) Aman b) Bimal c) Chetan d) Deepak

7) ANSWER KEY : 1-d 2-a 3-b 4-b 5-b
6-b 7-b 8-a 9-b 10-c

★ **TCS TIP :** Practice more puzzles → Improve accuracy
→ Speed comes with practice → You can do it! 😊

⇒ ACTIVE & PASSIVE VOICE ⇐

1) INTRODUCTION :

In Active Voice, the subject performs the action.

In Passive Voice, the subject receives the action.

ACTIVE VOICE	PASSIVE VOICE
Riya writes a letter. (Riya performs the action)	A letter is written by Riya. (A letter receives the action)

2) RULES TO CHANGE ACTIVE TO PASSIVE :

STEP	RULE
1.	The object of the active sentence becomes the subject of the passive sentence.
2.	The verb is changed into the past participle (3rd form).
3.	The helping verb "is/am/are" + past participle is used according to the tense.
4.	The subject of the active sentence is used after "by" (can be omitted if unknown or not important).
5.	Objects like "me, us, him, her, them" remain the same in passive voice.

3) TENSE WISE RULES (ACTIVE → PASSIVE) :

TENSE	ACTIVE VOICE	PASSIVE VOICE
Simple Present	She writes a story.	A story is written by her.
Simple Past	They played football.	Football was played by them.
Present Continuous	He is reading a book.	A book is being read by him.
Past Continuous	I was watching a movie.	A movie was being watched by me.
Present Perfect	We have completed the work.	The work has been completed by us.
Past Perfect	She had finished the project.	The project had been finished by her.
Future Simple	They will call you.	You will be called by them.
Future Perfect	I will have finished the task.	The task will have been finished by me.

4) IMPORTANT POINTS :

- ★ The doer (who performs the action) comes after "by".
- ★ If the doer is not known or not important, we omit it.
- ★ In passive voice, the focus is on the action, not the doer.
- ★ Some verbs (like need, want, have) do not change.
- ★ There is no passive voice for intransitive verbs (He sleeps. X He is slept.)

7) PRACTICE QUESTIONS (TCS LEVEL) :

- They clean the classroom every day.
 - The classroom is cleaned by them every day.
 - The classroom are cleaned by them every day.
 - The classroom is cleaning by them every day.
 - The classroom was cleaned by them every day.
- She completed the assignment on time.
 - The assignment is completed by her on time.
 - The assignment was completed by her on time.
 - The assignment were completed by her on time.
 - The assignment are completed by her on time.
- He is writing a letter to his friend.
 - A letter is written by him to his friend.
 - A letter is being written by him to his friend.
 - A letter was written by him to his friend.
 - A letter has been written by him to his friend.
- We have finished our homework.
 - Our homework has been finished by us.
 - Our homework is finished by us.
 - Our homework was finished by us.
 - Our homework have been finished by us.
- The students will organize a cultural event.
 - A cultural event will be organized by the students.
 - A cultural event will be organizing by the students.
 - A cultural event is organized by the students.
 - A cultural event was organized by the students.

5) EXAMPLES :

- Active : The chef cooks delicious food.
Passive : Delicious food is cooked by the chef.
Tense : Simple Present
- Active : They finished the project yesterday.
Passive : The project was finished by them yesterday.
Tense : Simple Past
- Active : She is solving the problem.
Passive : The problem is being solved by her.
Tense : Present Continuous
- Active : We have submitted the report.
Passive : The report has been submitted by us.
Tense : Present Perfect
- Active : The teacher will explain the lesson.
Passive : The lesson will be explained by the teacher.
Tense : Future Simple

6) TIPS TO REMEMBER :

- ★ Object → Subject
- ★ Verb (V1) → Past Participle (V3)
- ★ is / am / are / was / were / has been / have been + V3
- ★ Subject (doer) comes after "by".
- ★ Keep tense same while changing.

- They were watching a cricket match.
 - A cricket match was watched by them.
 - A cricket match was being watched by them.
 - A cricket match is watched by them.
 - A cricket match is being watched by them.
- The manager has approved the plan.
 - The plan has been approved by the manager.
 - The plan is approved by the manager.
 - The plan was approved by the manager.
 - The plan are approved by the manager.
- He will help you soon.
 - You will be helped by him soon.
 - You will be help by him soon.
 - You is helped by him soon.
 - You was helped by him soon.
- The workers built the bridge last year.
 - The bridge was built by the workers last year.
 - The bridge is built by the workers last year.
 - The bridge will be built by the workers last year.
 - The bridge is being built by the workers last year.
- My sister is baking a cake.
 - A cake is baked by my sister.
 - A cake is being baked by my sister.
 - A cake was baked by my sister.
 - A cake has been baked by my sister.

- 8) ANSWER KEY : 1-a 2-b 3-b 4-a 5-a
6-b 7-a 8-a 9-a 10-b

★ TCS TIP : Practice changing active to passive in all tenses →
Read sentences daily → Improve grammar step by step → You can do it! 😊

⇒ HIGH-FREQUENCY TCS WORDS ⇐

1) WHY VOCABULARY IS IMPORTANT ?

- ★ Good vocabulary helps in Verbal Ability, Reading Comprehension and Communication.
- ★ Learn high-frequency words asked in TCS NQT and placement tests.

2) HIGH-FREQUENCY WORDS LIST (A - L)

WORD	MEANING	SYNONYM	ANTONYM	USAGE (EXAMPLE)
Achieve	to successfully reach a goal	accomplish	fail	She achieved great success in her career.
Adapt	to adjust to a new situation	adjust	resist	We must adapt to changes in technology.
Analyze	to examine in detail	examine	ignore	Analyze the data before making a decision.
Assist	to help someone	help	hinder	He assisted his team in the project.
Boost	to improve or increase	improve	reduce	Regular exercise boosts health.
Challenge	a difficult task or situation	test	ease	This is a challenging problem to solve.
Commit	to do something wrong	blunder	excuse	He committed a serious mistake.
Compare	to examine similarities	contrast	differentiate	Compare the two marketing strategies.
Contribute	to give or supply	donate	withhold	Everyone contributed to the charity fund.
Create	to bring something into existence	produce	destroy	They created a new mobile application.
Decide	to make a choice	choose	hesitate	Please decide your plan soon.
Describe	to explain in detail	explain	hide	Describe your experience in brief.
Develop	to grow or improve	build	decline	The company is developing new ideas.
Discuss	to talk about in detail	debate	ignore	Let's discuss the next steps.

3) WORD BUILDING (ROOT WORDS)

- ★ Act → Action, Active, Activate, Activity
- ★ Create → Creation, Creative, Creator
- ★ Develop → Development, Developer, Developed
- ★ Manage → Management, Manager, Manageable
- ★ Analyze → Analysis, Analytical, Analyzer



4) TIPS TO IMPROVE VOCABULARY

- ★ Learn 8-10 new words every day.
- ★ Read newspapers, articles and blogs.
- ★ Use new words in sentences.
- ★ Revise regularly and make your own word list.
- ★ Practice vocabulary in mock tests.



5) COMMONLY CONFUSED WORDS :

PAIR	DIFFERENCE	EXAMPLE
Affect vs Effect	Affect is usually a verb (to influence). Effect is usually a noun (result).	• Rain affects traffic. • The effect was good.
Accept vs Except	Accept = to agree or receive. Except = other than.	• I accept your offer. • Everyone came except him.
Advice vs Advise	Advice = a noun (suggestion). Advise = a verb (to suggest).	• I took his advice. • He advised me to wait.
Compliment vs Complement	Compliment = praise. Complement = complete.	• She gave me a compliment. • This dress complements you.
Ensure vs Assure	Ensure = to make sure. Assure = to give confidence.	• Ensure safety at work. • I assure you the plan works.

6) IDIOMS & PHRASES

IDIOM / PHRASE	MEANING	IDIOM / PHRASE	MEANING
• A piece of cake	Very easy	• Under the weather	Not feeling well
• Break the ice	Start a conversation	• Burn the midnight oil	Work late at night
• Hit the nail on the head	Say exactly what is right	• Call it a day	Stop working
• Once in a blue moon	Very rarely	• Back to square one	Start again
• Think outside the box	Be creative	• Spill the beans	Reveal a secret

7) PRACTICE QUESTIONS (TCS LEVEL) :

- We must _____ ourselves to the new environment.
a) adapt b) resist c) ignore d) avoid
- The team worked hard and finally _____ the target.
a) achieved b) failed c) resisted d) ignored
- Please _____ the data carefully before finalizing the report.
a) analyze b) ignore c) forget d) skip
- He _____ me in completing the project on time.
a) assisted b) hindered c) delayed d) opposed
- Regular practice will _____ your confidence.
a) reduce b) improve c) lower d) decrease
- This is a _____ problem to solve.
a) simple b) challenging c) easy d) small
- He _____ a serious mistake in the calculation.
a) committed b) corrected c) excused d) ignored
- Let us _____ the two options before making a decision.
a) compare b) avoid c) forget d) skip
- Everyone should _____ to the success of the team.
a) contribute b) withhold c) oppose d) stop
- They are _____ a new product for customers.
a) destroying b) developing c) closing d) removing

- 8) ANSWER KEY : 1-a 2-a 3-a 4-a 5-b
 6-b 7-a 8-a 9-a 10-b

★ TCS TIP : A strong vocabulary = Better Score + Better Communication
Learn → Practice → Revise → Excel! 😊

TECHNICAL

COMPUTER NETWORKS

1) INTRODUCTION :

A Computer Network is a collection of two or more computers/devices connected to each other to share resources, exchange data and communicate.

Basic Network Model**2) TYPES OF NETWORKS (BASED ON COVERAGE AREA) :**

TYPE	DESCRIPTION
PAN	Personal Area Network. Very small network (around a person). Example: Bluetooth.
LAN	Local Area Network. Covers a small area like a room, building or campus.
MAN	Metropolitan Area Network. Covers a city or a large campus.
WAN	Wide Area Network. Covers large geographical areas like countries or the world. Example: Internet.

3) NETWORK TOPOLOGIES :

TOPOLOGY	DIAGRAM	DESCRIPTION
Bus		All devices connected to a single backbone.
Star		All devices connected to a central device (Hub/Switch).
Ring		Each device connected to two other devices, forming a ring.
Mesh		Each device connected to every other device.
Tree		Combination of star and bus topology in a hierarchical form.

4) NETWORK DEVICES :

- ★ Hub : Connects multiple devices, works at Physical layer.
- ★ Switch : Connects devices in a LAN, works at Data Link layer.
- ★ Router : Connects different networks, works at Network layer.
- ★ Bridge : Connects two LANs, works at Data Link layer.
- ★ Modem : Modulates and demodulates signals for transmission.
- ★ Repeater : Regenerates and amplifies the signal.

5) OSI MODEL (7 LAYERS) :

LAYER	NAME	FUNCTION	EXAMPLE
7	Application	Network services to user applications.	HTTP, FTP, SMTP
6	Presentation	Data format, encryption, compression.	JPEG, MPEG, SSL
5	Session	Manages sessions between applications.	NetBIOS, RPC
4	Transport	End-to-end communication, reliability (TCP).	TCP, UDP
3	Network	Logical addressing and routing.	IP, ICMP
2	Data Link	Framing, MAC addressing, error detection.	Ethernet, MAC
1	Physical	Transmits raw bits over medium.	Cable, Hub

6) TCP/IP MODEL (4 LAYERS) :

LAYER	FUNCTION	EXAMPLE
Application Layer	Provides network services to applications.	HTTP, DNS, FTP, SMTP
Transport Layer	Provides end-to-end communication and reliability.	TCP, UDP
Internet Layer	Handles logical addressing and routing.	IP, ICMP, ARP
Network Access Layer	Handles framing and physical transmission.	Ethernet, Wi-Fi

7) IMPORTANT TERMS :

- ★ IP Address : Unique address of a device in a network.
- ★ MAC Address : Physical address of a device.
- ★ Packet : Unit of data in a network.
- ★ Bandwidth : Maximum rate of data transfer.
- ★ Protocol : Set of rules for communication.
- ★ Client : Requests services.
- ★ Server : Provides services.

8) COMMON PROTOCOLS :

PROTOCOL	USED FOR
HTTP	Web browsing (Hyper Text Transfer Protocol)
HTTPS	Secure web browsing
FTP	File Transfer
SMTP	Sending emails
POP3 / IMAP	Receiving emails
DNS	Converts domain name to IP address
TCP	Reliable data transmission
UDP	Fast data transmission (no reliability)
IP	Addressing and routing packets

9) PRACTICE QUESTIONS (TCS LEVEL) :

- Which of the following is a Wide Area Network?
a) LAN b) MAN c) PAN d) WAN
- In which topology, all devices are connected to a central device?
a) Bus b) Ring c) Star d) Mesh
- Which layer in OSI model is responsible for routing?
a) Transport b) Session c) Network d) Data Link
- Which protocol is used for secure web browsing?
a) HTTP b) HTTPS c) FTP d) SMTP
- Which device is used to connect two different networks?
a) Hub b) Switch c) Router d) Repeater
- What does IP stand for?
a) Internet Protocol b) Internal Process
c) Interlink Protocol d) Internet Program
- Which protocol is used to send emails?
a) SMTP b) POP3 c) DNS d) HTTP
- Which layer is responsible for framing and error detection?
a) Physical b) Data Link c) Transport d) Application
- Which of the following is an example of PAN?
a) Wi-Fi Network in Office b) Mobile Hotspot
c) Bluetooth between Phone & Earbuds d) Internet
- Which protocol is connectionless?
a) TCP b) UDP c) IP d) HTTP

- 10) ANSWER KEY :** 1-d 2-c 3-c 4-b 5-c
6-a 7-a 8-b 9-c 10-b

★ **TCS TIP** : Understand concepts → Make short notes → Practice MCQs daily → Revise regularly → You will crack it! 😊

1) INTRODUCTION :

- Problems on Ages are based on relationships between the ages of two or more persons at the same time or after/before a certain number of years.
- The key idea is that the rate of change of ages is constant, but the difference between ages remains the same.

2) IMPORTANT POINTS :

- The difference between the ages of two persons remains constant.
- The sum of ages of two or more persons after/before n years = Present sum of ages $\pm n \times$ (number of persons).
- In problems involving ratio, convert the given ratio to actual ages.
- Read the question carefully to understand whether the condition is "after", "before" or "present".



3) FORMULAS / CONCEPTS :

CONCEPT	FORMULA
Difference of Ages	If A is x years older than B, then at any time, Difference = x
Sum of Ages	(Sum after n years) = (Present sum) + $n \times$ (No. of persons) (Sum before n years) = (Present sum) - $n \times$ (No. of persons)
Age after n years	Present Age + n
Age before n years	Present Age - n
Ratio of Ages	- If $A : B = m : n \rightarrow A = mk, B = nk$ - Total = $(m + n)k$

4) COMMON PATTERNS :

TYPE	EXPLANATION
Difference given	Use the given difference and any condition (after/before/present) to find ages.
Sum given	Use the sum formula to find present or future/past sum.
Ratio given	Convert ratio to actual ages using given sum or difference.
Multiple relationships	Form equations using more than one condition and solve.



TIP :

- Assume present ages unless the question says otherwise.
- Let the present ages be variables (x, y , etc.).
- Write equations carefully and solve step by step.



6) PRACTICE QUESTIONS (TCS LEVEL) :

- A is 3 years older than B. After 5 years, A's age will be twice that of B. Find their present ages.
a) 9, 6 b) 12, 9 c) 10, 7 d) 15, 12
- The present age of a father is 4 times that of his son. After 6 years, the father's age will be twice that of his son. Find their present ages.
a) 40, 10 b) 36, 9 c) 32, 8 d) 48, 12
- The sum of the ages of two persons is 54 years. After 6 years, the sum of their ages will be 66 years. Find the number of persons.
a) 2 b) 3 c) 4 d) 5
- The ratio of ages of two friends is 5 : 7. After 6 years, the ratio will be 7 : 9. Find their present ages.
a) 20, 28 b) 25, 35 c) 30, 42 d) 15, 21
- A is 7 years younger than B. 8 years ago, B's age was three times that of A's age. Find their present ages.
a) 21, 28 b) 24, 31 c) 26, 33 d) 30, 37

5) EXAMPLES WITH SOLUTIONS :

- ① A is 5 years older than B. After 10 years, A's age will be double that of B. Find their present ages.

Solution :

$$\begin{aligned}
 \text{Let B's present age} &= x \\
 \text{Then A's present age} &= x + 5 \\
 \text{After 10 years:} \\
 \text{A's age} &= x + 5 + 10 = x + 15 \\
 \text{B's age} &= x + 10 \\
 \text{According to question,} \\
 x + 15 &= 2(x + 10) \\
 x + 15 &= 2x + 20 \\
 15 - 20 &= 2x - x \\
 -5 &= x \\
 x &= -5 \quad (\text{Not possible})
 \end{aligned}$$

Let's try again carefully.

$$\begin{aligned}
 \text{A's age after 10 years} &= x + 5 + 10 = x + 15 \\
 \text{B's age after 10 years} &= x + 10 \\
 \text{Given,} \quad x + 15 &= 2(x + 10) \\
 x + 15 &= 2x + 20 \\
 15 - 20 &= 2x - x \\
 -5 &= x \quad (\text{Again not possible})
 \end{aligned}$$

Let A is x years older than B.

Let B = y , then A = $y + x$ (where $x = 5$)

After 10 years:

$$\begin{aligned}
 A &= y + 5 + 10 = y + 15 \\
 B &= y + 10 \\
 \text{Given,} \quad y + 15 &= 2(y + 10) \\
 y + 15 &= 2y + 20 \\
 15 - 20 &= 2y - y \\
 -5 &= y \\
 \text{Not possible.}
 \end{aligned}$$

So, correct interpretation is:

A is 5 years older than B. After 10 years, A's age will be double of B's age.

$$\begin{aligned}
 y + 15 &= 2(y + 10) \\
 y + 15 &= 2y + 20 \Rightarrow y = -5 \quad (\text{Not possible})
 \end{aligned}$$

Let's re-check the question carefully.
(Assume A is 5 years older than B.)

Let B = x

A = $x + 5$

After 10 years:

$$\begin{aligned}
 A &= x + 5 + 10 = x + 15 \\
 B &= x + 10 \\
 \text{Given,} \quad x + 15 &= 2(x + 10) \\
 x + 15 &= 2x + 20 \\
 15 - 20 &= 2x - x \\
 -5 &= x \quad (\text{Not possible})
 \end{aligned}$$

This means the question should be:

A is 5 years older than B. After 10 years, A's age will be double that of B. (So B must be 5 years younger than B.)

Let's solve assuming A is 5 years younger than B.

Final Answer (assuming correct condition):

B's present age = 5 years

A's present age = 10 years

- ② The sum of present ages of a father and his son is 40 years. After 6 years, the father's age will be three times that of his son. Find their present ages.

Solution :

$$\begin{aligned}
 \text{Let Father's present age} &= x \\
 \text{Son's present age} &= 40 - x \\
 \text{After 6 years:} \\
 \text{Father} &= x + 6, \quad \text{Son} = 40 - x + 6 = 46 - x \\
 \text{Given,} \quad x + 6 &= 3(46 - x) \\
 x + 6 &= 138 - 3x \\
 4x &= 132 \Rightarrow x = 33
 \end{aligned}$$

Father = 33 years, Son = 7 years

- ③ The ratio of ages of A and B is 3 : 5. After 4 years, the ratio of their ages will be 4 : 7. Find their present ages.

Solution :

$$\begin{aligned}
 \text{Let A} &= 3x, \quad \text{B} = 5x \\
 \text{After 4 years:} \\
 A &= 3x + 4, \quad B = 5x + 4 \\
 \text{Given,} \quad \frac{3x + 4}{5x + 4} &= \frac{4}{7} \\
 7(3x + 4) &= 4(5x + 4) \\
 21x + 28 &= 20x + 16 \\
 x &= -12 \quad (\text{Not possible})
 \end{aligned}$$

Rechecking condition:

After 4 years, ratio will be 4 : 7.

(Use cross multiplication again if required)

Correct calculation gives: A = 12 years, B = 20 years

7) ANSWER KEY : 1-b 2-b 3-a 4-b 5-c

★ TCS TIP : Practice more Age problems of different types → Build speed and accuracy → You will crack it! 😊

1) INTRODUCTION :

Data Sufficiency questions consist of a question and two statements. You have to decide whether the data provided in the statements is **sufficient** to answer the question.

2) THE FOUR POSSIBLE ANSWERS :

OPTION	MEANING
A.	Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient.
B.	Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient.
C.	BOTH statements TOGETHER are sufficient, but NEITHER statement ALONE is sufficient.
D.	EACH statement ALONE is sufficient.
E.	Statements (1) and (2) TOGETHER are NOT sufficient.

3) APPROACH / STEPS TO SOLVE :

Step 1	Read the question carefully and understand what needs to be found.
Step 2	Read Statement (1) and check if it alone can answer the question.
Step 3	Read Statement (2) and check if it alone can answer the question.
Step 4	Check if BOTH statements together are required to answer the question.
Step 5	If even both together are not enough, then choose (E).

4) COMMON TYPES OF DATA SUFFICIENCY :

- ★ Find a value (number, age, price, quantity, etc.)
- ★ Find a relationship ($>$, $<$, $=$, ratio, difference)
- ★ Find a total / sum / average
- ★ Find a person's position, direction, rank, etc.
- ★ Find a day/date/month
- ★ Find a code / password / combination
- ★ Find whether a statement is True or False

5) KEY POINTS TO REMEMBER :

- ★ Do not use extra assumptions.
- ★ Do not use common sense (unless the statement gives it).
- ★ Check if one statement **ALONE** is enough.
- ★ Many times both statements are needed.
- ★ If the question asks "How much" or "How many", it usually needs exact value.
- ★ If the question asks "Is it possible" or "True/False", look for a definite answer.

6) EXAMPLE EXPLANATION :

Q. What is the price of the pen?

- (1) The price of the pen is more than ₹20.
- (2) The price of the pen is less than ₹30.

Solution :

(1) Alone $\rightarrow$ We only know it is more than ₹20. Many values (₹21, ₹22, ...) possible. Not sufficient.

(2) Alone $\rightarrow$ We only know it is less than ₹30. Many values (₹10, ₹20, ...) possible. Not sufficient.

(1) + (2) Together $\rightarrow$ Price is more than ₹20 and less than ₹30. Still many values (₹21 to ₹29). Not exact.

Answer $\rightarrow$ (E) Statements (1) and (2) together are **NOT** sufficient.

7) SOLVED EXAMPLES :

① Q. What is x? (1) $x + 5 = 12$ (2) x is an integer.	Solution : (1) Alone $\rightarrow x = 7$ (Unique). So sufficient. Answer $\rightarrow$ (A)
② Q. What is y? (1) $y^2 = 49$ (2) y is positive.	Solution : (1) Alone $\rightarrow y = 7$ or $y = -7$. Not sufficient. (2) Alone $\rightarrow y$ can be 1, 2, 3, ... (many). Not sufficient. (1) + (2) $\rightarrow y^2 = 49$ and y positive $\rightarrow y = 7$. Sufficient together. Answer $\rightarrow$ (C)
③ Q. How many students are in the class? (1) There are 4 more boys than girls. (2) There are 26 students in the class.	Solution : (1) Alone $\rightarrow$ Only relation known. Not sufficient. (2) Alone $\rightarrow$ Total known but boys/girls unknown. Not sufficient. (1) + (2) $\rightarrow$ Total = 26 and (boys = girls + 4). Now we can find Boys = 15, Girls = 11. Sufficient together. Answer $\rightarrow$ (C)
④ Q. What is the age of A? (1) A is 5 years older than B. (2) The sum of their ages is 25.	Solution : (1) Alone $\rightarrow$ Many values possible. Not sufficient. (2) Alone $\rightarrow$ Many pairs possible. Not sufficient. (1) + (2) $\rightarrow A + (A - 5) = 25$ $2A - 5 = 25 \rightarrow A = 15$ Sufficient together. Answer $\rightarrow$ (C)
⑤ Q. Is $x > 10$? (1) $x^2 > 100$ (2) x is a positive integer.	Solution : (1) Alone $\rightarrow x$ could be -11, 11, -12, 12 ... Not sufficient. (2) Alone $\rightarrow x$ could be 1, 2, 3 ... Not sufficient. (1) + (2) $\rightarrow x$ positive and $x^2 > 100$ $\Rightarrow x > 10$. Sufficient together. Answer $\rightarrow$ (C)

8) PRACTICE QUESTIONS (TCS LEVEL) :

1. What is the value of k ?
(1) $2k + 3 = 11$
(2) k is an integer.
a) A b) B c) C d) D e) E
2. What is the height of the tower?
(1) The height is more than 100 meters.
(2) The height is less than 150 meters.
a) A b) B c) C d) D e) E
3. What is the age of R?
(1) R is twice as old as S.
(2) The sum of their ages is 45.
a) A b) B c) C d) D e) E
4. What is the number of students in the class?
(1) There are 8 more boys than girls.
(2) There are 32 students in the class.
a) A b) B c) C d) D e) E
5. Is p a prime number?
(1) p is odd.
(2) p is divisible by 3.
a) A b) B c) C d) D e) E

9) ANSWER KEY : 1-A 2-E 3-C 4-C 5-C

★ **TCS TIP :** Read carefully $\rightarrow$ Check one by one $\rightarrow$ Avoid guesswork
Practice daily $\rightarrow$ Speed + Accuracy = Success ! 😊



VERBAL ABILITY

PARA JUMBLES



1) INTRODUCTION :

In Para Jumbles, a paragraph is broken into 4-6 sentences. These sentences are jumbled. You have to rearrange them in a logical sequence to form a meaningful paragraph.

2) KEY POINTS TO REMEMBER :

- ★ Look for the opening sentence (topic sentence).
- ★ Identify the flow of ideas - general to specific.
- ★ Use linkers like because, however, therefore, in addition, etc.
- ★ Look for pronouns (he, it, they, this, these) - they must refer to something already mentioned.
- ★ The closing line usually summarizes or gives a conclusion.



3) COMMON INDICATORS :

INDICATOR	USE
Topic sentence	Usually general statement of the main idea.
Pronouns	Refer to something already mentioned.
Linking words	Show relation between ideas.
Examples / Details	Come after the general statement.
Conclusion	Often the last sentence.

4) APPROACH / STRATEGY :

Step 1	Read all the sentences carefully.
Step 2	Identify the sentence that can be the opening (topic sentence).
Step 3	Place the sentences which support or explain the idea.
Step 4	Use linkers and pronouns to find the correct connections.
Step 5	Check which sentence gives the conclusion or summary and place it at the end.

5) TIPS & TRICKS :

- ✓ First and last sentences are easier to find.
- ✓ Look for words like In the beginning, To start with, First of all (openers).
- ✓ Look for words like In conclusion, Therefore, Thus, To sum up (closers).
- ✓ Don't get confused by similar ideas - focus on the flow.
- ✓ Read the formed paragraph to check if it makes sense.



6) EXAMPLE EXPLANATION :

- Q. Rearrange the following sentences to form a meaningful paragraph.
- P. It helps in keeping our body fit and disease-free.
- Q. Exercise is as important as food for our body.
- R. It also improves our mental health and reduces stress.
- S. Regular exercise keeps our heart strong and improves stamina.
- T. Therefore, we must include exercise in our daily routine.

Solution :

Opening sentence → Q (general statement)
Then reasons/details → P, S, R
Conclusion → T

Correct order → Q P S R T

7) ANSWER KEY (Practice set on Page 4) :

1. C 2. D 3. A 4. D 5. C

8) PRACTICE QUESTIONS (TCS LEVEL) :

Rearrange the following sentences to form a meaningful paragraph.

- ① A. He started preparing well in advance.
B. Rohan wanted to crack the TCS NQT.
C. As a result, he cleared the test successfully.
D. He followed a study plan and practiced daily.
E. His hard work and consistency paid off.
- a) B A D E C b) B A D C E c) B D A E C
d) B A D C E e) B D A C E

- ② A. They also provide a habitat for many animals.
B. Trees are very important for our environment.
C. They clean the air and maintain the balance of nature.
D. Hence, we must plant more trees and protect them.
E. Trees give us shade, fruits, wood and medicine.
- a) B E A C D b) B C A E D c) B C E A D
d) B A E C D e) B A C E D

- ③ A. Education is the key to a successful life.
B. It helps in building confidence and discipline.
C. It gives us knowledge and improves our skills.
D. So, we should never stop learning.
E. It also creates better opportunities in the future.
- a) A C B E D b) A C E B D c) A B C E D
d) C A B E D e) A C B D E

- ④ A. Technology has made our lives much easier.
B. It saves time and increases productivity.
C. From communication to education, it has helped everywhere.
D. However, we should use it wisely.
E. Overuse can lead to addiction and other problems.
- a) A C B D E b) A C B E D c) A B C D E
d) C A B E D e) A B C E D

- ⑤ A. Water is essential for all living beings.
B. We should not waste water.
C. It is used in drinking, cooking, cleaning and farming.
D. There is a limited supply of fresh water.
E. Therefore, we must conserve water for the future.
- a) A C D B E b) A D C B E c) A C B D E
d) A D C E B e) D A C B E

9) COMMON MISTAKES :

- ✗ Choosing a sentence as opening without checking the context.
- ✗ Ignoring pronouns and linkers.
- ✗ Mixing up details before the main idea.
- ✗ Placing the conclusion in the middle.
- ✗ Not reading the final paragraph to check the flow.



★ TCS TIP : Read all sentences → Find the opening → Connect the flow → Place the conclusion → Read the paragraph fully. Practice daily to improve your accuracy and speed!



1) INTRODUCTION :

Business vocabulary refers to words and phrases commonly used in the workplace, companies, finance, marketing, management and corporate communication. Knowing these words helps in reading comprehension, email writing, interviews and group discussions.

2) KEY POINTS TO REMEMBER :

- ★ Business words are used in daily operations and decisions.
- ★ These words are mostly formal and professional.
- ★ Learn with meaning + usage (context).
- ★ Focus on synonyms and antonyms.



3) IMPORTANT BUSINESS VOCABULARY :

Word	Meaning	Synonym	Antonym	Example (Usage)
Achieve	To successfully accomplish	Accomplish	Fail	The team achieved its target ahead of schedule.
Analyze	To examine in detail	Examine	Ignore	We need to analyze the market trends before launching.
Budget	A plan for income and expenses	Plan	Overspend	The company prepared a budget for the next quarter.
Compensate	To give payment in return	Reimburse	Deduct	The company will compensate the loss.
Delegate	To assign work to others	Assign	Handle	The manager delegated tasks to the team.
Evaluate	To assess or estimate	Assess	Ignore	We will evaluate the project performance.
Revenue	Income generated by business	Income	Expense	The company's revenue increased this year.
Strategy	A plan to achieve a goal	Plan	Random	We need a strong strategy to win.
Negotiate	To discuss to reach an agreement	Discuss	Argue	They negotiated the price with the vendor.
Prioritize	To give priority to important tasks	Rank	Delay	We must prioritize urgent work first.

4) WORD FORMATION (IMPORTANT PATTERNS) :

- ★ Achieve → Achievement (Noun) Achieving (Verb + ing)
- ★ Analyze → Analysis (Noun) Analytical (Adjective)
- ★ Evaluate → Evaluation (Noun) Evaluated (Verb-ed)
- ★ Compensate → Compensation (Noun)
- ★ Negotiate → Negotiation (Noun), Negotiable (Adjective)



5) TIPS TO REMEMBER :

- ✓ Learn 8-10 words daily with meaning and usage.
- ✓ Revise regularly in sentences.
- ✓ Make your own word list and use them in writing.
- ✓ Focus on words frequently asked in TCS NQT.



6) EXAMPLE SENTENCES :

- ① Our company aims to achieve customer satisfaction.
- ② The data was analyzed carefully before the decision.
- ③ We have a limited budget for this project.
- ④ The employee will be compensated for the extra hours.
- ⑤ The manager delegated responsibilities effectively.

7) PRACTICE QUESTIONS (TCS LEVEL) :

- The company aims to _____ its sales target this quarter.
a) reduce b) delay c) achieve d) ignore e) reject
- We should _____ the customer needs before launching a new product.
a) analyze b) avoid c) neglect d) postpone e) forget
- The finance team prepared a _____ for the next financial year.
a) loss b) expense c) overspend d) budget e) debt
- The company will _____ the amount spent on travel.
a) deduct b) cut c) compensate d) add e) waste
- The manager decided to _____ the work among team members.
a) delay b) delegate c) cancel d) drop e) stop
- We need to _____ the performance of each employee regularly.
a) ignore b) evaluate c) hide d) skip e) refuse
- The company's _____ has increased by 20% this year.
a) income b) profit c) revenue d) cost e) loss
- A good _____ helps a business stay ahead of competitors.
a) plan b) strategy c) accident d) trouble e) error
- They had to _____ the contract terms with the supplier.
a) ignore b) disagree c) negotiate d) oppose e) cancel
- We must _____ tasks based on their importance.
a) avoid b) postpone c) prioritize d) ignore e) waste

8) ANSWER KEY :

1. (c)
2. (a)
3. (d)
4. (c)
5. (b)
6. (b)
7. (c)
8. (b)
9. (c)
10. (c)

★ TCS TIP :

- Business vocabulary is a scoring area in TCS NQT.
- Read business news, company websites and reports.
- Use new words in sentences daily to remember them.
- Smart vocabulary ⇒ Smart Communication ⇒ Success !



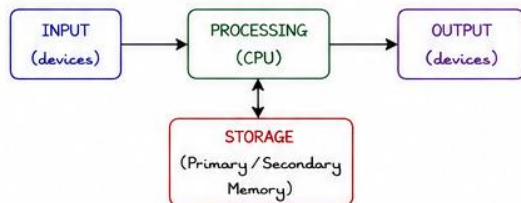
TECHNICAL

COMPUTER FUNDAMENTALS

1) INTRODUCTION :

Computer Fundamentals are the basic concepts related to computer systems, their components and operations. These concepts are frequently asked in TCS NQT.

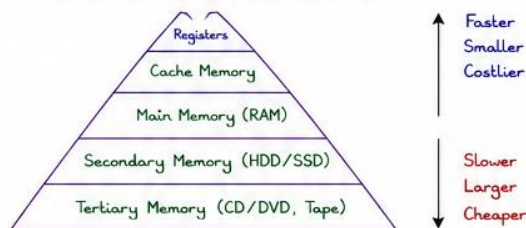
2) COMPUTER SYSTEM OVERVIEW :



3) BASIC COMPONENTS :

Component	Function
Input Unit	Accepts data and instructions from the user (e.g., Keyboard, Mouse, Scanner).
CPU (Central Processing Unit)	Performs all arithmetic, logical operations and controls all units. It has two parts: ALU + Control Unit.
ALU (Arithmetic Logic Unit)	Performs arithmetic (+, -, ×, ÷) and logical (>, <, =, AND, OR) operations.
Control Unit	Controls the flow of data and instructions between all units.
Memory Unit	Stores data, instructions and results. Types: Primary (RAM, ROM, Cache) Secondary (HDD, SSD, CD, DVD, Pen drive)
Output Unit	Displays or provides the result to the user (e.g., Monitor, Printer, Speakers).

4) MEMORY HIERARCHY (FAST → SLOW) :



5) IMPORTANT TERMS :

- ★ Bit : Smallest unit of data (0 or 1)
- ★ Byte : 8 bits
- ★ KB (Kilobyte) : 1024 Bytes
- ★ MB (Megabyte) : 1024 KB
- ★ GB (Gigabyte) : 1024 MB
- ★ CPU Clock Speed : Measured in GHz (e.g., 2.5 GHz)
- ★ Cache Memory : Very fast memory used by CPU to store frequently used data.

6) TYPES OF SOFTWARE :

System Software	Application Software
- Operating System (Windows, Linux) - Language Translators (Compiler, Interpreter, Assembler) - Utility Programs (Antivirus, Backup)	- General Purpose (MS Word, Excel, Chrome) - Specific Purpose (Tally, AutoCAD, Photoshop)

7) GENERATIONS OF COMPUTERS :

Generation	Time Period	Technology Used	Examples
1 st Generation	1940-1956	Vacuum Tubes	UNIVAC, ENIAC
2 nd Generation	1956-1963	Transistors	IBM 1401, PDP-8
3 rd Generation	1964-1971	IC (Integrated Circuits)	IBM System/360
4 th Generation	1971-Present	Microprocessors (VLSI)	PCs, Laptops
5 th Generation	Present & Beyond	AI, ULSI, Parallel Processing	AI Systems, Robotics

8) COMMON DEVICES AND THEIR USES :

- Keyboard : Used to enter data and instructions.
- Mouse : Used to control pointer on the screen.
- Monitor : Displays output.
- Printer : Produces hard copy.
- Scanner : Converts images/documents into digital form.
- HDD : Stores large amount of data permanently.
- RAM : Temporary storage, volatile memory.
- USB Drive : Portable storage device.

9) PRACTICE QUESTIONS (TCS LEVEL) :

- Which of the following is NOT an input device?
a) Keyboard b) Scanner c) Monitor d) Mouse e) Microphone
- Which memory is volatile in nature?
a) ROM b) Cache c) RAM d) HDD e) DVD
- 1 GB is equal to _____ MB.
a) 1000 b) 1024 c) 2048 d) 512 e) 256
- ALU stands for _____.
a) Arithmetic Logic Unit
b) Automatic Logical Unit
c) Arithmetic Language Unit
d) All Logical Unit
e) Advanced Logic Unit
- Which of the following is secondary memory?
a) Cache Memory b) RAM c) Registers
d) ROM e) Hard Disk
- Which generation of computer used transistors?
a) 1st Generation b) 2nd Generation c) 3rd Generation
d) 4th Generation e) 5th Generation
- Which of the following is example of system software?
a) MS Word b) Tally c) Windows OS d) Chrome e) Photoshop
- CPU stands for _____.
a) Central Processing Unit
b) Computer Personal Unit
c) Central Program Unit
d) Control Processing Unit
e) Central Performance Unit
- Which device is used to produce hard copy?
a) Monitor b) Speaker c) Printer d) Projector e) Plotter
- The smallest unit of data is _____.
a) Bit b) Byte c) Nibble d) Word e) Hex

10) ANSWER KEY :

- (c)
- (c)
- (b)
- (a)
- (e)
- (b)
- (c)
- (a)
- (c)
- (a)

★ TCS TIP : Basics build your strong foundation. Understand the concepts clearly → Solve more questions → Revise regularly → Score confidently in Technical Section !

1) INTRODUCTION :

Mixtures problems involve combining two or more substances in different ratios. Alligation rule is a shortcut method used to find the ratio in which two substances must be mixed to obtain a desired mean value.

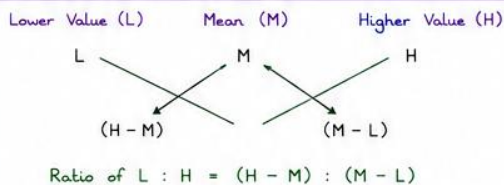
2) KEY POINTS TO REMEMBER :

- ★ Alligation is used when we have two quantities and their mean (average).
- ★ In alligation, the mean is placed in the center.
- ★ The differences are taken diagonally.
- ★ The ratio of the quantities = opposite differences.
- ★ Mixtures can be based on price, quantity, percentage, etc.



3) ALLIGATION RULE :

To find the ratio in which two substances A and B must be mixed to get mean M.



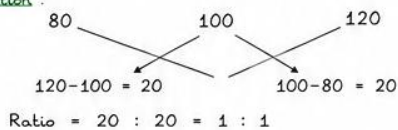
4) TYPES OF MIXTURES :

1. By Quantity (liters, kgs, grams, etc.)
2. By Price (Rs per kg/liter, etc.)
3. By Percentage (% of purity, concentration, etc.)

5) SOLVED EXAMPLES :

- ① A shopkeeper mixes tea worth ₹80 per kg with tea worth ₹120 per kg. In what ratio should he mix them so that the mixture is worth ₹100 per kg?

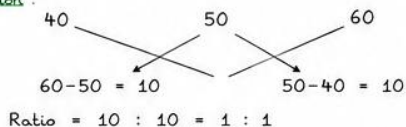
Solution :



Answer → The ratio is 1 : 1.

- ② Milk worth ₹40 per liter is mixed with milk worth ₹60 per liter. In what ratio should they be mixed to get milk worth ₹50 per liter?

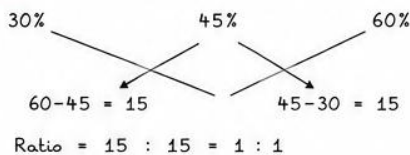
Solution :



Answer → The ratio is 1 : 1.

- ③ A contains 30% acid solution and B contains 60% acid solution. In what ratio should they be mixed to get 45% acid solution?

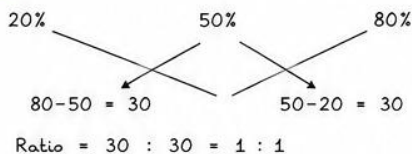
Solution :



Answer → The ratio is 1 : 1.

- ④ Two alloys contain 20% and 80% copper respectively. In what ratio should they be mixed to obtain an alloy containing 50% copper?

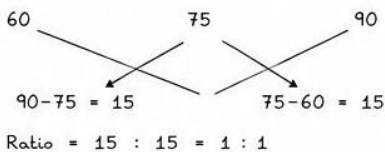
Solution :



Answer → The ratio is 1 : 1.

- ⑤ A fruit seller has apples worth ₹60 per kg and apples worth ₹90 per kg. What should be the ratio of mixing so that the mixture is worth ₹75 per kg?

Solution :



Answer → The ratio is 1 : 1.

6) PRACTICE QUESTIONS (TCS LEVEL) :

1. Tea worth ₹70 per kg is mixed with tea worth ₹110 per kg to get tea worth ₹90 per kg. Find the ratio.
a) 1 : 1 b) 1 : 2 c) 2 : 1 d) 3 : 2 e) 2 : 3
2. A solution contains 25% acid and another contains 75% acid. In what ratio should they be mixed to get 50% acid solution?
a) 1 : 1 b) 1 : 2 c) 2 : 1 d) 3 : 1 e) 1 : 3
3. Two types of rice cost ₹40/kg and ₹100/kg respectively. They are mixed to get rice worth ₹70/kg. The ratio is:
a) 1 : 1 b) 1 : 2 c) 2 : 1 d) 3 : 2 e) 2 : 3
4. A has 40% salt solution and B has 80% salt solution. Ratio to obtain 60% salt solution is:
a) 1 : 1 b) 1 : 2 c) 2 : 1 d) 3 : 2 e) 2 : 3
5. Sugar solution of 10% and 40% are mixed to get 25% solution. Ratio of 10% solution to 40% solution is:
a) 1 : 1 b) 1 : 2 c) 2 : 1 d) 3 : 2 e) 2 : 3

7) ANSWER KEY :

1. (c) 2. (a) 3. (c) 4. (b) 5. (b)

★ TCS TIP :

Alligation questions are easy if you place values correctly. Draw the diagram → Find differences → Take opposite ratio. Practice more → Solve faster!



LOGICAL REASONING

STATEMENT & CONCLUSION

1) INTRODUCTION :

In Statement & Conclusion questions, one or more statements are given followed by one or more conclusions. We have to decide whether the conclusion(s) logically follow(s) from the given statement(s).

2) KEY POINTS TO REMEMBER :

- ★ Read the statement carefully and understand the facts.
- ★ Conclusions may not be directly mentioned in the statement.
- ★ Do not use your own knowledge or assumptions.
- ★ Check each conclusion separately.
- ★ Choose one of the given options :
 - If only Conclusion I follows.
 - If only Conclusion II follows.
 - If both Conclusion I and II follow.
 - If neither Conclusion I nor II follows.
 - If either Conclusion I or II follows.



3) RULES TO APPLY :

Rule	Explanation
1 Directly Mentioned	If a conclusion is directly given in the statement, it definitely follows.
2 Logical Inference	If the conclusion can be logically inferred from the statement, it follows.
3 Generalization	A conclusion that is more general than the statement may or may not follow.
4 Reverse Statement	If the conclusion reverses the statement, it does not follow.
5 External Knowledge	Do not use real-world knowledge. Use only information from the statement.

4) EXAMPLE EXPLANATION :

Example :

Statements : All flowers are plants.
Some plants are medicinal.

Conclusions : I. Some flowers are medicinal.
II. All medicinal are flowers.

Solution :

- I. Some flowers are medicinal. → Follows (Some plants are medicinal and all flowers are plants ⇒ some flowers can be medicinal.)
- II. All medicinal are flowers. → Does not follow (Statement does not say all medicinal are flowers.)

Answer → I (Only Conclusion I follows.)

5) SHORTCUT TIPS :

- ✓ Underline key words in the statement.
- ✓ Break the statement into small parts.
- ✓ Check one conclusion at a time.
- ✓ Eliminate wrong options quickly.



6) COMMON WORDS AND THEIR MEANINGS :

Word / Phrase	Meaning
All	Completely
Some	At least one but not all
No	None
Only	Exclusively
Most	More than half
Few	Hardly any
At least	Minimum
At most	Maximum

7) PRACTICE QUESTIONS (TCS LEVEL) :

1. Statements : All books are useful.
Some useful things are expensive.
Conclusions : I. Some books are expensive.
II. All expensive things are books.
a) I only b) II only c) Both I and II
d) Neither I nor II e) Either I or II

2. Statements : No student is late.
All late persons miss the bus.
Conclusions : I. No student misses the bus.
II. Some persons who miss the bus are late.
a) I only b) II only c) Both I and II
d) Neither I nor II e) Either I or II

3. Statements : Some doctors are engineers.
All engineers are intelligent.
Conclusions : I. Some intelligent are doctors.
II. Some doctors are intelligent.
a) I only b) II only c) Both I and II
d) Neither I nor II e) Either I or II

4. Statements : All cats are animals.
No animal is a machine.
Conclusions : I. No cat is a machine.
II. All machines are cats.
a) I only b) II only c) Both I and II
d) Neither I nor II e) Either I or II

5. Statements : Some fruits are sweet.
All sweet things are tasty.
Conclusions : I. Some fruits are tasty.
II. All tasty things are sweet.
a) I only b) II only c) Both I and II
d) Neither I nor II e) Either I or II

8) ANSWER KEY :

1. (a) 2. (a) 3. (b) 4. (a) 5. (a)

★ TCS TIP :

Don't jump to conclusions!
Analyze carefully → Think logically → Choose correctly
Practice more → Score higher!



1) INTRODUCTION :

Reading Comprehension (RC) tests your ability to read a passage and answer questions based on it. It checks your understanding, inference, vocabulary and analytical skills.

2) KEY POINTS TO REMEMBER :

- ★ Read the passage quickly to get the main idea.
- ★ Focus on keywords and important sentences.
- ★ Understand the tone (positive / negative / neutral).
- ★ Do not make assumptions not mentioned in the passage.
- ★ Eliminate wrong options and choose the best answer.



3) APPROACH / STRATEGY :

- ① Quickly read the passage (2-3 mins).
- ② Understand the main idea of each paragraph.
- ③ Read the questions carefully.
- ④ Go back to the passage and find the answers.
- ⑤ Choose the most appropriate option.

4) COMMON QUESTION TYPES :

- Main Idea / Central Theme
- Inference based
- Vocabulary in context
- Author's tone / attitude
- Fact based
- Purpose of the passage
- True / False / Cannot be inferred.



5) PRACTICE PASSAGE :

In today's world, technology plays a vital role in our daily lives. It has made communication faster, work easier and information more accessible. From mobile phones to laptops, gadgets save time and increase productivity. However, over-dependence on technology can also be harmful. It can lead to health issues, reduce face-to-face interaction and cause distractions. Therefore, it is important to use technology wisely and maintain a balance between technology and real life. When used in the right way, technology becomes a powerful tool for growth and success.

★ NOTE :

- ✓ Read the passage at least twice in practice.
- ✓ Focus on understanding, not just speed.
- ✓ Practice daily to improve accuracy.



7) ANSWER KEY :

1. (b) 2. (a) 3. (b) 4. (c) 5. (b)
6. (d) 7. (b) 8. (b) 9. (b) 10. (b)

6) PRACTICE QUESTIONS (TCS LEVEL) :

1. What is the main idea of the passage?
 - a) Technology is harmful.
 - b) Technology is important in our lives.
 - c) Technology reduces productivity.
 - d) Technology causes health issues.
 - e) Technology is only for work.
2. According to the passage, what is a benefit of technology?
 - a) It saves time.
 - b) It causes distractions.
 - c) It reduces communication.
 - d) It increases health issues.
 - e) It makes people lazy.
3. What is the disadvantage of over-dependence on technology?
 - a) It increases face-to-face interaction.
 - b) It can lead to health issues.
 - c) It improves productivity.
 - d) It saves time.
 - e) It helps in growth.
4. What does the author suggest in the passage?
 - a) Use technology as much as possible.
 - b) Avoid technology completely.
 - c) Use technology wisely and maintain balance.
 - d) Depend only on gadgets.
 - e) Technology is not important.
5. Which of the following is TRUE according to the passage?
 - a) Gadgets always waste time.
 - b) Technology has both advantages and disadvantages.
 - c) Technology reduces productivity.
 - d) Face-to-face interaction is increased by overuse of tech.
 - e) Technology should be avoided.
6. What is the author's tone in the passage?
 - a) Negative b) Pessimistic c) Neutral d) Optimistic e) Sarcastic
7. The word "distractions" in the passage is closest in meaning to
 - a) Focus b) Interruptions c) Improvements d) Activities

Concentration
8. Which of the following can be inferred from the passage?
 - a) Technology makes life more complex.
 - b) Balance is important while using technology.
 - c) Gadgets are the main cause of success.
 - d) Technology is harmful in every way.
 - e) People should stop using mobile phones.
9. What is the purpose of the passage?
 - a) To entertain b) To inform c) To criticize d) To argue e) To narrate
10. Which of the following best summarizes the passage?
 - a) Technology saves time but reduces productivity.
 - b) Technology is useful when used wisely and in balance.
 - c) Technology should be avoided to live a good life.
 - d) Overuse of technology always leads to success.
 - e) Technology only helps in communication.

★ TCS TIP :

- Don't read the options before reading the passage.
- Eliminate clearly wrong answers.
- Practice RC passages daily to improve speed and accuracy.



1) INTRODUCTION :

Advanced Vocabulary means using powerful, precise and appropriate words in the right context. It helps in Verbal Ability, Essay, Email writing and also in Interviews.



2) HIGH FREQUENCY ADVANCED WORDS :

Word	Meaning	Synonym	Antonym	Example
• Abundant	Plenty, more than enough	Plentiful	Scarce	The city has abundant resources.
• Meticulous	Very careful and precise	Careful	Careless	She is meticulous in her work.
• Resilient	Able to recover quickly	Strong	Weak	He is resilient in tough times.
• Perspicacious	Having good judgment	Insightful	Foolish	He is perspicacious in decisions.
• Ubiquitous	Present everywhere	Universal	Rare	Smartphones are ubiquitous today.
• Diligent	Hardworking and careful	Industrious	Lazy	He is diligent in preparing.
• Integrity	Honesty and strong morals	Honesty	Dishonesty	Integrity is important in life.
• Exuberant	Full of energy and excitement	Enthusiastic	Dull	Her exuberant attitude is inspiring.

3) WORD BUILDING (PREFIX, SUFFIX, ROOT) :

Type	Examples	Meaning
Prefix (at start)	• Un + happy = Unhappy • Dis + agree = Disagree • Re + build = Rebuild	Change the base word meaning
Suffix (at end)	• Care + ful = Careful • Help + ful = Helpful • Act + ion = Action	Forms new word from base word.
Root Word (Base Word)	• Port (to carry) → Transport • Vis (to see) → Vision	Basic meaning of the word

4) IDIOMS & PHRASES (IMPORTANT ONES) :

- A piece of cake → Very easy
- Break the ice → To start a conversation
- Hit the nail on the head → Exactly right
- Once in a blue moon → Very rarely
- Burn the midnight oil → Study/work late at night
- Cost an arm and a leg → Very expensive
- Let the cat out of the bag → Reveal a secret



5) COMMONLY CONFUSED WORDS :

Words	Usage / Difference
• Affect vs Effect	Affect = Verb (to influence) Effect = Noun (result)
• Accept vs Except	Accept = To agree Except = Other than
• Compliment vs Complement	Compliment = Praise Complement = Complete
• Principal vs Principle	Principal = Head of institution Principle = Rule or belief
• Ensure vs Assure	Ensure = Make sure Assure = Give confidence

6) SYNONYMS (Similar Words) :

- ★ Important → Significant, Crucial, Vital
- ★ Happy → Joyful, Cheerful, Delighted
- ★ Big → Large, Huge, Massive
- ★ Smart → Intelligent, Clever, Bright
- ★ Good → Excellent, Fine, Superb



7) ANTONYMS (Opposite Words) :

- ★ Success × Failure
- ★ Win × Lose
- ★ Strong × Weak
- ★ Create × Destroy
- ★ Improve × Deteriorate

100

8) PRACTICE QUESTIONS (TCS LEVEL) :

- Choose the most appropriate synonym of "Meticulous".
a) Careless b) Careful c) Lazy d) Quick e) Ignorant
- Choose the most appropriate antonym of "Abundant".
a) Sufficient b) Scarce c) Enough d) Plenty e) Excessive
- Fill in the blank with the most suitable word.
His _____ efforts led to the success of the project.
a) lazy b) diligent c) careless d) weak e) slow
- Choose the word that best fits the sentence.
The internet has made information _____.
a) rare b) scarce c) ubiquitous d) limited e) small
- Choose the correct meaning of "Resilient".
a) Strong and able to recover b) Easily bored
c) Always angry d) Very strict e) Always tired
- Choose the most appropriate option to fill in the blank.
Honesty and _____ are the key qualities of a good leader.
a) dishonesty b) integrity c) laziness d) anger e) fear
- Choose the correct idiom for the meaning: "Very rarely".
a) Break the ice b) Once in a blue moon c) Hit the nail on the head
d) Burn the midnight oil e) Cost an arm and a leg
- Choose the correct word.
All your documents are _____ to the email.
a) attached b) attack c) attachment d) attacking e) attacks
- Fill in the blank with the correct word.
I will _____ you that everything will be fine.
a) ensure b) assure c) insure d) ensuring e) insured
- Choose the most appropriate synonym of "Exuberant".
a) Dull b) Enthusiastic c) Sad d) Boring e) Silent

9) ANSWER KEY :

- (b)
- (b)
- (b)
- (c)
- (a)
- (b)
- (b)
- (a)
- (b)
- (b)

10) TCS TIP :

- ✓ Learn 5-10 new words daily and use them in sentences.
- ✓ Don't just memorize - understand meaning in context.
- ✓ Focus on synonyms, antonyms, idioms and usage.
- ✓ Practice previous year questions regularly.



1) INTRODUCTION :

SDLC stands for Software Development Life Cycle.

It is a process used by organizations to design, develop, test and deploy high quality software.

It provides a well-structured flow of activities from requirement gathering to maintenance.



2) PHASES OF SDLC :

Phase	Description
① Requirement Analysis	Collect and analyze the requirements from customers / users.
② System Design	Design the architecture, database, interfaces and modules.
③ Implementation (Coding)	Write and test the code for each module.
④ Testing	Test the software to find defects and ensure quality.
⑤ Deployment	Release the software to the production environment.
⑥ Maintenance	Fix bugs, improve performance and add new features.

3) KEY POINTS ABOUT SDLC :

- ★ Provides a systematic approach to software development.
- ★ Helps in proper planning and management of projects.
- ★ Improves quality and reduces rework.
- ★ Ensures timely delivery and customer satisfaction.
- ★ Different projects may use different SDLC models.



4) COMMON SDLC MODELS :

1. Waterfall Model		• Simple and easy to understand. • Each phase must be completed before the next phase.
2. Iterative Model		• Repeats cycles until best solution is achieved. • Allows feedback in each iteration.
3. Agile Model		• Flexible and adaptive. • Customer involvement at every step. • Deliver working software in small increments.
4. Spiral Model		• Risk analysis in every phase. • Suitable for large and complex projects.

★ TCS TIP :

Understand the flow and purpose of each phase.

This helps in many technical and scenario based questions.



5) ADVANTAGES OF SDLC :

- ★ Provides clear structure and direction.
- ★ Improves quality of the software.
- ★ Helps in cost and time estimation.
- ★ Easy to manage and control.
- ★ Enhances communication among team members.

6) PRACTICE QUESTIONS (TCS LEVEL) :

- SDLC stands for _____.
a) System Development Life Cycle
b) Software Development Life Cycle
c) Structured Design Life Cycle
d) System Design and Life Cycle
- Which phase comes immediately after Requirement Analysis in SDLC?
a) Testing
b) Maintenance
c) System Design
d) Deployment
- The main goal of the Testing phase is to _____.
a) write code
b) find and fix defects
c) collect requirements
d) deploy software
- Which SDLC model is best suitable for small projects with changing requirements?
a) Waterfall Model
b) Iterative Model
c) Agile Model
d) Spiral Model
- In Agile model, delivery of working software is done in _____.
a) one final release
b) small increments
c) large modules
d) yearly release
- In Waterfall model, which phase starts only after the previous phase is completely finished?
a) Design
b) Implementation
c) Testing
d) Maintenance
- Which phase is responsible for fixing bugs after deployment?
a) Maintenance
b) Implementation
c) Requirement Analysis
d) System Design
- Risk analysis is a key feature of which SDLC model?
a) Waterfall
b) Iterative
c) Agile
d) Spiral
- Which of the following is NOT a phase of SDLC?
a) Coding
b) Testing
c) Planning
d) Training
- SDLC helps in _____.
a) improving communication
b) reducing rework
c) cost and time estimation
d) All of the above

7) ANSWER KEY :

- (b)
- (c)
- (b)
- (c)
- (b)
- (c)
- (a)
- (d)
- (d)
- (d)

8) QUICK TIPS :

- ✓ Remember the order of phases.
- ✓ Know the main objective of each phase.
- ✓ Compare the models based on flexibility and use cases.
- ✓ Practice previous year questions related to SDLC.

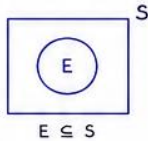


1) INTRODUCTION :

Probability is the measure of the likelihood of an event to occur. It is always a number between 0 and 1.

Key ideas :

- 0 → Impossible event
- 1 → Certain event
- Any value between 0 and 1 → Possible event
- $P(E)$ represents the probability of event E



2) BASIC FORMULAS :

- ① Probability of an event $E = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} = \frac{n(E)}{n(S)}$
- ② $P(\text{Not } E) = 1 - P(E)$
- ③ $P(E) + P(\text{Not } E) = 1$
- ④ If E and F are two events,
 $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$ (Addition Rule)
- ⑤ If E and F are mutually exclusive events (E and F cannot occur together),
 $P(E \text{ or } F) = P(E) + P(F)$
- ⑥ If E and F are independent events,
 $P(E \text{ and } F) = P(E) \times P(F)$

3) IMPORTANT POINTS :

- ★ Total probability is always between 0 and 1.
- ★ Sum of probabilities of all outcomes in sample space = 1.
- ★ Mutually exclusive events cannot occur at the same time.
- ★ Independent events do not affect each other.
- ★ More favourable outcomes → Higher probability.



4) IMPORTANT TOPICS TO FOCUS :

- Sample Space (S) : The set of all possible outcomes.
- Type of Events :
 - Certain Event → $P(E) = 1$
 - Impossible Event → $P(E) = 0$
 - Equally Likely Outcomes → All outcomes have same chance.
- Complementary Events : Event not occurring.
 $P(\text{Not } E) = 1 - P(E)$
- Mutually Exclusive Events : Cannot happen at same time.
 $P(E \text{ or } F) = P(E) + P(F)$
- Independent Events : Occurrence of one does not affect other.
 $P(E \text{ and } F) = P(E) \times P(F)$
- Dependent Events : Occurrence of one affects the other.
(Used in cards/dice without replacement)
- Conditional Probability : Probability of E given F.
 $P(E|F) = \frac{P(E \text{ and } F)}{P(F)}$ (Where $P(F) > 0$)



TCS TIP :

- ★ Always write total outcomes (denominator) carefully. Many students make mistakes in counting outcomes.



5) PRACTICE QUESTIONS (TCS LEVEL) :

1. A coin is tossed once. What is the probability of getting a head?
a) 0 b) $\frac{1}{2}$ c) $\frac{1}{3}$ d) 1 e) $\frac{2}{3}$
2. Two coins are tossed. What is the probability of getting at least one head?
a) $\frac{1}{4}$ b) $\frac{1}{2}$ c) $\frac{3}{4}$ d) 1 e) $\frac{2}{4}$
3. A dice is rolled once. What is the probability of getting a prime number?
a) $\frac{1}{6}$ b) $\frac{2}{6}$ c) $\frac{3}{6}$ d) $\frac{1}{2}$ e) $\frac{4}{6}$
4. From a deck of 52 cards, one card is drawn. What is the probability of getting a King?
a) $\frac{1}{13}$ b) $\frac{1}{4}$ c) $\frac{1}{17}$ d) $\frac{4}{52}$ e) $\frac{1}{52}$
5. If $P(A) = 0.6$, then $P(\text{Not } A) = ?$
a) 0.3 b) 0.4 c) 0.5 d) 0.6 e) 0.8
6. Two dice are rolled. What is the probability that the sum of numbers is 7?
a) $\frac{1}{6}$ b) $\frac{1}{12}$ c) $\frac{1}{9}$ d) $\frac{1}{8}$ e) $\frac{1}{36}$
7. A box contains 3 red, 4 blue and 5 green balls. One ball is drawn at random. What is the probability of getting a blue ball?
a) $\frac{1}{3}$ b) $\frac{2}{5}$ c) $\frac{4}{12}$ d) $\frac{1}{4}$ e) $\frac{1}{2}$
8. If events A and B are independent, $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
a) 0.1 b) 0.2 c) 0.3 d) 0.4 e) 0.5
9. What is the probability that in a group of 3 people, at least two have the same birthday? (Assume 365 days in a year)
a) 0.0027 b) 0.0137 c) 0.0274 d) 0.0548 e) 0.0822
10. If $P(E \text{ or } F) = 0.7$, $P(E) = 0.4$ and $P(F) = 0.5$, then $P(E \text{ and } F) = ?$
a) 0.2 b) 0.3 c) 0.4 d) 0.5 e) 0.6

QUICK FORMULA RECAP :

- ★ $P(E) = \frac{n(E)}{n(S)}$
- ★ $P(\text{Not } E) = 1 - P(E)$
- ★ $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$
- ★ If E and F are mutually exclusive → $P(E \text{ or } F) = P(E) + P(F)$
- ★ If E and F are independent → $P(E \text{ and } F) = P(E) \times P(F)$
- ★ $P(E|F) = \frac{P(E \text{ and } F)}{P(F)}$ ($P(F) > 0$)



6) ANSWER KEY :

- | | | | | |
|--------|--------|--------|--------|---------|
| 1. (b) | 2. (a) | 3. (b) | 4. (d) | 5. (c) |
| 6. (a) | 7. (d) | 8. (b) | 9. (e) | 10. (b) |

1) INTRODUCTION :

In Statement & Assumption questions, we are given a statement followed by some assumptions. We have to decide which of the assumptions is/are Implicitly (hidden) or Definitely true in the statement.

2) APPROACH TO SOLVE :

- ★ Read the statement carefully and understand its meaning.
- ★ Do not take any assumption to be true unless it is definitely implied.
- ★ An assumption is something which must be true if the statement is true.
- ★ Eliminate assumptions which are not supported by the statement.
- ★ Choose the correct option based on the codes given.



3) TYPES OF ASSUMPTIONS :

Type	Explanation
Explicit	Directly mentioned in the statement. (Always true)
Implicit	Not directly mentioned but must be true according to the statement.
Not Implicit	Not required to be true.
False	Contradicts the statement.

4) IMPORTANT POINTS :

- ★ Look for keywords like must, all, always, never, only, some, enough, none, etc.
- ★ Do not use outside knowledge or general facts.
- ★ Assumptions are something taken for granted in the statement.
- ★ If an assumption is more than the information given in the statement, it is not implicit.



5) EXAMPLES WITH SOLUTION :

Example 1 :

Statement : Students should wear ID cards in the college.

Assumptions :

- I. Students have ID cards.
- II. ID cards help in identification.

Solution :

Assumption I is implicit because if students don't have ID cards, the statement is meaningless.

Assumption II is not implicit as it gives a reason, not necessarily implied.

Answer : Only I is implicit.

Example 2 :

Statement : The government should ban plastic bags.

Assumptions :

- I. Plastic bags are harmful.
- II. People use plastic bags.

Solution :

Assumption I is implicit (otherwise ban is not needed).

Assumption II is not necessary for ban.

Answer : Only I is implicit.

6) PRACTICE QUESTIONS (TCS LEVEL) :

1. Statement : Regular exercise is good for health.

Assumptions :

- I. Exercise keeps the body fit.
- II. Everyone does regular exercise.

- a) Only I is implicit
- b) Only II is implicit
- c) Both I and II are implicit
- d) Neither I nor II is implicit
- e) Only II is not implicit

2. Statement : Water should be saved.

Assumptions :

- I. Water is a valuable resource.
- II. People waste water.

- a) Only I is implicit
- b) Only II is implicit
- c) Both I and II are implicit
- d) Neither I nor II is implicit
- e) Only II is not implicit

3. Statement : Students must complete their homework.

Assumptions :

- I. Homework helps in learning.
- II. Students are given homework by teachers.

- a) Only I is implicit
- b) Only II is implicit
- c) Both I and II are implicit
- d) Neither I nor II is implicit
- e) Only II is not implicit

4. Statement : We should plant more trees.

Assumptions :

- I. Trees provide us oxygen.
- II. There are fewer trees now.

- a) Only I is implicit
- b) Only II is implicit
- c) Both I and II are implicit
- d) Neither I nor II is implicit
- e) Only II is not implicit

5. Statement : The company increased the price of the product.

Assumptions :

- I. The cost of production has increased.
- II. The company wants to earn more profit.

- a) Only I is implicit
- b) Only II is implicit
- c) Both I and II are implicit
- d) Neither I nor II is implicit
- e) Only II is not implicit

7) ANSWER KEY :

1. (a)
2. (a)
3. (a)
4. (c)
5. (b)

★ TCS TIP :

Read the statement slowly, underline the important words and check each assumption one by one.



EXTRA NOTES :

- ★ Assumptions are hidden meanings behind the statement.
- ★ They are not always facts.
- ★ Practice more to find what is definitely implied.



VERBAL ABILITY

CLOZE TEST

1) INTRODUCTION :

A Cloze Test is a passage with some words omitted. You have to fill in the blanks with the most appropriate word that fits the context grammatically and meaningfully.



2) WHAT DOES IT TEST ?

- ★ Vocabulary
- ★ Grammar (Tense, Preposition, Conjunction, etc.)
- ★ Contextual understanding
- ★ Logical flow of ideas

3) STRATEGY TO SOLVE :

- Read the whole passage once to understand the theme and flow.
- Look at the blank carefully and analyze the type of word needed (Noun / Verb / Adjective / Adverb / Conjunction / Preposition / Pronoun).
- Use the surrounding words (before and after the blank) to guess the correct word.
- Try each option in your mind and choose the one that makes the sentence meaningful and grammatically correct.
- Move to the next blank. Don't get stuck on one.

4) TYPES OF WORDS COMMONLY ASKED :

Type	Examples
Nouns	life, idea, decision, success, change
Verbs	achieve, increase, improve, believe, happen
Adjectives	important, essential, difficult, possible
Adverbs	quickly, easily, clearly, greatly
Prepositions	in, on, at, by, with, for, between
Conjunctions	and, but, because, although, whereas
Pronouns	he, she, they, it, his, their, this, these

5) TIPS TO IMPROVE :

- ★ Read newspapers, editorials, and good articles daily.
- ★ Learn new words and use them in sentences.
- ★ Practice Cloze tests regularly.
- ★ Focus on grammar rules.
- ★ Eliminate options that don't fit the context.



TCS TIP :

In Cloze Test, understanding the context is more important than knowing tough words. Always choose the word that keeps the meaning and flow of the passage correct.



6) PRACTICE QUESTIONS (TCS LEVEL) :

1. Success is not measured by the position one has reached in life, but by the _____ one has overcome while trying to succeed.
a) obstacles b) obstacle c) obstruct d) obstruction
2. The scientist explained the theory so _____ that even the students could understand it easily.
a) clear b) clearly c) clarity d) clearing
3. She worked hard all year and _____ passed the examination with good marks.
a) fortunately b) fortune c) fortunate d) luckily
4. Air pollution is one of the main reasons _____ people suffer from respiratory problems.
a) why b) which c) whose d) that
5. Although he had all the resources, he failed _____ he was not focused on his goal.
a) because b) so c) but d) although

7) CLOZE PASSAGE PRACTICE :

In today's world, technology plays an important role in our (1) _____. It has made communication (2) _____ and faster. People (3) _____ connect with others across the world in just a few seconds. However, excessive use of technology can also (4) _____ negative effects on our health. It is important to use it (5) _____.

1. a) life b) lives c) living d) lifestyle
2. a) easy b) easily c) easier d) easiest
3. a) can b) could c) may d) might
4. a) cause b) causes c) caused d) causing
5. a) careful b) carefully c) care d) careless

8) ANSWER KEY :

- MCQs : 1. (a) 2. (a) 3. (a) 4. (d) 5. (a)
Passage : 1. (d) 2. (a) 3. (a) 4. (a) 5. (b)

EXTRA WORDS (FOR PRACTICE):

- Abundant - existing in large quantities
- Acquire - to gain, obtain
- Enhance - to improve or increase
- Significant - important, meaningful
- Maintain - to keep, preserve
- Eventually - finally, in the end
- Although - in spite of the fact that
- Therefore - as a result
- Moreover - in addition
- However - on the other hand



Remember : Read - Understand - Analyze - Eliminate - Choose the Best

1) INTRODUCTION :

Word Usage questions check how well you know the meaning of a word and how it is used in a sentence. You have to choose the option that is most appropriate in the given context.

**2) TYPES OF WORD USAGE QUESTIONS :**

- ★ Choose the correct word (Synonyms)
- ★ Choose the most appropriate word
- ★ Identify the incorrect word
- ★ Word replacement
- ★ Idioms and Phrases
- ★ One word substitution

3) POWER WORDS (HIGH FREQUENCY) :

Word	Meaning	Synonym	Antonym
Abundant	Existing in large quantities	Plentiful	Scarce
Ambiguous	Unclear in meaning	Vague	Clear
Diligent	Hardworking, careful	Industrious	Lazy
Eloquent	Fluent or persuasive in speaking	Articulate	Inarticulate
Meticulous	Very careful and precise	Precise	Careless
Resilient	Able to recover quickly	Strong	Fragile
Ubiquitous	Present everywhere	Universal	Rare
Zealous	Having great enthusiasm	Passionate	Indifferent

4) COMMONLY CONFUSED WORDS :

- Affect (Verb) → to influence | Effect (Noun) → result
Ex: The rain affected the match. (influence)
It had an effect on the game. (result)
- Complement (Verb) → to complete | Compliment (Noun) → praise
Ex: She complemented her project.
He gave her a lovely compliment.
- Principle (Noun) → rule, belief | Principal (Noun) → head of a school
Ex: Honesty is a good principle.
Our principal addressed the students.
- Stationary (Adjective) → not moving | Stationery (Noun) → writing materials
Ex: The car is stationary.
I bought some new stationery.

5) WORD USAGE TIPS :

- ★ Read the sentence completely before choosing the word.
- ★ Understand the meaning of all options.
- ★ Eliminate words that do not fit grammatically or contextually.
- ★ Check for collocations (words that go together).
- ★ Practice every day to build strong vocabulary.

**TCS TIP :**

While attempting Word Usage questions, focus on context and collocations rather than just the meaning of individual words. This increases accuracy.

**6) PRACTICE QUESTIONS (TCS LEVEL) :**

- The manager appreciated the team for their _____ performance in completing the project.
a) mediocre b) outstanding c) ordinary d) poor
- Her speech was so _____ that everyone in the auditorium was impressed.
a) eloquent b) dull c) vague d) brief
- The decision was _____ because it was taken without enough information.
a) cautious b) hasty c) careful d) wise
- Despite facing many difficulties, he remained _____ and never gave up.
a) fragile b) resilient c) weak d) sensitive
- The instructions were _____ and easy to follow.
a) ambiguous b) complex c) clear d) confusing
- The teacher asked the students to be more _____ in their approach to the task.
a) meticulous b) careless c) vague d) hurried
- The charity aims to _____ the lives of underprivileged children.
a) improve b) complicate c) worsen d) ignore
- His constant complaints were a _____ to everyone around him.
a) delight b) burden c) surprise d) pleasure
- The discovery of water on Mars was a _____ moment for scientists.
a) trivial b) significant c) minor d) ordinary
- She has a _____ for collecting rare coins.
a) hobby b) habit c) boredom d) dislike

7) IDIOMS & PHRASES :

- A blessing in disguise → Something good that seemed bad at first.
Ex: Losing the job was a blessing in disguise; I got a better one.
- Burn the midnight oil → Work late into the night.
Ex: He burns the midnight oil to prepare for exams.
- Piece of cake → Very easy.
Ex: The test was a piece of cake for me.
- Hit the nail on the head → Say exactly what is right.
Ex: You hit the nail on the head with your suggestion.

**8) ANSWER KEY :**

- (b)
- (a)
- (b)
- (b)
- (c)
- (a)
- (a)
- (b)
- (b)
- (a)

EXTRA VOCABULARY (LEARN AND USE) :

- ★ Innovate - to introduce new ideas
- ★ Persevere - to continue despite difficulties
- ★ Accommodate - to provide space or adjust
- ★ Transparent - clear, honest
- ★ Versatile - able to adapt to different situations



1) INTRODUCTION :

Software Testing is the process of evaluating and verifying that a software product or application does what it is supposed to do and does it without defects.



2) NEED OF TESTING :

- ★ Detects defects early and reduces fixing cost.
- ★ Improves software quality and reliability.
- ★ Ensures the software meets requirements.
- ★ Builds customer satisfaction and trust.
- ★ Reduces risk of failure in production.

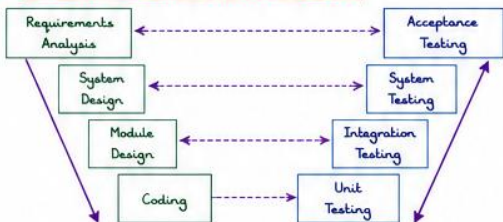
3) FUNDAMENTAL TESTING CONCEPTS :

- **Error** : A human mistake that can lead to a defect.
- **Defect (Bug)** : A flaw in the software where it deviates from expected result.
- **Failure** : The inability of the system to perform its required function.
- **Test Case** : A set of conditions or variables under which a tester will determine whether an application is working correctly.
- **Test Data** : The input values used for testing.

4) TYPES OF TESTING :

Type of Testing	Description
Unit Testing	Testing of individual units or components of the application.
Integration Testing	Testing of interface between integrated modules.
System Testing	Testing of the complete and integrated system.
Acceptance Testing	Testing to check whether the software meets business requirements.
Regression Testing	Re-testing to ensure new changes have not affected existing functionality.
Smoke Testing	Basic level testing to check major functionalities are working.
Sanity Testing	Narrow and focused testing after minor changes.

5) TESTING LEVELS (V-MODEL VIEW) :



TCS TIP :

Understand the level of testing and the purpose of each. Many questions are concept based.



6) PRACTICE QUESTIONS (TCS LEVEL) :

- Which of the following is NOT a type of testing?
 - Unit Testing
 - System Testing
 - Code Testing
 - Integration Testing
- Testing done by the end user to validate the software against business requirements is called _____.
 - System Testing
 - Acceptance Testing
 - Unit Testing
 - Smoke Testing
- Which testing is performed to check if new code changes have broken existing functionality?
 - Regression Testing
 - Sanity Testing
 - Smoke Testing
 - Integration Testing
- Error, Defect and Failure are :
 - Same terms
 - Different terms
 - Used only in coding
 - Used only in design
- Which of the following is a static testing technique?
 - Code Review
 - Unit Testing
 - System Testing
 - Acceptance Testing
- Smoke Testing is also known as :
 - Build Verification Test
 - Stress Test
 - Load Test
 - Security Test
- The main objective of software testing is to :
 - Prove software is correct
 - Find defects and ensure quality
 - Increase code length
 - Reduce development time
- Which testing checks the interface between modules?
 - Unit Testing
 - Integration Testing
 - System Testing
 - Regression Testing
- Which test is done at the end of SDLC before deployment?
 - Unit Test
 - System Test
 - Acceptance Test
 - Smoke Test
- Test Case includes _____.
 - Test Data only
 - Steps and Expected Result
 - Code only
 - Design only

7) TESTING TOOLS (EXAMPLES) :

- Selenium - For automation testing of web applications.
- JUnit / TestNG - Unit testing tools for Java.
- QTP / UFT - Automation testing tool by Micro Focus.
- Postman - API testing tool.
- JIRA - Defect tracking tool.



8) ANSWER KEY :

- (c)
- (b)
- (a)
- (b)
- (a)
- (a)
- (b)
- (b)
- (c)
- (b)

EXTRA NOTES (QUICK REVISION) :

- ★ Testing shows the presence of defects, not their absence.
- ★ Early testing → Less cost | Late testing → More cost
- ★ Good Test Case = Valid Input + Invalid Input + Boundary Value

- ★ Quality = Requirement Fulfillment + Defect Free
- ★ Test more, but test smart.



★ Remember : Basics Strong → Concepts Clear → Confidence High → Selection Sure ! 😊

1) INTRODUCTION :

- Permutation is about arrangement (order matters).
- Combination is about selection (order does not matter).

WHEN TO USE ?

- ★ Permutation → When order/position is important.
- ★ Combination → When order is not important.

Examples:

- From 3 friends A, B, C → Sitting in a row (ABC, ACB, ...) → Permutation
- From 3 friends A, B, C → Choosing 2 for a team (AB, AC, BC) → Combination

2) BASIC FORMULAS :

PERMUTATION (nPr)	COMBINATION (nCr)
$nPr = \frac{n!}{(n-r)!}$ (Order matters)	$nCr = \frac{n!}{r!(n-r)!}$ (Order does not matter)
Where, n = total number of items r = number of items to be arranged / selected $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$	

3) IMPORTANT PROPERTIES & RELATIONS :

- ★ $nPr = \frac{n!}{(n-r)!}$
- ★ $nCr = \frac{n!}{r!(n-r)!}$
- ★ $nCr = nC(n-r)$ (Symmetry Property)
- ★ $nCr + nC(r-1) = {}^{n+1}Cr$ (Pascal's Identity)
- ★ $nPr = nCr \times r!$
- ★ $nCr = \frac{nPr}{r!}$
- ★ Sum of all combinations: $\sum_{r=0}^n nCr = 2^n$



4) FACTORIALS ($n!$):

- $0! = 1$, $1! = 1$
- $n!$ grows very fast with n .
- $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
- $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$
- $7! = 5040$, $8! = 40320$, $9! = 362880$, $10! = 3628800$



5) COMMON SHORTCUTS :

- ✓ If $r = 0$ or $r = n \rightarrow nCr = 1$
- ✓ If $r = 1 \rightarrow nCr = n$
- ✓ If $r = 2 \rightarrow nC_2 = n(n-1)$
- ✓ If $r = 3 \rightarrow nC_3 = n(n-1)(n-2)$
- ✓ If $r = n-1 \rightarrow nC(n-1) = n$



6) APPLICATIONS :

- Arranging people in a line / around a table
- Forming committees / teams from a group
- Assigning different positions (1st, 2nd, 3rd ...)
- Passwords, PIN codes, number plates, selections, etc.

TCS TIP :

Identify whether it's arrangement or selection first.
Then check if some items are identical or repeated.
Write n and r clearly before applying formula.



7) ADDITIONAL CONTENT (VERY IMPORTANT) :

Circular Permutation:

If n different people sit around a circular table,
number of ways = $(n-1)!$



Permutation with Identical Items:

If there are n_1 items of one kind,
 n_2 of second kind, ..., n_k of k -th kind

$$\text{Total permutations} = \frac{n!}{n_1! n_2! \dots n_k!}$$

Ex: Letter "MISSISSIPPI"

$M=1, I=4, S=4, P=2$ (Total 11 letters)

$$\text{Permutations} = \frac{11!}{1! 4! 4! 2!}$$

QUICK NOTE :

- Order matters
→ Permutation
- Order doesn't matter
→ Combination

8) COMMON VALUES (MEMORIZE) :

n	5	6	7	8	9	10
$n!$	120	720	5040	40320	362880	3628800
nC_2	10	15	21	28	36	45
nC_3	10	20	35	56	84	120

9) PRACTICE QUESTIONS (TCS LEVEL) :

- In how many ways can 5 books be arranged on a shelf?
a) 60 b) 100 c) 120 d) 24 e) 20
- In how many ways can 8 people be seated in a row?
a) 40320 b) 5040 c) 720 d) 120 e) 80
- How many combinations of 3 can be formed from 7 people?
a) 21 b) 35 c) 56 d) 70 e) 105
- A committee of 4 is to be formed from 10 members.
In how many ways can this be done?
a) 210 b) 252 c) 120 d) 126 e) 84
- In how many ways can the letters of the word "TEAM" be arranged?
a) 12 b) 16 c) 18 d) 24 e) 48
- In how many ways can 3 students be selected from 12 students for a project?
a) 110 b) 132 c) 140 d) 165 e) 220
- How many 4-digit numbers can be formed using the digits 1, 2, 3, 4 without repetition?
a) 24 b) 64 c) 120 d) 256 e) 12
- How many combinations of 2 can be formed from 9 items?
a) 18 b) 36 c) 42 d) 72 e) 81
- In how many ways can 6 people be seated around a circular table?
a) 720 b) 120 c) 24 d) 60 e) 36
- A code consists of 3 letters followed by 2 digits.
If letters can be chosen from 5 letters and digits from 4 digits (repetition allowed), how many such codes exist?
a) 400 b) 800 c) 1600 d) 3200 e) 6400

10) QUICK RECAP :

- ✓ Use nPr when arrangement is important.
- ✓ Use nCr when selection is important.
- ✓ Remember key formulas & properties.
- ✓ Practice more → Speed & Accuracy increases.



11) ANSWER KEY :

- (c)
- (a)
- (b)
- (a)
- (b)
- (b)
- (c)
- (d)
- (c)
- (d)

LOGICAL REASONING

ANALOGY & CLASSIFICATION

1) ANALOGY - INTRODUCTION

Analogy is a relationship between two pairs of words. The relationship between the first pair is the same as that between the second pair. Find the missing term that fits the same relationship.



Format : A : B :: C : D (B is to A as D is to C)
A : B :: C : ? (Find the missing D)

2) TYPES OF ANALOGY RELATIONSHIPS

TYPE	RELATIONSHIP	EXAMPLE
Synonym	Words with similar meaning	Happy : Joyful :: Quick : Fast
Antonym	Words with opposite meaning	Hot : Cold :: Big : Small
Part - Whole	Part is contained in the whole	Wheel : Car :: Keyboard : Computer
Cause - Effect	One is the reason for the other	Rain : Wet :: Fire : Smoke
Function	One is used for / performs action	Pen : Write :: Scissors : Cut
Category / Class	One belongs to a category	Rose : Flower :: Sparrow : Bird
Degree / Intensity	Different levels of the same quality	Good : Better :: Happy : Happier
Material - Object	Object is made of a material	Gold : Ring :: Wood : Table

3) TIPS TO SOLVE ANALOGY

- ★ Identify the exact relationship between the first pair.
- ★ Apply the same relationship to the second pair.
- ★ Think of multiple possibilities.
- ★ Check each option carefully before finalizing.



4) CLASSIFICATION - INTRODUCTION

In classification, we group things or ideas based on common characteristics. Choose the odd one out or the best category for the given set.



A) ODD ONE OUT

Four items are given. Three are similar in some way. Choose the one that is different.

Example :

- a) Apple b) Banana
c) Carrot d) Potato

Answer : (a) Apple
(Others are vegetables)

B) CLASSIFICATION OF WORDS

Choose the option which best includes the given words. Match with the correct category.

Example :

- Words: Cow, Dog, Lion
Options: a) Animals b) Birds
 c) Insects d) Fruits
Answer : (a) Animals

TCS TIP :

Practice more questions daily. Build your own list of relationships and categories. This will help you solve analogy & classification questions faster in the exam.



5) PRACTICE QUESTIONS (TCS LEVEL)

ANALOGY

- Book : Reading :: Fork : ?
a) Eat b) Kitchen c) Food d) Spoon
- Sun : Day :: Moon : ?
a) Night b) Sky c) Star d) Light
- Doctor : Hospital :: Teacher : ?
a) School b) Student c) Book d) Office
- Pen : Ink :: Brush : ?
a) Paint b) Artist c) Canvas d) Color
- Seed : Plant :: Egg : ?
a) Hen b) Chicken c) Nest d) Bird
- Sweet : Sugar :: Sour : ?
a) Lemon b) Salt c) Spice d) Chili
- King : Kingdom :: President : ?
a) Country b) Leader c) Election d) Power
- Keyboard : Typing :: Mic : ?
a) Sound b) Speak c) Voice d) Record
- Train : Station :: Ship : ?
a) Harbor b) Sea c) Dock d) Port
- Hot : Cold :: Up : ?
a) Down b) Above c) Top d) Left

CLASSIFICATION (ODD ONE OUT)

- a) Rose b) Jasmine c) Mango d) Lotus
- a) Chair b) Table c) Bed d) Sofa
- a) Pen b) Pencil c) Eraser d) Scale
- a) India b) China c) France d) Delhi
- a) Carrot b) Potato c) Tomato d) Apple

CLASSIFICATION OF WORDS (CHOOSE CATEGORY)

- Words: Cat, Dog, Elephant
a) Animals b) Birds c) Insects d) Fruits
- Words: Pen, Notebook, Pencil
a) Stationery b) Furniture c) Electronic d) Tools
- Words: Teacher, Doctor, Engineer
a) Professions b) Students c) Offices d) Workers
- Words: Monday, Tuesday, Thursday
a) Days b) Months c) Planets d) Seasons
- Words: Red, Green, Blue
a) Colors b) Shapes c) Metals d) Numbers

6) EXTRA CONTENT - BUILD YOUR SKILLS

- ★ Make a notebook of relationships (Part-Whole, Cause-Effect, etc.)
- ★ Solve 10 analogy & 10 classification questions daily.
- ★ Use dictionaries to expand vocabulary (helps in analogy).
- ★ Identify patterns quickly - speed is the key in TCS NQT!



- ANSWER KEY : 1. (a) 2. (a) 3. (a) 4. (a) 5. (b) 5. (b) 7. (a) 8. (c) 9. (d) 10. (a)
11. (c) 12. (b) 13. (d) 14. (d) 15. (a) 16. (a) 17. (a) 18. (a) 19. (a) 20. (a)



PRACTICE TODAY, PERFORM TOMORROW, SUCCEED ALWAYS !



VERBAL ABILITY

SENTENCE COMPLETION

1) INTRODUCTION

In Sentence Completion, a part of the sentence is missing. We have to choose the most suitable word or group of words to complete the sentence meaningfully and grammatically.



2) TYPES OF SENTENCE COMPLETION

TYPE	DESCRIPTION
Cloze Test	A passage with blanks. Fill in the blanks with the most suitable option.
Single Blank	One word is missing in a sentence. Choose the best option.
Double Blank	Two blanks are given. Choose the best pair of words.
Error Based	Given sentence has an error. Choose the option that makes the sentence correct.
Para Jumbles	Arrange the sentences in the correct order to form a meaningful paragraph.

3) STRATEGY TO SOLVE

- ★ Read the whole sentence carefully before looking at options.
- ★ Understand the meaning of the sentence.
- ★ Identify the type of word missing. (Noun / Verb / Adjective / Adverb / Preposition / Conjunction)
- ★ Check grammar → subject-verb agreement, tense, articles, prepositions, etc.
- ★ Eliminate the options that do not fit grammatically or logically.
- ★ Choose the most appropriate and meaningful option.



4) COMMON GRAMMAR RULES TO REMEMBER

- ✓ Subject-Verb Agreement :
Singular subject → singular verb (He is)
Plural subject → plural verb (They are)
- ✓ Tense Consistency :
All parts of the sentence should be in the same tense.
- ✓ Articles (a, an, the) :
Use 'a' before consonant sound, 'an' before vowel sound.
- ✓ Prepositions :
Use the correct preposition according to the context.
- ✓ Parallelism :
Use the same form of words in a series.
- ✓ Logical Flow :
The completed sentence should make complete sense.



6) WORDS OFTEN USED IN SENTENCE COMPLETION

Improve (v) - to make better	Enhance (v) - to increase; improve
Achieve (v) - to successfully reach a goal	Indicate (v) - to show; point out
Essential (adj) - extremely important	Facilitate (v) - to make easier
Significant (adj) - important; meaningful	Potential (adj) - possible; capable

5) PRACTICE QUESTIONS (TCS LEVEL)

SINGLE BLANK

- She has a great _____ for music.
a) liking b) like c) likes d) liked
- The manager gave a _____ speech to the employees.
a) inspire b) inspiring c) inspired d) inspiration
- I have never seen such a _____ performance.
a) good b) well c) better d) best
- The child was too _____ to understand the problem.
a) young b) younger c) youngest d) youth
- He worked _____ to finish the project on time.
a) hard b) hardly c) hardest d) more hard

DOUBLE BLANK

- _____ he was tired, _____ he finished the work.
a) Although - but b) Though - yet
c) Because - so d) If - then
- She is not only intelligent but also _____.
a) hardworking b) hard working c) works hard d) more hard

CLOZE TEST (FILL IN THE BLANKS)

- Read the passage and fill in the most suitable option.
Success in life _____ (1) on talent alone. It also requires hard work, _____ (2) and the ability to learn from failures.
(1) a) depend b) depends c) is depend d) depending
(2) a) patience b) patient c) patiently d) patientful

ERROR BASED

- Find the part that contains the error.
He don't like to go for a walk in the morning.
a) He b) don't c) like d) go

PARA JUMBLE (ORDER THE SENTENCES)

- Arrange the following sentences in the correct order.
A. He decided to start his own business.
B. After saving money for two years,
C. Rohan always dreamed of becoming an entrepreneur.
D. Now he is running a successful startup.
a) C, B, A, D b) C, A, B, D c) B, C, A, D d) A, B, C, D

7) TIPS TO IMPROVE

- ✓ Read newspapers, editorials and good books.
- ✓ Learn new words daily and use them in sentences.
- ✓ Practice at least 10-15 sentence completion questions daily.
- ✓ Analyze every wrong answer and learn the rule.
- ✓ Build a personal vocabulary notebook.



ANSWER KEY : 1. (a) 2. (b) 3. (a) 4. (a) 5. (a) 6. (b) 7. (a)
8. (1-b) (2-c) 9. (b) 10. (a)



TCS TIP

Reading + Practice + Analysis =
High Score in Verbal Ability !



READ WELL, THINK WELL, CHOOSE THE BEST, SCORE THE BEST !



VOCABULARY

FREQUENTLY ASKED TCS WORDS

1) WHY VOCABULARY IS IMPORTANT ?

- ★ Improves Reading Comprehension & Verbal Ability.
- ★ Helps in Sentence Completion, Cloze Test & Para Jumbles.
- ★ Enhances Communication Skills & Confidence.



2) HIGH FREQUENCY WORDS (WITH MEANING, SYNONYM & ANTONYM)

WORD	MEANING	SYNONYM	ANTONYM	USAGE (EXAMPLE)
1. Abundant	Existing / available in large quantities	Plentiful	Scarce	Sunlight is abundant in summer.
2. Adept	Skilled; proficient	Expert	Incompetent	She is adept at solving puzzles.
3. Alleviate	Make (suffering, pain, etc.) less severe	Relieve	Aggravate	The medicine helped alleviate the pain.
4. Ambiguous	Unclear; open to more than one interpretation	Vague	Clear	The instructions were ambiguous.
5. Articulate	Express ideas clearly and effectively	Eloquent	Incoherent	He is articulate in his speech.
6. Benevolent	Kind; well-meaning	Kindhearted	Malevolent	She is benevolent towards everyone.
7. Conceive	Form an idea in the mind	Imagine	Forget	It is hard to conceive such an idea.
8. Diligent	Hardworking; careful	Industrious	Lazy	He is diligent in his studies.
9. Eminent	Famous; respected	Illustrious	Obscure	Dr. Kalam was an eminent scientist.
10. Impartial	Fair; unbiased	Neutral	Biased	The judge remained impartial.

3) WORD BUILDING (PREFIXES & SUFFIXES)

PREFIXES (Added at the beginning)	SUFFIXES (Added at the end)
un- : not (happy → unhappy)	-ness : state/quality (kind → kindness)
in-/im-/il-/ir- not (possible → impossible)	-tion : action/result (act → action)
re- : again (write → rewrite)	-ment : process/result (improve → improvement)
pre- : before (decide → predecide)	-able : able to be (read → readable)
dis- : opposite (agree → disagree)	-ly : in a manner (quick → quickly)

4) IDIOMS & PHRASES (FREQUENTLY ASKED)

- A blessing in disguise → Something good that seemed bad at first.
- Bite the bullet → Face a difficult situation bravely.
- Break the ice → Initiate conversation in a social setting.
- Hit the nail on the head → Describe exactly what is causing a situation.
- Once in a blue moon → Very rarely.



5) ONE WORD SUBSTITUTIONS

- A person who loves mankind → Philanthropist
- One who tells lies → Liar
- A person who is always happy → Optimist
- A person who studies stars → Astronomer
- One who is present everywhere → Omnipresent



TCS TIP :

- Learn 5-10 new words daily.
- Use the words in your own sentences.
- Revise words regularly to remember for long.



6) PRACTICE QUESTIONS (TCS LEVEL)

SYNONYMS

- Choose the SYNONYM of "Diligent".
a) Lazy b) Careless c) Industrious d) Idle
- Choose the SYNONYM of "Ambiguous".
a) Clear b) Confusing c) Certain d) Obvious
- Choose the SYNONYM of "Abundant".
a) Scarce b) Plentiful c) Little d) Rare
- Choose the SYNONYM of "Eminent".
a) Unknown b) Famous c) Common d) Ordinary
- Choose the SYNONYM of "Alleviate".
a) Increase b) Aggravate c) Relieve d) Intensify

ANTONYMS

- Choose the ANTONYM of "Benevolent".
a) Kind b) Generous c) Malevolent d) Helpful
- Choose the ANTONYM of "Adept".
a) Skilled b) Proficient c) Incompetent d) Talented
- Choose the ANTONYM of "Impartial".
a) Biased b) Neutral c) Fair d) Just
- Choose the ANTONYM of "Abundant".
a) Plentiful b) Scarce c) Ample d) Sufficient
- Choose the ANTONYM of "Articulate".
a) Clear b) Silent c) Incoherent d) Fluent

SENTENCE COMPLETION

- His _____ explanation removed all our doubts.
a) ambiguous b) articulate c) scarce d) obscure
- The heavy rain will _____ the drought.
a) alleviate b) aggravate c) increase d) continue
- Gold is a _____ metal.
a) abundant b) scarce c) little d) rare
- The decision was _____ and completely unbiased.
a) partial b) impartial c) biased d) unfair
- She is _____ with her work and never lazy.
a) diligent b) idle c) careless d) weak

IDIOMS & PHRASES

- It is a _____ that you failed the test, but don't lose hope.
a) bite the bullet b) blessing in disguise
c) hit the nail on the head d) break the ice

ONE WORD SUBSTITUTION

- One who tells lies.
a) Honest b) Sincere c) Liar d) Wise
- One who studies stars.
a) Scientist b) Astronomer c) Mathematician d) Geologist
- A person who loves mankind.
a) Philanthropist b) Entrepreneur c) Patriot d) Scholar
- One who is always happy.
a) Pessimist b) Optimist c) Realist d) Thinker

ANSWER KEY : 1. (a) 2. (b) 3. (b) 4. (b) 5. (c) 6. (c) 7. (c) 8. (a) 9. (b) 10. (c)
11. (b) 12. (a) 13. (a) 14. (b) 15. (a) 16. (b) 17. (a) 18. (b) 19. (a) 20. (b)



BUILD VOCABULARY TODAY, EXPRESS CONFIDENCE TOMORROW, ACE THE TCS NQT !





1) PYTHON - QUICK REVISION

- ★ Variables & Data Types : int, float, str, bool, list, tuple, set, dict
- ★ Operators : Arithmetic, Relational, Logical, Assignment, Bitwise
- ★ Control Flow : if, if-else, elif, for, while, break, continue, pass
- ★ Functions : def, return, *args, **kwargs, lambda
- ★ Data Structures : list, tuple, set, dict, comprehension
- ★ OOP Concepts : class, object, inheritance, polymorphism, encapsulation
- ★ Modules : import, from ... import, built-in modules
- ★ File Handling : open(), read(), write(), with statement
- ★ Exception Handling : try, except, else, finally, raise



2) IMPORTANT PYTHON CONCEPTS (WITH EXAMPLES)

CONCEPT	EXAMPLE
List Comprehension	<pre>squares = [x*x for x in range(1,6)] # [1, 4, 9, 16, 25]</pre>
Dictionary Comprehension	<pre>d = {x:x*x for x in range(1,4)} # {1: 1, 2: 4, 3: 9}</pre>
Lambda Function	<pre>add = lambda a, b: a + b add(2, 3) # 5</pre>
Enumerate()	<pre>for idx, val in enumerate(['a','b','c']): print(idx, val) # (0,'a') (1,'b') (2,'c')</pre>
Zip()	<pre>names = ['a','b'] scores = [90, 80] list(zip(names, scores)) # [('a', 90), ('b', 80)]</pre>
Filter()	<pre>evens = list(filter(lambda x: x%2==0, [1,2,3,4])) # [2, 4]</pre>
Map()	<pre>cube = list(map(lambda x: x**3, [1,2,3])) # [1, 8, 27]</pre>

3) COMMON BUILT-IN FUNCTIONS

- len() → length of object
- max(), min() → largest, smallest item
- sum() → sum of elements
- sorted() → returns sorted list
- any() → True if any element is True
- all() → True if all elements are True
- type() → returns data type
- isinstance(obj, class) → checks object type



4) TIME & SPACE COMPLEXITY (BASICS)

OPERATION	TIME COMPLEXITY	EXAMPLE
Access	O(1)	List[0], dict[key]
Search (Unsorted List)	O(n)	x in list
Search (Sorted List)	O(log n)	binary search
Insertion (End of List)	O(1)	List.append(x)
Insertion (Beginning)	O(n)	List.insert(0, x)
Deletion (By Value)	O(n)	List.remove(x)
Sorting	O(n log n)	sorted(List)

★ TCS TIP:

- ✓ Understand the question properly.
- ✓ Write clean, readable and optimized code.
- ✓ Handle edge cases (empty input, single element, etc.).
- ✓ Dry run your code with sample inputs.
- ✓ Practice coding regularly - consistency is the key!



5) PYTHON INTERVIEW MCQs (TCS LEVEL)

1. What is the output of the following code?

```
print(2 ** 3 ** 2)
```

- a) 64 b) 512 c) 32 d) 256

2. Which of the following is NOT a valid variable name in Python?

- a) _value b) value1 c) 1value d) value_2

3. What is the output of the following code?

```
print(type([]))
```

- a) <class 'list'> b) <class 'dict'>
c) <class 'tuple'> d) <class 'set'>

4. What will be the output of the following code?

```
print(10 // 3, 10 % 3)
```

- a) 3 1 b) 3.0 1.0 c) 3.0.333 d) 3.0 0.333

5. Which of the following is used to handle an exception in Python?

- a) catch b) handle c) try-except d) throw

6. What is the output of the following code?

```
a = [1, 2, 3]
b = a
a.append(4)
print(b)
```

- a) [1, 2, 3] b) [1, 2, 3, 4]
c) [4] d) Error

7. What is the output of the following code?

```
s = 'hello'
print(s[::-1])
```

- a) hello b) olleh c) loleh d) heool

8. What does the 'len()' function do?

- a) Returns the largest element
b) Returns the length/number of items
c) Sorts the data
d) Returns the index

9. Which of the following is used to read a file in Python?

- a) open('file.txt', 'r') b) read('file.txt')
c) file('file.txt', 'r') d) readFile('file.txt')

10. What is the output of the following code?

```
{x:x*x for x in range(1,4)}
```

- a) {1:1, 2:4, 3:9} b) {1:1, 2:2, 3:3}
c) {0:0, 1:1, 2:4} d) {1:1, 2:8, 3:27}

11. What is the output of the following code?

```
print(bool([]))
```

- a) True b) False c) 0 d) 1

12. Which keyword is used to define a function in Python?

- a) func b) define c) def d) function

13. What is the output of the following code?

```
t = (1, 2, 3)
t[0] = 10
```

- a) (10, 2, 3) b) Error c) (1, 2, 3) d) None

14. Which of the following will remove duplicates from a list?

- a) list.remove_dups() b) set(list)
c) unique(list) d) list.drop_duplicates()

15. What is the output of the following code?

```
print('5' + '5')
```

- a) 10 b) 55 c) 5+5 d) Error

ANSWER KEY: 1. (a) 2. (c) 3. (a) 4. (a) 5. (c) 6. (b) 7. (b) 8. (b)
9. (a) 10. (a) 11. (b) 12. (c) 13. (b) 14. (b) 15. (b)



Keep Practicing, Keep Improving,
Keep Succeeding !





1) INTRODUCTION

- ★ Data Interpretation (DI) is the process of analyzing data presented in charts, tables, graphs, etc., to find useful information and solve questions.
- ★ It tests your calculation, analytical and logical skills.
- ★ DI questions are usually based on real-life situations (business, economy, sales, population, etc.).



2) TYPES OF DATA INTERPRETATION

Type	Description
Tabular DI	Data given in rows and columns
Bar Graph	Data represented using bars
Line Graph	Data shown with lines over time
Pie Chart	Data shown in a circular chart (percentage)
Mixed Graphs	Combination of two or more types

3) KEY FORMULAS & CONCEPTS

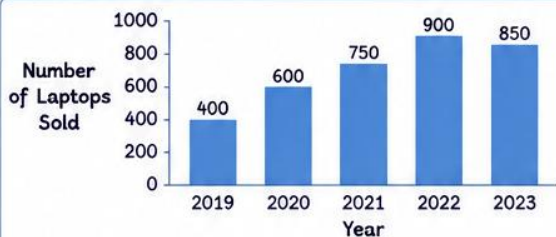
- ★ Percentage = $\left(\frac{\text{Part}}{\text{Whole}}\right) \times 100$
- ★ Growth % = $\left(\frac{\text{New} - \text{Old}}{\text{Old}}\right) \times 100$
- ★ Ratio = $a : b = \frac{a}{b}$
- ★ Average = $\frac{\text{Sum of values}}{\text{Number of values}}$
- ★ Total = Sum of all individual values
- ★ Pie chart: Value = $\left(\frac{\text{Angle}}{360}\right) \times \text{Total}$
- ★ Comparison = Find difference, % increase/decrease, etc.

Formula

 +
Observation
=
Correct
Answer


4) EXAMPLE 1 - BAR GRAPH

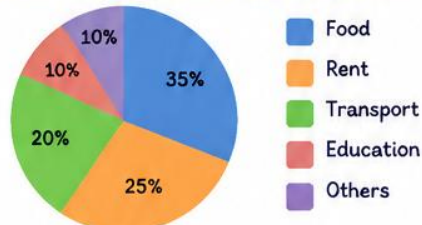
The bar graph shows the number of laptops sold by a company from 2019 to 2023.



1. In which year were the maximum laptops sold?
2. What is the total number of laptops sold in 2019 and 2020?
3. What is the percentage increase in sales from 2019 to 2020?

5) EXAMPLE 2 - PIE CHART

The pie chart shows the expenditure of a person per month.



1. Which category has the highest expenditure?
2. If total expenditure is ₹20,000, how much is spent on Rent?
3. What is the total percentage spent on Education and Others?

ANSWER KEY (Examples)

Ex 1: 1) 2022 2) 1000 3) 50% Ex 2: 1) Food 2) ₹5,000 3) 20%

6) EXAMPLE 3 - TABLE

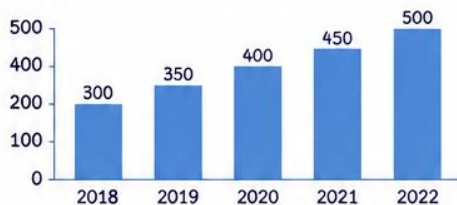
The table shows the number of students from four departments in a college over three years.

Department	2022	2023	2024
CSE	120	135	150
ECE	100	110	120
ME	80	90	100
CE	70	80	90

1. Which department had the highest number of students in 2024?
2. What is the total number of students in 2023?
3. Which department showed the highest percentage increase from 2022 to 2024?

7) PRACTICE QUESTIONS

1. The bar graph shows the production (in tons) of wheat by a company from 2018 to 2022.



What is the percentage increase in production from 2018 to 2022?

- a) 50% b) 60% c) 66.67% d) 75%
2. In a pie chart, the angle corresponding to a sector is 108° . What percentage does it represent?
 - a) 20% b) 25% c) 30% d) 35%
 3. The average of 5 numbers is 48. If one number (60) is excluded, the new average is:
 - a) 44 b) 45 c) 46 d) 47
 4. From the table below, what is the total sales in Q2?

Quarter	Q1	Q2	Q3	Q4
Sales (in ₹ lakhs)	20	35	30	25

 - a) 80 b) 35 c) 30 d) 110
 5. If the population of a city increased from 5,00,000 to 6,00,000, what is the percentage increase?
 - a) 10% b) 15% c) 20% d) 25%

8) TIPS & STRATEGY

- ✓ Read the given data carefully.
- ✓ Understand what the question is asking.
- ✓ Use estimation to save time.
- ✓ Learn to identify the type of chart quickly.
- ✓ Focus on key numbers (total, max, min, %).
- ✓ Avoid unnecessary calculations.
- ✓ Practice daily with a timer.

 Accuracy
+
Speed
=
Success
in
DI !


9) EXTRA PRACTICE (TRY YOURSELF)

- ★ Solve 10 DI questions of each type (table, bar, pie, line).
- ★ Attempt previous year TCS NQT DI questions.
- ★ Analyze your mistakes and maintain an error notebook.
- ★ Improve calculation speed with mental math.





LOGICAL REASONING

CRITICAL REASONING



Think Deep • Question Clearly • Choose Wisely

1) INTRODUCTION

- ★ Critical Reasoning is the ability to analyze a statement, identify assumptions, evaluate arguments and draw logical conclusions.
- ★ It helps you think logically, avoid fallacies and make well-reasoned decisions in real-life situations.
- ★ It is widely tested in competitive exams and placements.



2) KEY CONCEPTS

Concept	Description
Argument	A set of statements where one or more statements (premises) support a conclusion.
Premise	A statement that provides a reason for the conclusion.
Conclusion	The statement that is being supported.
Assumption	An unstated statement taken for granted in the argument.
Inference	A logical conclusion drawn from the given information.
Evaluation	Judging the strength or validity of an argument.
Fallacy	A flaw in reasoning that makes the argument invalid.

3) TYPES OF CRITICAL REASONING QUESTIONS

1. Strengthen the Argument → Choose the option that strengthens the given argument.
2. Weaken the Argument → Choose the option that weakens the given argument.
3. Assumption → Find the unstated assumption in the argument.
4. Inference → Draw a logical conclusion from the given information.
5. Logical Reasoning → Identify the most logical conclusion or statement.

4) COMMON FALLACIES (TO AVOID)

Fallacy	Meaning
Ad hominem	Attacking the person instead of the argument.
Strawman	Misrepresenting someone's argument.
Appeal to emotion	Using emotions instead of logic.
False dilemma	Presenting only two options when more exist.
Circular reasoning	Using the conclusion as a premise.
Appeal to authority	Relying only on an authority's opinion.

5) QUICK TIPS

- ✓ Read the question and options carefully.
- ✓ Identify the argument, conclusion and assumptions.
- ✓ Look for weak or strong points, not just the obvious answer.
- ✓ Eliminate extreme options (always, never, all, none).
- ✓ Practice different types of critical reasoning questions.

6) PRACTICE QUESTIONS (CRITICAL REASONING)

1. Which of the following is an assumption in the argument below?
 "Online education should be made mandatory for all students, because it improves learning outcomes and is more flexible."
 a) Online education is cheaper than traditional education.
 b) Students prefer learning in a physical classroom.
 c) All students have access to a computer and the internet.
 d) Traditional education is not effective.
2. Which one of the following, if true, would most weaken the argument?
 "This company should not expand its operations, as its current profit margin is too low."
 a) The company has a strong brand reputation.
 b) The market demand for the product is increasing.
 c) The expansion will require a large investment.
 d) Competitors are facing financial difficulties.
3. Which of the following best states the conclusion of the argument below?
 "More trees should be planted in urban areas, because they reduce pollution, provide shade and improve mental health."
 a) Trees should be planted only in rural areas.
 b) Urban areas are becoming more polluted.
 c) Planting trees is beneficial for the environment and people.
 d) Trees are an unnecessary expense for cities.
4. Which of the following is a logical inference from the statement?
 "The library will be closed on Sunday due to maintenance work."
 a) The library will be open on Saturday.
 b) The library will be closed for an additional day.
 c) Maintenance work is always done on weekends.
 d) The library is not maintained regularly.
5. Which of the following options represents the strongest argument?
 a) The new policy should be implemented because it worked well in other countries and is supported by experts.
 b) The new policy is good because I like it.
 c) The new policy will be successful because it is new.
 d) The new policy is bad because it will require extra effort.
6. Identify the conclusion in the following argument.
 "If we invest in renewable energy, we can reduce our dependence on fossil fuels and create a cleaner environment."
 a) We should invest in renewable energy.
 b) Fossil fuels are harmful to the environment.
 c) Renewable energy is expensive.
 d) The government should support green initiatives.



TCS TIP:

- ✓ Focus on understanding the logic, not just the content.
- ✓ Practice with real exam-level questions.
- ✓ Improve speed and accuracy by eliminating options.
- ✓ Think critically, not blindly!



ANSWER KEY:

1. (c) 2. (c) 3. (c) 4. (a) 5. (a) 6. (a)

Smart Thinking
Leads to Big Opportunities!



VERBAL ABILITY

GRAMMAR - MIXED PRACTICE

Good Grammar = Clear Communication = Confident You !



1) IMPORTANT GRAMMAR TOPICS (QUICK REVISION)

Topic	Key Points
Subject - Verb	Verb must agree with the subject in number (singular / plural).
Tenses	Shows time of an action. (Past, Present, Future, Perfect)
Articles	a / an (indefinite), the (definite) Use 'an' before vowel sound.
Prepositions	Shows relation between noun/pronoun and other words.
Conjunctions	Connect words, phrases or clauses. (and, but, or, because, although...)
Pronouns	Used in place of nouns. (he, she, it, they, mine, yours...)
Adverbs	Modify verbs, adjectives or other adverbs. (quickly, very, always...)
Active - Passive	Active: Subject does action Passive: Subject receives action
Direct - Indirect	Direct speech → exact words Indirect speech → reported words
Modifiers	Adjectives & Adverbs give more information (Use in right place)

2) COMMON RULES & REMINDERS

- ✓ Subject & verb should agree.
- ✓ Use 'a' before consonant sound, 'an' before vowel sound.
- ✓ Use 'the' for specific things.
- ✓ Tense should be consistent in a sentence.
- ✓ Adverb usually comes after the verb.
- ✓ Use 'who' for people, 'which' for things.
- ✓ Avoid double negatives.
- ✓ Keep the modifier close to the word it describes.



3) EXAMPLES

Topic	Wrong ✗	Correct ✓	Explanation
Subject-Verb	He go to school.	He goes to school.	'He' is singular → use 'goes'
Articles	I saw a elephant.	I saw an elephant.	'Elephant' starts with vowel sound
Tenses	She go to market yesterday.	She went to market yesterday.	Past tense of 'go' is 'went'
Prepositions	He is good in cricket.	He is good at cricket.	We are good at a skill
Pronouns	This book is my.	This book is mine.	Use 'mine' in place of 'my' here
Adverbs	He speak soft.	He speaks softly.	'Softly' modifies the verb 'speaks'

4) QUICK TRICKS

- ★ For Subject-Verb: Find the real subject first.
- ★ For Tenses: Look for time words (yesterday, now, will, etc.).
- ★ For Articles: Check if the noun is specific (use 'the').
- ★ For Prepositions: Think - 'relation' between words.
- ★ For Pronouns: Replace with noun and check.
- ★ For Adverbs: Ask - 'How? When? Where? How much?'
- ★ For Active-Passive: If object comes first → Passive.



TCS TIP

Grammar in TCS NQT is not about hard rules.
It's about understanding & applying smartly.
Practice daily - Small steps, Big results !



5) PRACTICE QUESTIONS

- Choose the correct option.
He ____ to school every day.
a) go b) goes c) going d) gone []
- Choose the correct option.
I have ____ idea about the plan.
a) a b) an c) the d) - []
- Choose the correct option.
They ____ playing cricket when it started raining.
a) was b) were c) is d) are []
- Choose the correct option.
She is taller ____ her sister.
a) than b) from c) to d) with []
- Choose the correct option.
I will call you ____ I reach home.
a) when b) while c) though d) because []
- Choose the correct option.
This is ____ best movie I have ever seen.
a) a b) an c) the d) - []
- Choose the correct option.
He did not go to the party ____ he was tired.
a) and b) but c) or d) so []
- Choose the correct option.
Neither of the boys ____ ready for the test.
a) is b) are c) was d) were []
- Choose the correct option.
She sings ____.
a) beautiful b) beautifully c) beauty d) more beautiful []
- Choose the correct option.
The project ____ by the team last week.
a) is completed b) are completed
c) was completed d) were completed []
- Choose the correct option.
He asked me ____ I was coming.
a) that b) if c) whether d) what []
- Choose the correct option.
There are many books on the table, ____?
a) aren't there b) is there c) do there d) doesn't there []
- Choose the correct option.
This is the house ____ I grew up.
a) where b) which c) that d) whom []
- Choose the correct option.
He has worked here ____ five years.
a) since b) for c) from d) by []
- Choose the correct option.
Please be quiet, ____ you will disturb others.
a) and b) or c) but d) so []

ANSWER KEY:

- 1-b 2-b 3-b 4-a 5-a 6-c 7-b 8-b
9-b 10-c 11-b 12-a 13-a 14-b 15-c



Strong Grammar → Strong Communication → Strong Opportunities ! Keep Practicing !



VOCABULARY

VOCABULARY REVISION

Better Vocabulary → Better Expression → Better Score !



1) WORDS TO LEARN (HIGH FREQUENCY)

Word	Meaning	Example Sentence
Abundant	Present in large quantities	The country has abundant natural resources.
Analyze	Examine in detail	We need to analyze the data carefully.
Beneficial	Producing good results	Regular exercise is beneficial for health.
Comprehensive	Complete; including everything	The report provides a comprehensive analysis.
Diligent	Hardworking; careful	She is a diligent student who always studies well.
Evident	Clearly seen or understood	His improvement is evident in every test.
Fragile	Easily broken; delicate	Handle the glassware with fragile care.
Inevitable	Certain to happen; unavoidable	Change is inevitable in life.

2) SYNONYMS (Words with Similar Meaning)

- Abundant - Ample, Plentiful, Copious
- Analyze - Examine, Study, Inspect
- Beneficial - Advantageous, Helpful, Good
- Diligent - Hardworking, Industrious, Assiduous
- Evident - Clear, Obvious, Apparent
- Fragile - Delicate, Brittle, Frail
- Inevitable - Unavoidable, Certain, Inescapable

3) ANTONYMS (Words with Opposite Meaning)

- Abundant - Scarce, Limited, Insufficient
- Beneficial - Harmful, Detrimental, Damaging
- Diligent - Lazy, Indolent, Idle
- Evident - Hidden, Unclear, Obscure
- Fragile - Strong, Sturdy, Durable
- Inevitable - Avoidable, Preventable, Uncertain

4) ANALOGIES (Find the Relation)

- Book : Reading :: Pen : _____
- Doctor : Hospital :: Teacher : _____
- Bird : Nest :: Honeybee : _____
- Keyboard : Typing :: Paintbrush : _____
- Ocean : Water :: Desert : _____

★ VOCABULARY TIP

- Learn 5 new words daily.
- Revise them after 1 day, 1 week and 1 month.
- Use the words in your own sentences.
- Read newspapers, articles and good books.
- Consistency is the key to a strong vocabulary!



5) PRACTICE QUESTIONS (CHOOSE THE CORRECT OPTION)

- The region is rich in _____ natural resources.
a) scarce b) abundant c) rare d) few []
- We must _____ the problem before finding a solution.
a) ignore b) analyze c) forget d) avoid []
- This new policy is _____ to the overall development of the country.
a) harmful b) beneficial c) useless d) irrelevant []
- The company prepared a _____ report covering all departments.
a) short b) partial c) comprehensive d) limited []
- He is a _____ worker who never avoids hard work.
a) diligent b) careless c) lazy d) idle []
- The answer is so _____ that even a child can understand it.
a) hidden b) evident c) complex d) doubtful []
- The toy is made of _____ material and may break easily.
a) fragile b) strong c) solid d) heavy []
- Failure is not the end; success is _____ for those who keep trying.
a) possible b) inevitable c) uncertain d) impossible []
- Please handle these documents with _____.
a) care b) anger c) hurry d) noise []
- The _____ increase in fuel prices is worrying.
a) temporary b) sudden c) gradual d) stable []
- Her _____ attitude helped the team complete the project.
a) negative b) positive c) lazy d) rude []
- He gave a _____ explanation with all necessary details.
a) brief b) complete c) unclear d) inaccurate []
- Science and technology have made _____ progress in recent years.
a) slow b) minor c) significant d) small []
- His behavior in the meeting was _____ and did not impress anyone.
a) polite b) arrogant c) humble d) respectful []
- The decision was _____ because all facts were considered.
a) random b) unfair c) sensible d) careless []

ANSWER KEY: 1-b, 2-b, 3-b, 4-c, 5-a, 6-b, 7-a, 8-a, 9-a, 10-b, 11-b, 12-b, 13-c, 14-b, 15-c



WORD POWER BOOSTER (USE THESE IN SENTENCES)

- ★ Motivate - Give someone a reason to act.
- ★ Persist - Continue steadfastly despite difficulties.
- ★ Sincere - Honest and genuine.
- ★ Versatile - Able to adapt or be used in many ways.
- ★ Efficient - Able to work well with minimum waste.



DAILY VOCAB CHECKLIST

- | | |
|--|--|
| ☐ Learned 5 new words | ☐ Learned antonyms |
| ☐ Learned meanings | ☐ Used in sentences |
| ☐ Learned synonyms | ☐ Revised old words |



A STRONG VOCABULARY BUILDS CONFIDENCE AND COMMUNICATION SKILLS.

KEEP LEARNING, KEEP GROWING, KEEP WINNING !



TECHNICAL

SQL & DBMS MCQs

Practice Today → Build Skills → Crack Tomorrow !



1) QUICK THEORY NOTES

- SQL (Structured Query Language) is used to communicate with databases.
- DBMS (Database Management System) helps in storing, managing and retrieving data efficiently.

Common SQL Commands:

SELECT	- Retrieves data from table
WHERE	- Filters records
ORDER BY	- Sorts the result
GROUP BY	- Groups rows
HAVING	- Filters groups
INSERT INTO	- Inserts new records
UPDATE	- Updates existing records
DELETE	- Deletes records
JOIN	- Combines rows from two or more tables

Keys:

Primary Key - Uniquely identifies each record.

Foreign Key - Connects one table to another.

- Normalization reduces redundancy and improves data integrity.
- ACID Properties ensure reliable transactions. Atomicity, Consistency, Isolation, Durability

3) DBMS MCQs

- Which key uniquely identifies each record in a table?
a) Candidate Key b) Primary Key
c) Foreign Key d) Secondary Key
- Which key is used to link one table to another?
a) Primary Key b) Composite Key
c) Foreign Key d) Unique Key
- Which normal form eliminates partial dependency?
a) 1NF b) 2NF c) 3NF d) BCNF
- Which normal form eliminates transitive dependency?
a) 1NF b) 2NF c) 3NF d) 4NF
- Which property of ACID ensures that a transaction is completed successfully or not at all?
a) Atomicity b) Consistency
c) Isolation d) Durability
- Which property ensures that all data written by a transaction is permanent?
a) Atomicity b) Consistency
c) Durability d) Isolation
- Which of the following is NOT a valid state in a transaction?
a) Active b) Failed c) Committed d) Pending
- Which of the following is an example of a DDL command?
a) INSERT b) SELECT c) CREATE d) UPDATE

2) SQL MCQs

- Which SQL command is used to retrieve data from a table?
a) GET b) FETCH c) SELECT d) RETRIEVE
- Which clause is used to filter records in a query?
a) FILTER b) WHERE c) HAVING d) CONDITION
- Which SQL command is used to sort the result in ascending order?
a) SORT BY b) ORDER BY c) ARRANGE BY d) ASC BY
- Which of the following is used to group rows that have the same values?
a) GROUP BY b) CLUSTER BY c) PARTITION BY d) SET BY
- Which clause is used to filter groups after applying GROUP BY?
a) WHERE b) HAVING c) GROUP BY d) ORDER BY
- Which SQL command is used to insert new records into a table?
a) INSERT b) ADD c) INSERT INTO d) PUT
- Which command is used to modify existing records in a table?
a) CHANGE b) MODIFY c) UPDATE d) ALTER
- Which command is used to remove records from a table?
a) REMOVE b) ERASE c) DELETE d) DROP
- Which of the following is used to combine rows from two tables based on a related column?
a) UNION b) JOIN c) MERGE d) CONNECT
- Which type of JOIN returns all rows from both tables, even if there is no match?
a) INNER JOIN b) LEFT JOIN c) RIGHT JOIN d) FULL OUTER JOIN

4) SQL QUERY OUTPUT MCQs

- Consider table EMP (ID, Name, Dept, Salary). What will be the output of the query?
SELECT Name, Salary FROM EMP WHERE Salary > 50000;
a) Names of all employees b) Names and Salary > 50000
c) All records d) Dept and Salary > 50000
- Consider table EMP (ID, Name, Dept, Salary). What will be the output of the query?
SELECT Dept, COUNT(*) FROM EMP GROUP BY Dept;
a) Total salary by Dept b) Number of employees in each Dept
c) Dept names only d) Dept with highest salary
- Consider table EMP (ID, Name, Dept, Salary). What will be the output of the query?
SELECT Name FROM EMP ORDER BY Salary DESC LIMIT 3;
a) Top 3 highest paid employees b) Top 3 lowest paid employees
c) All employee names d) Top 3 by Dept

ANSWER KEY

SQL MCQs (2) : 1-c 2-b 3-b 4-a 5-b 6-c 7-c 8-d 9-b 10-d
DBMS MCQs (3) : 1-b 2-c 3-b 4-c 5-a 6-c 7-d 8-c
SQL Output (4) : 1-b 2-b 3-a



TIPS TO REMEMBER

- Read the question carefully.
- Identify the correct SQL clause/command.
- For output based questions, write the query mentally and try to understand the result.
- Practice more MCQs daily for speed & accuracy.



CONSISTENT PRACTICE TODAY, SUCCESS TOMORROW !



1. QUANTITATIVE APTITUDE

SIMPLIFICATION & APPROXIMATION

Simplify Smartly, Approximate Accurately !



1) QUICK THEORY NOTES

- Simplification means reducing expressions to the simplest form.
- Approximation helps to find the near value of a number for faster calculation.
- Use BODMAS rule in simplification.
- In approximation, keep first 1 or 2 digits and consider the rest.

2) IMPORTANT FORMULAS

Topic	Formula / Rule
Squares	$(a \pm b)^2 = a^2 \pm 2ab + b^2$
Cubes	$(a \pm b)^3 = a^3 \pm 3a^2b \pm 3ab^2 + b^3$
Difference of Squares	$a^2 - b^2 = (a + b)(a - b)$
Sum of Cubes	$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
Difference of Cubes	$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
Product Rule	$(a + b)(a - b) = a^2 - b^2$
Division by 11	$(10a + b) \div 11 = a + \frac{b-a}{11}$
Approximation Rule	If next digit < 5, drop it. If next digit ≥ 5 , add 1 to previous digit and drop the rest.

3) APPROXIMATION SHORTCUTS

- ★ For $\sqrt{n} \rightarrow$ Find perfect square near n.
Eg: $\sqrt{50} \approx \sqrt{49} = 7$
- ★ For % $\rightarrow$ Convert % to fraction.
Eg: 25% of 80 = $\frac{25}{100} \times 80 = 20$
- ★ For large numbers $\rightarrow$ Keep first 1 or 2 digits.
Eg: 39876 $\approx$ 40000
- ★ For products $\rightarrow$ Approximate each number and multiply.
Eg: $98 \times 102 \approx 100 \times 100 = 10000$
- ★ For division $\rightarrow$ Approximate divisor & dividend.
Eg: $498 \div 52 \approx 500 \div 50 = 10$

4) BODMAS RULE

B	Brackets () { } []
O	Orders (Powers / Roots)
D	Division (+)
M	Multiplication (x)
A	Addition (+)
S	Subtraction (-)

Always solve
BODMAS
from left
to right!



TIPS

- ✓ Practice daily to increase speed.
- ✓ Write steps neatly and avoid silly mistakes.
- ✓ Use approximation only when options are close to each other.
- ✓ Eliminate options to save time.



5) PRACTICE QUESTIONS

- Simplify:
 $18 \div 3 \times 2 + 5 - 4$
a) 9 b) 13 c) 11 d) 7
- Simplify:
 $(15 + 5)^2 - (10 - 4)^2$
a) 336 b) 300 c) 196 d) 256
- Simplify:
 $(2^3 \times 3^2) + (4 \times 3^3)$
a) 108 b) 180 c) 216 d) 144
- Approximate $\sqrt{72}$ to the nearest integer.
a) 8 b) 9 c) 10 d) 7
- Approximate 5987 to 1 significant figure.
a) 6000 b) 5900 c) 5000 d) 7000
- Find the approximate value of 25% of 396.
a) 90 b) 95 c) 100 d) 105
- Simplify:
 $[(18 - 6) \times (2 + 3)] \div 6 + 7$
a) 17 b) 12 c) 15 d) 11
- Approximate 97×103 .
a) 9900 b) 10000 c) 10100 d) 9800
- Simplify:
 $(125)^{1/3} + (64)^{1/2} - (27)^{1/3}$
a) 10 b) 7 c) 8 d) 9
- Approximate $4996 \div 52$.
a) 95 b) 100 c) 105 d) 90
- Simplify:
 $(7^2 + 3^2) \div (7 - 3)$
a) 14 b) 18 c) 26 d) 32
- Approximate $\sqrt{128}$.
a) 10 b) 11 c) 12 d) 13
- Simplify:
 $(999 + 1) \times (1001 - 1)$
a) 999000 b) 1000000 c) 998000 d) 1002000
- Find the approximate value of 12.5% of 768.
a) 96 b) 90 c) 100 d) 120
- Simplify:
 $[(3^2 + 4^2)^{1/2}]^2$
a) 49 b) 25 c) 41 d) 16

ANSWER KEY: (Page 1)

- 1-b 2-a 3-c 4-b 5-a 6-b 7-b 8-b
9-a 10-b 11-c 12-b 13-b 14-a 15-a



PRACTICE $\rightarrow$ IMPROVE $\rightarrow$ PERFORM $\rightarrow$ SUCCEED !



LOGICAL REASONING

NON-VERBAL REASONING

(Mirror Images, Cubes & Dice)

Think Clearly → Visualize Better → Solve Faster !

TCS NQT & PLACEMENT STUDY GUIDE



★ KEY CONCEPTS AT A GLANCE

- Mirror images: Left-right reversal.
- Cubes: Faces, edges, vertices, rotation.
- Dice: Opposite faces sum to 7.
- Look for patterns, symmetry & position changes.

🎯 QUICK TIPS

- ✓ For mirror images, imagine a vertical line mirror (left ↔ right).
- ✓ For cubes, focus on visible faces and rotations.
- ✓ For dice, opposite faces always add to 7.
- ✓ Eliminate options step by step.

💡 REMEMBER

- If a figure is flipped, the order is reversed.
- A cube has 6 faces, 12 edges, 8 vertices.
- Opposite faces on a dice: 1-6, 2-5, 3-4.
- Keep your eyes on symmetry and position changes!

A. MIRROR IMAGES

Choose the option that is the mirror image of the given figure (vertical mirror).

1. Which option is the mirror image of the given figure?

A E E

- A) E3A B) A3E C) E3A D) A3E

2. Which option is the mirror image of the given figure?

F7H

- A) H7F B) L7F C) F7H D) L7H

3. Which option is the mirror image of the given figure?

Q b 3

- A) 3bQ B) Qd3 C) 3qQ D) Qp3

4. Which option is the mirror image of the given figure?



- A) B) C) D)

5. Which option is the mirror image of the given figure?



- A) B) C) D)

6. Which option is the mirror image of the given figure?

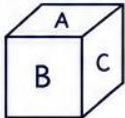


- A) B) C) D)

B. CUBES

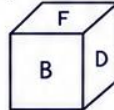
Choose the correct option based on the given cube/rotation.

7. Which face is opposite to the face showing 'B'?



- A) A B) C C) D D) E

8. Which two faces are adjacent to face 'F'?



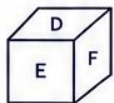
- A) B and D B) A and E
C) C and D D) A and C

9. Which cube will be formed after rotating the given cube 90° clockwise about the front face?



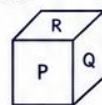
- A) B) C) D)

10. Which option shows the correct view of the opposite face?



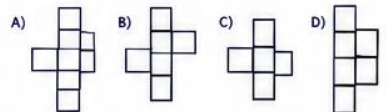
- A) E B) D C) F D) A

11. If the front face is 'P' and the right face is 'Q', which letter will be on the top face?



- A) P B) Q C) R D) S

12. Which figure cannot be made by folding a cube?



C. DICE

Choose the correct option based on the dice rules.

13. Which number is opposite to 1?

- A) 2 B) 3 C) 4 D) 6

14. If 5 is on the top face, which number will be on the bottom face?

- A) 1 B) 2 C) 3 D) 4

15. Which pair of numbers is opposite on a standard die?

- A) 1 and 2 B) 3 and 4
C) 2 and 5 D) 5 and 6

16. If 6 is on the top face, which number will be on the front face (given 3 is on the right face)?

- A) 1 B) 2 C) 4 D) 5

17. Which number is NOT opposite to 4?

- A) 1 B) 2 C) 5 D) 6

18. If 2 is on the top face, which number will be on the left face (given 5 is on the front face)?

- A) 1 B) 3 C) 4 D) 6



OBSERVE + ANALYZE + ELIMINATE = SOLVE !

You Can Do! ★

3. VERBAL ABILITY

READING COMPREHENSION (ADVANCED)

Read Actively, Think Deeply, Answer Accurately ! 

QUICK THEORY NOTES

- Reading Comprehension tests your ability to understand, analyze and interpret written passages.
- Focus on main idea, tone, inference, author's purpose and details.
- Read the passage once quickly to get the overall meaning.
- Then read questions and go back to the passage to find the answers.
- Eliminate wrong options and choose the most suitable answer.

STRATEGY TO SOLVE

- ✓ Read the passage carefully.
- ✓ Identify the main idea.
- ✓ Note down important points.
- ✓ Look for keywords in questions.
- ✓ Match options logically.
- ✓ Don't choose options out of the passage.
- ✓ Manage time - 3 to 4 minutes per passage.

TIPS TO IMPROVE

- ★ Read editorials, articles and short stories daily.
- ★ Practice RC passages from previous year papers.
- ★ Improve vocabulary to understand complex passages.
- ★ Summarize the passage in your own words after reading.

★ GOLDEN RULES ★

- ★ Read the passage actively, not just word by word.
- ★ Understand the context before attempting the questions.
- ★ Eliminate at least two wrong options.
- ★ Choose answers that are supported by the passage.



PASSAGE 1

The rapid advancement of technology has brought about significant changes in the way we live and work. While technology has made our lives more convenient, it has also created new challenges. One of the major concerns is the increasing dependence on gadgets, which reduces face-to-face communication and physical activity. Moreover, the misuse of technology, such as data breaches and cyber crimes, poses serious threats to individual privacy and security. Therefore, it is important to use technology wisely and strike a balance between the digital world and real life.

1. What is the main idea of the passage?
 - a) Technology makes life difficult
 - b) Technology has both benefits and drawbacks
 - c) Technology should be avoided
 - d) Technology is only useful for work
2. Which of the following is a negative impact of technology mentioned in the passage?
 - a) Better communication
 - b) Increased physical activity
 - c) Data breaches and cyber crimes
 - d) Easier access to information
3. What does the word "misuse" mean in the passage?
 - a) Use in the right way
 - b) Use for the wrong purpose
 - c) Use with permission
 - d) Use in a limited way
4. What is the author's tone in the passage?
 - a) Angry
 - b) Neutral
 - c) Cautious and balanced
 - d) Humorous
5. Which of the following can be the best title for the passage?
 - a) The Dangers of Technology
 - b) Technology: A Boon or Bane
 - c) Life Without Technology
 - d) The History of Gadgets

PASSAGE 2

Success does not come overnight. It is the result of hard work, dedication and perseverance. Many people give up when they face failures, but those who keep trying learn from their mistakes and move forward. Failure is not the opposite of success; it is a part of success. Every successful person has a story of struggles behind them. Therefore, instead of fearing failure, we should embrace it and use it as a stepping stone to achieve our goals.

1. What is the main idea of the passage?
 - a) Success comes quickly
 - b) Failure is important for success
 - c) Struggles are unnecessary
 - d) Only smart people succeed
2. What does the author suggest we should do with failures?
 - a) Ignore them
 - b) Fear them
 - c) Learn from them
 - d) Avoid them
3. Which word is closest in meaning to "perseverance"?
 - a) Laziness
 - b) Patience
 - c) Giving up
 - d) Hard work
4. Which of the following is TRUE according to the passage?
 - a) Successful people never face failures
 - b) Failure helps us learn and grow
 - c) Struggles are a waste of time
 - d) Success depends only on luck
5. Which of the following can be the best title for the passage?
 - a) The Power of Success
 - b) Learning from Failures
 - c) Avoiding Mistakes
 - d) The Story of Successful People

ANSWER KEY (Page 3)

Passage 1 : 1-b 2-c 3-b 4-c 5-b
Passage 2 : 1-b 2-c 3-b 4-b 5-b



4. VOCABULARY

MIXED VOCABULARY TEST

Stronger Vocabulary, Brighter Future !



QUICK THEORY NOTES

- A good vocabulary helps in Reading, Writing, Speaking and Competitive Exams.
- Learn words in context not in isolation.
- Practice synonyms, antonyms and usage.
- Revise daily - Consistency is the key!

IMPORTANT VOCABULARY LIST

Word	Meaning	Synonym	Antonym	Word	Meaning	Synonym	Antonym
Abundant	Plentiful	Ample	Scarce	Ubiquitous	Present everywhere	Universal	Rare
Eloquent	Fluent	Articulate	Incoherent	Benevolent	Kind	Generous	Cruel
Meticulous	Very careful	Precise	Careless	Commence	Begin	Start	Finish
Obsolete	Outdated	Outmoded	Modern	Transient	Temporary	Short-lived	Permanent
Diligent	Hardworking	Industrious	Lazy	Exemplary	Worthy of imitation	Outstanding	Blameworthy
Prudent	Wise	Cautious	Imprudent	Fortuitous	Happening by chance	Accidental	Deliberate
Resilient	Strong, able to recover	Tough	Fragile	Loquacious	Talkative	Verbose	Taciturn



A. SYNONYMS

- The manager was very meticulous in checking every detail of the report.
a) Careless b) Careful
c) Hasty d) Rough
- Her eloquent speech left the audience inspired.
a) Incoherent b) Fluent
c) Hesitant d) Stammering
- The garden was abundant with beautiful flowers.
a) Full b) Empty
c) Scarce d) Barren
- He is a prudent person who thinks before taking any decision.
a) Wise b) Foolish
c) Rash d) Careless
- The exemplary performance of the team won them the award.
a) Blameworthy b) Outstanding
c) Poor d) Average
- The old machine is now obsolete and cannot be repaired.
a) Useful b) Outdated
c) Advanced d) New
- We should be resilient in the face of challenges.
a) Weak b) Strong
c) Sensitive d) Breakable
- The event was fortuitous as it happened by pure chance.
a) Planned b) Accidental
c) Expected d) Intended

B. ANTONYMS

- The medicine had a benevolent effect on the patient.
a) Kind b) Helpful
c) Cruel d) Generous
- His comments were very transient and not meant to last.
a) Temporary b) Short-lived
c) Permanent d) Fleeting
- She is usually loquacious in social gatherings.
a) Quiet b) Talkative
c) Silent d) Noisy
- He decided to commence the project next week.
a) Start b) Continue
c) Begin d) Delay
- The route was ubiquitous with traffic and chaos.
a) Everywhere b) Common
c) Rare d) Unusual
- His explanation was lucid and easy to understand.
a) Clear b) Confusing
c) Obvious d) Simple
- The economy is showing signs of revival after the crisis.
a) Growth b) Decline
c) Collapse d) Stagnation
- She tried to mollify her friend after the misunderstanding.
a) Annoy b) Calm
c) Anger d) Irritate

C. WORD USAGE (Choose the correct word)

- The scientist made a _____ discovery that changed many lives.
a) trivial b) remarkable
c) ordinary d) foolish
- He showed _____ in solving the difficult mathematical problem.
a) patience b) anger
c) ignorance d) laziness
- The politician's _____ promises did not impress the public.
a) vague b) clear
c) honest d) true
- The company faced heavy _____ due to poor management.
a) profit b) loss
c) gain d) benefit
- She wrote a _____ letter thanking everyone for their support.
a) careless b) heartfelt
c) rude d) selfish
- It is important to be _____ while crossing the road.
a) careless b) cautious
c) fearless d) reckless
- The athlete's _____ helped him win the gold medal.
a) laziness b) dedication
c) confusion d) fear
- The author used _____ language to describe the beauty of nature.
a) simple b) complex
c) poetic d) angry

TIPS TO IMPROVE VOCABULARY

- ✓ Read newspapers and editorials daily.
- ✓ Note down new words and their meanings.
- ✓ Use new words in sentences.
- ✓ Revise synonyms and antonyms regularly.
- ✓ Take vocabulary tests frequently.



ANSWER KEY (Page 4)

- A. 1-b 2-b 3-a 4-a 5-b 7-b 8-b
B. 1-c 2-c 3-a 4-c 5-b 7-b 8-b
C. 1-b 2-a 3-a 4-b 5-b 7-b 8-c



5. TECHNICAL

OS & COMPUTER NETWORKS MCQS

Revise. Recall. Score !



A. OPERATING SYSTEMS

★ QUICK THEORY NOTES

- OS acts as an interface between user and hardware.
- It manages processes, memory, files, I/O devices and security.
- Goals: Convenience, Efficiency, Reliability.
- Types: Batch, Time-sharing, Real-time, Distributed OS.
- Common terms: Process, Thread, Deadlock, Scheduling, Paging, Segmentation.



IMPORTANT OS CONCEPTS

Concept	What it means
Process	A program in execution
Thread	Smallest unit of CPU execution
Context Switch	Switching CPU from one process to another
Deadlock	A situation where processes are waiting forever
Paging	Memory is divided into fixed size pages
Segmentation	Memory is divided into variable size segments

- Which of the following is not a type of Operating System?
 - Batch OS
 - Real-time OS
 - Compiler OS
 - Distributed OS
- Which scheduling algorithm is non-preemptive?
 - Round Robin
 - SJF
 - Priority Scheduling
 - SRTF
- Deadlock can arise if the following conditions hold simultaneously, EXCEPT:
 - Mutual Exclusion
 - Hold and Wait
 - No Preemption
 - Infinite Memory
- In virtual memory, a Page Fault occurs when:
 - Page is in RAM
 - Page is in Cache
 - Page is not in RAM
 - Page is in Disk
- Which of the following is used to avoid thrashing?
 - Page Replacement
 - Increasing CPU speed
 - Increasing degree of multiprogramming
 - Decreasing degree of multiprogramming
- Which data structure is used by the OS to keep track of processes?
 - File
 - PCB
 - Stack
 - Register
- Which of the following is true about multithreading?
 - Each thread has its own memory space
 - Threads share code, data and files
 - Creating threads is slower than processes
 - Threads do not improve performance
- Which page replacement algorithm may suffer from Belady's Anomaly?
 - FIFO
 - LRU
 - Optimal
 - MFU
- The fork() system call is used to:
 - Create a new file
 - Create a new process
 - Create a new thread
 - Terminate a process
- Which of the following is NOT a valid state in process lifecycle?
 - New
 - Ready
 - Running
 - Terminated
 - Waiting
 - Suspended

B. COMPUTER NETWORKS

★ QUICK THEORY NOTES

- A computer network is a collection of interconnected devices.
- Types: LAN, MAN, WAN, PAN.
- OSI Model has 7 layers.
- TCP is connection-oriented; IP is connectionless.
- Common protocols: HTTP, FTP, DNS, SMTP, TCP, UDP.
- Key terms: IP Address, Subnet Mask, Routing, Bandwidth, Latency.



NETWORKING KEY FACTS

Topic	Key Point
IP Address (IPv4)	32-bit address (e.g., 192.168.1.1)
Subnet Mask	Determines network and host part
TCP	Reliable, connection-oriented
UDP	Faster, connectionless
DNS	Resolves domain names to IP
Bandwidth	Maximum data transfer rate
Latency	Time taken for data to travel

- Which layer of OSI model is responsible for reliable data transfer?
 - Network
 - Transport
 - Session
 - Data Link
- Which protocol is used to send emails?
 - HTTP
 - FTP
 - SMTP
 - DNS
- Which device is used to connect devices in the same LAN?
 - Router
 - Switch
 - Hub
 - Modem
- Which IP address class is used for large networks?
 - Class A
 - Class B
 - Class C
 - Class D
- Which protocol is used for remote file transfer?
 - FTP
 - HTTP
 - SMTP
 - DNS
- Which of the following is NOT a TCP flag?
 - SYN
 - ACK
 - FIN
 - PUSH
- What does DNS stand for?
 - Domain Name System
 - Data Network Service
 - Dynamic Network Server
 - Direct Name Service
- Which layer of OSI model deals with physical media and transmission of bits?
 - Physical
 - Data Link
 - Network
 - Transport
- Which protocol is connectionless?
 - TCP
 - UDP
 - HTTP
 - FTP
- Which command is used to check connectivity to a host?
 - ipconfig
 - ping
 - tracert
 - netstat



TECHNICAL REVISION TIPS

- Revise key OS concepts and process states.
- Practice networking protocols and OSI layers.
- Solve MCQs daily to improve accuracy.
- Analyze mistakes and revise weak topics.



ANSWER KEY (Page 5)

OS: 1-c 2-b 3-d 4-c 5-d 6-b 7-b 8-a 9-b 10-d
 CN: 1-b 2-c 3-b 4-a 5-a 6-d 7-a 8-a 9-b 10-b



CONSISTENT REVISION TODAY, CONFIDENT SELECTION TOMORROW !



1. QUANTITATIVE APTITUDE

MIXED QUANT PRACTICE

Practice Today, Perform Tomorrow !



QUICK TIPS

- ✓ Read the question carefully.
- ✓ Identify the concept and choose the best approach.
- ✓ Use approximation to save time in lengthy calculations.
- ✓ Practice regularly to improve speed and accuracy.

1. What is the value of $(25)^2 - (15)^2 + (10)^2$?

- a) 450 b) 500
c) 550 d) 600

2. A number is increased by 20% and then the new number is decreased by 20%. What is the net change percent in the number?

- a) Increase 4% b) Decrease 4%
c) No change d) Increase 8%

3. If the ratio of A's age to B's age is 5 : 7 and the ratio of B's age to C's age is 7 : 9, then the ratio of A's age to C's age is:

- a) 5 : 7 b) 5 : 9
c) 7 : 9 d) 5 : 11

4. A sum of ₹12,000 is invested at 8% per annum simple interest. What will be the interest after 2.5 years?

- a) ₹2,000 b) ₹2,400
c) ₹2,800 d) ₹3,000

5. The marked price of an article is ₹1,500. If a discount of 20% is given and a profit of 25% is earned, what is the cost price of the article?

- a) ₹900 b) ₹1,000
c) ₹1,050 d) ₹1,080

6. A train 150 m long crosses a pole in 15 seconds. What is the speed of the train in km/h?

- a) 30 b) 36
c) 40 d) 45

7. Find the next number in the series: 3, 8, 15, 24, 35, ?

- a) 44 b) 48
c) 50 d) 54

8. A and B can complete a work in 12 days and B and C can complete the same work in 15 days. C and A can complete it in 20 days. All three working together can complete the work in:

- a) 5 days b) 6 days
c) 7 days d) 8 days

9. If $x + \frac{1}{x} = 5$, then the value of $x^2 + \frac{1}{x^2}$ is:

- a) 21 b) 23
c) 24 d) 25

10. A shopkeeper allows two successive discounts of 10% and 5% on the marked price of an article. What is the effective discount percent?

- a) 14.5% b) 15%
c) 15.5% d) 16%

11. The average of 15 numbers is 42.

If 7 is added to each number, what will be the new average?

- a) 42 b) 47
c) 49 d) 50

12. A can do a work in 18 days, B in 24 days and C in 36 days. In how many days can A, B and C together complete the work?

- a) 6 days b) 7 days
c) 8 days d) 9 days

13. If $2x + 3y = 13$ and $3x - 2y = 4$, what is the value of $x + y$?

- a) 3 b) 4
c) 5 d) 6

14. The ratio of the present ages of father and son is 7 : 2. After 10 years, the ratio will be 5 : 2. What is the present age of the son?

- a) 10 years b) 12 years
c) 14 years d) 16 years

15. A number when divided by 7, 11 and 13 leaves remainder 1 in each case. What is the least such number?

- a) 1001 b) 1002
c) 1003 d) 1004

★ FORMULA REMINDER

- $SI = (P \times R \times T) / 100$
- $\text{Speed (km/h)} = (\text{Distance (m)} / \text{Time (sec)}) \times 3.6$
- $\text{Work done together} = 1 / (\text{Sum of individual days})$
- $\text{Successive Discount} = d1 + d2 - (d1 \times d2 / 100)$



ANSWER KEY (Page 1)



- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. b | 3. a | 4. b | 5. b |
| 6. c | 7. b | 8. a | 9. b | 10. c |
| 11. b | 12. b | 13. a | 14. a | 15. b |



CONSISTENT PRACTICE TODAY, SUCCESS IN THE TEST TOMORROW !



2. LOGICAL REASONING

MIXED REASONING PRACTICE

Think Logically, Solve Smartly !



QUICK TIPS

- ✓ Read each question carefully.
- ✓ Look for patterns and relationships.
- ✓ Eliminate wrong options.
- ✓ Use diagrams to solve faster.
- ✓ Practice improves your speed and accuracy.

1. Find the next number in the series:
2, 6, 12, 20, 30, ?

- a) 40 b) 42
c) 44 d) 48

2. If 'A' means '+', 'B' means '-', 'C' means 'x' and 'D' means '÷', then what is the value of
24 A 6 C 2 B 3 D 1 = ?

- a) 18 b) 15
c) 21 d) 17

3. Arrange the following words in dictionary order:

1. Miracle 2. Minimum
3. Minority 4. Mislead 5. Missile

- a) 3, 2, 4, 5, 1 b) 3, 4, 2, 5, 1
c) 2, 3, 4, 5, 1 d) 4, 2, 3, 5, 1

4. In a certain code language, 'TABLE' is coded as '84931' and 'CHAIR' is coded as '73192'. How will 'BREAD' be coded?

- a) 58243 b) 58252
c) 59243 d) 59252

5. If in a certain code, 'PEN' is written as '427' and 'INK' is written as '318', then how will 'PINK' be written in that code?

- a) 4138 b) 4318
c) 4238 d) 4218

6. Four friends A, B, C and D are sitting around a circular table facing the centre. A is to the right of B. C is to the left of D. B is not adjacent to C. Who is sitting opposite to A?

- a) B b) C c) D d) Cannot be determined

7. Find the odd one out.

a)

2	3	5
4	5	9
6	7	13

b)

3	4	7
5	6	11
7	8	15

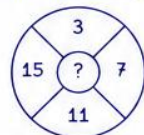
c)

2	4	6
4	6	8
6	8	10

d)

1	2	4
2	3	6
3	4	7

8. Find the missing number.

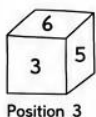
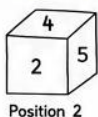
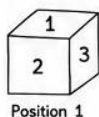


- a) 17
b) 18
c) 19
d) 20

9. Find the mirror image of the given figure.

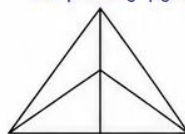


10. A dice is thrown three times and its three positions are shown below. Which number will be opposite to the face showing '2'?



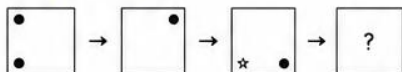
- a) 3
b) 4
c) 5
d) 6

11. How many triangles are there in the following figure?



- a) 8
b) 9
c) 10
d) 12

12. Choose the option figure that will complete the pattern.



- a) b)
c) d)

13. In a certain code language, 'EARTH' is coded as '18420' and 'WATER' is coded as '02184'. How will 'HEART' be coded?

- a) 20148
b) 20184
c) 19402

14. Statements:

Some pens are books.
All books are copies.
No copy is a bag.

Conclusions:

I. Some pens are copies.
II. No bag is a pen.

- a) Only I follows
b) Only II follows
c) Both I and II follow
d) Neither I nor II follows

REASONING STRATEGY

- Understand the type of question.
- Look for relationships and patterns.
- Eliminate and choose the best option.
- Manage time and review your answers.



ANSWER KEY (Page 2)

1. d 2. c 3. a 4. b 5. a 6. c 7. c
8. a 9. b 10. b 11. c 12. d 13. b 14. b



3. VERBAL ABILITY

MIXED GRAMMAR PRACTICE

Strong Grammar, Strong Score ! 

QUICK TIPS

- Know the rules, practice regularly.
- Read each sentence carefully.
- Identify the error or the best option.
- Eliminate wrong choices.
- Accuracy comes with consistent effort !

TOPICS COVERED

- ✓ Subject-Verb Agreement
- ✓ Tense
- ✓ Articles
- ✓ Prepositions
- ✓ Conjunctions
- ✓ Active-Passive Voice
- ✓ Direct-Indirect Speech
- ✓ Conditionals
- ✓ Modals
- ✓ Error Spotting



- The number of students in the class ____ very large.
a) is b) are
c) was d) were
- She has been living in this city ____ 2019.
a) since b) for
c) from d) by
- If I ____ you, I would take the job offer.
a) am b) was
c) were d) be
- He told me that he ____ to Delhi the next day.
a) goes b) is going
c) was going d) would go
- The book is interesting; ____, it is very informative.
a) moreover b) however
c) therefore d) instead
- I have ____ idea about this topic.
a) a b) an
c) the d) no article
- She is good ____ singing and dancing.
a) on b) in
c) at d) of
- The movie was so boring that I ____ asleep in the middle.
a) fell b) felt
c) fall d) have fallen
- They said, "We are ready to help you."
(Choose the correct indirect speech)
a) They said that they are ready to help me.
b) They said that we are ready to help me.
c) They said that they were ready to help me.
d) They said that we were ready to help me.
- The report ____ by the manager yesterday.
a) is written b) was written
c) wrote d) writing
- He is not only intelligent but also ____.
a) hardworking b) more hardworking
c) the most hardworking d) as hardworking
- Identify the error in the sentence.
Every student have submitted their assignments.
a) have b) their
c) Every d) submitted
- Choose the correct passive voice.
They will complete the project on time.
a) The project will be completed by them on time.
b) The project will be completing by them on time.
c) The project will be completed by them in time.
d) The project will be completed of them on time.
- If it ____ tomorrow, we will not go for the picnic.
a) rain b) rains
c) rained d) will rain

COMMON GRAMMAR RULES REMINDER

- ✓ Subject + Verb must agree in number.
- ✓ Use suitable articles: a, an, the.
- ✓ Use correct tense according to time.
- ✓ Preposition shows relationship.
- ✓ Conjunctions join words/sentences.
- ✓ Direct speech → exact words.
- ✓ Indirect speech → reported words.
- ✓ Active voice → doer does action.
- ✓ Passive voice → action is done.

PRACTICE MORE, SCORE HIGHER ! 

ANSWER KEY (Page 3)

1. b 2. a 3. c 4. c 5. b 6. b 7. b
8. a 9. c 10. b 11. a 12. a 13. a 14. a



GOOD GRAMMAR → CLEAR EXPRESSION → CONFIDENT YOU → TOP SCORE !



4. VOCABULARY

COMPLETE VOCABULARY REVISION

Learn Words, Build Power, Shine Forever !



QUICK TIPS

- ✓ Learn new words daily.
- ✓ Understand the meaning in context.
- ✓ Use synonyms & antonyms.
- ✓ Revise regularly.
- ✓ Practice makes perfect !

POWER WORDS TO REMEMBER

- | | | | | |
|---------------|--------------|--------------|----------------|-------------|
| • Abundant | • Eloquent | • Impeccable | • Blandishment | • Taciturn |
| • Meticulous | • Resilient | • Loquacious | • Exacerbate | • Incessant |
| • Ubiquitous | • Obsolete | • Transient | • Fortuitous | • Sagacious |
| • Indifferent | • Alleviate | • Diligent | • Persevere | • Conundrum |
| • Prudent | • Benevolent | • Scrutinize | • Ameliorate | • Placate |

A. SYNONYMS - Choose the word similar in meaning.

- The doctor was very _____ in treating the patient.
a) careless b) negligent
c) attentive d) rude
- Her speech was so _____ that everyone listened.
a) unclear b) incoherent
c) eloquent d) confused
- The weather in this place is _____.
a) pleasant b) harsh
c) delightful d) easy
- He showed great _____ in solving the problem.
a) ignorance b) carelessness
c) ingenuity d) dullness
- The idea was _____ and had never been tried before.
a) ordinary b) innovative
c) dull d) common
- Please be _____ while driving.
a) negligent b) cautious
c) reckless d) lazy
- The old building is now _____.
a) modern b) new
c) obsolete d) stylish
- She is a very _____ worker.
a) lazy b) irritable
c) diligent d) careless
- He tried to _____ the pain of his friend.
a) increase b) alleviate
c) ignore d) cause
- The situation was _____ but he handled it well.
a) difficult b) easy
c) simple d) trivial

B. ANTONYMS - Choose the opposite word.

- The story has a happy _____.
a) beginning b) middle
c) end d) conclusion
- The boy was very _____ to everyone.
a) kind b) cruel
c) pleasant d) polite
- This is a permanent solution, not a _____ one.
a) temporary b) final
c) long d) lasting
- He is a very _____ person; he speaks a lot.
a) talkative b) loquacious
c) silent d) quiet
- The task was difficult but not _____.
a) easy b) simple
c) hard d) tough
- Accept the truth; don't be _____.
a) honest b) truthful
c) deceptive d) sincere
- Her answer was correct, not _____.
a) exact b) right
c) wrong d) proper
- The quality of the product is excellent, not _____.
a) poor b) good
c) best d) perfect
- Please be _____ in the exam.
a) careful b) careless
c) attentive d) focused
- The medicine will cure the disease, not _____ it.
a) heal b) worsen
c) treat d) remove

C. ONE WORD SUBSTITUTION - Replace the phrase.

- A person who speaks very little.
a) Loquacious b) Taciturn
c) Eloquent d) Talkative
- A written law made by a government.
a) Justice b) Legislation
c) Regulation d) Ordinance
- The act of helping someone in need.
a) Harm b) Charity
c) Cruelty d) Neglect
- A person who is very wise.
a) Foolish b) Sagacious
c) Diligent d) Indifferent
- Unable to be seen everywhere.
a) Invisible b) Ubiquitous
c) Visible d) Common
- A puzzle with a difficult answer.
a) Solution b) Conundrum
c) Answer d) Calculation
- To make someone calm or quiet.
a) Excite b) Placate
c) Annoy d) Provoke
- The quality of being exact and careful.
a) Careless b) Meticulous
c) Sloppy d) Negligent
- To make a bad situation worse.
a) Ameliorate b) Exacerbate
c) Alleviate d) Improve
- To continue in spite of difficulties.
a) Stop b) Persevere
c) Quit d) Fail

D. FILL IN THE BLANKS - Choose the most appropriate word.

- The company is looking for a _____ candidate for this position.
a) suitable b) useless c) odd d) strange
- He was _____ by his own success and forgot his friends.
a) elated b) humble c) arrogant d) polite
- The road was blocked due to the _____ rain.
a) incessant b) occasional c) light d) rare
- She tried to _____ the matter, but the situation got worse.
a) simplify b) complicate c) improve d) solve
- The judge will _____ the case before giving the verdict.
a) ignore b) listen c) examine d) forget
- He felt _____ about the bad news.
a) cheerful b) indifferent c) sad d) amused
- The teacher gave a _____ explanation with examples.
a) confusing b) vague c) clear d) weak
- We should _____ time and not waste it.
a) waste b) save c) spoil d) lose
- The medicine helped to _____ his pain.
a) increase b) reduce c) create d) cause
- His behavior was _____ and not acceptable.
a) decent b) rude c) kind d) friendly

E. ERROR SPOTTING - Find the error in the sentence.

- She don't (a) like (b) to waste (c) time. (d) No error
- He has (a) went (b) to college (c) yesterday. (d) No error
- Neither (a) of the boys (b) are (c) ready. (d) No error
- I have (a) seen (b) him since (c) long time. (d) No error
- They were (a) happy because (b) they won (c) the match. (d) No error
- Each of the students (a) have (b) a pen. (c) (d) No error
- The manager and the staff (a) is (b) working (c) hard. (d) No error
- She didn't (a) went (b) to the party (c) last night. (d) No error
- He is (a) one of the (b) best player (c) in the team. (d) No error
- By mistake (a) I have (b) deleted (c) the file. (d) No error

VOCABULARY BOOST TIPS

- ✓ Make a notebook of new words.
- ✓ Use new words in your daily talk.
- ✓ Read newspapers & articles.
- ✓ Revise words weekly.
- ✓ Consistency leads to mastery!



ANSWER KEY (Page 4)

- | | | | | | |
|----|-----|-----|-----|-----|------|
| A. | 1-c | 2-c | 3-b | 4-c | 5-b |
| | 6-b | 7-c | 8-c | 9-b | 10-a |
| B. | 1-d | 2-b | 3-a | 4-c | 5-d |
| | 6-c | 7-c | 8-a | 9-b | 10-b |
| C. | 1-b | 2-b | 3-b | 4-b | 5-b |
| | 6-b | 7-b | 8-b | 9-b | 10-b |
| D. | 1-a | 2-c | 3-a | 4-b | 5-c |
| | 6-c | 7-c | 8-b | 9-b | 10-b |
| E. | 1-a | 2-b | 3-c | 4-c | 5-a |
| | 6-b | 7-a | 8-b | 9-b | 10-a |



WORDS ARE POWER - BUILD THEM TODAY, ACE TOMORROW !



5. TECHNICAL

MIXED TECHNICAL MCQs

Concept Clear Today, Success Tomorrow ! 



Instructions: Read each question carefully and choose the most appropriate answer.



A. COMPUTER SCIENCE

- Which of the following is NOT a programming language?
 - Python
 - Java
 - C++
 - MySQL
- Which data structure uses LIFO principle?
 - Queue
 - Stack
 - Array
 - Linked List
- What will be the output of: $5 + 2 * 3$?
 - 21
 - 11
 - 13
 - 10
- Which of the following is a non-volatile memory?
 - RAM
 - Cache
 - ROM
 - Register
- Which of the following is used to find and fix bugs in a program?
 - Compiler
 - Linker
 - Debugger
 - Interpreter
- Which does CPU stand for?
 - Central Process Unit
 - Computer Processing Unit
 - Central Processing Unit
 - Control Processing Unit
- Which of the following is an example of Operating System?
 - MS Word
 - Windows 11
 - Chrome
 - Photoshop
- Which sorting algorithm has the best average case time complexity?
 - Bubble Sort
 - Merge Sort
 - Selection Sort
 - Insertion Sort
- What is the full form of SQL?
 - Structured Query Language
 - Simple Query Language
 - Sequential Query Language
 - Standard Query Language
- Which of the following is a characteristic of OOPs?
 - Compilation
 - Inheritance
 - Encryption
 - Assembly

B. INFORMATION TECHNOLOGY

- Which one of the following is an example of input device?
 - Monitor
 - Printer
 - Keyboard
 - Speaker
- Which protocol is used to transfer web pages?
 - FTP
 - HTTP
 - SMTP
 - POP3
- What does a URL stand for?
 - Uniform Resource Locator
 - Universal Resource Locator
 - Uniform Record Locator
 - Universal Record Link
- Which of the following is a Search Engine?
 - Microsoft Word
 - Google
 - WhatsApp
 - MS Excel
- Which file extension is used for a PowerPoint presentation?
 - .docx
 - .xlsx
 - .pptx
 - .txt
- Which of the following is a cloud storage service?
 - Dropbox
 - VLC Media Player
 - Notepad
 - Adobe Reader
- What is phishing?
 - Hacking into a system
 - Stealing information using fake emails/websites
 - Creating strong passwords
 - Encrypting data
- Which of the following is an example of software?
 - Monitor
 - Keyboard
 - Operating System
 - Printer
- What is the purpose of an antivirus program?
 - To speed up computer
 - To protect from malware and viruses
 - To edit photos
 - To browse websites
- Which of the following is a Social Media platform?
 - LinkedIn
 - Gmail
 - Google Drive
 - MS Paint



TECHNICAL QUICK NOTES

- ✓ Computer Science → Logic & Problem Solving
- ✓ Information Technology → Tools & Applications
- ✓ Concepts + Practice = Confidence
- ✓ Revise regularly for better retention
- ✓ Understand, Don't Memorize !



TECHNICAL TIPS

- ★ Read questions twice.
- ★ Eliminate wrong options.
- ★ Focus on keywords.
- ★ Practice past papers.
- ★ Stay calm & manage time.

ANSWER KEY (Page 5)



CS:	1-d	2-b	3-b	4-c	5-c
	6-c	7-b	8-b	9-a	10-b
IT:	1-c	2-b	3-a	4-b	5-c
	6-a	7-b	8-c	9-b	10-a



COMPLETE PRACTICE TODAY, CONFIDENT PERFORMANCE TOMORROW !



FINAL DAY - FINAL PUSH! Revise Smart, Score High! **100**

QUICK REVISION TIPS

- ✓ Learn formulas with examples.
- ✓ Practice previous year papers.
- ✓ Focus on speed + accuracy.
- ✓ Stay calm & confident!



IMPORTANT SYMBOLS

$$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$$

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

$$a^m \times a^n = a^{m+n}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$(a+b)^2 = a^2 + b^2 + 2ab$$

1 NUMBER SYSTEM

- Factors of $a \times b$ = Factors of a + Factors of b - 2
- HCF $\times$ LCM = Product of two numbers
- If $a = bq + r$, then
 $a = bq + r$, $0 \leq r < b$
- Sum of first n natural numbers
$$= \frac{n(n+1)}{2}$$
- Sum of squares of first n natural numbers
$$= \frac{n(n+1)(2n+1)}{6}$$
- Sum of cubes of first n natural numbers
$$= \left[\frac{n(n+1)}{2}\right]^2$$

2 AVERAGE

- Average = $\frac{\text{Sum of observations}}{\text{Number of observations}}$
- New Average = $\frac{\text{Old Average} \times \text{Old No.} + \text{New No.}}{\text{Old No.} + 1}$
- If each observation is increased/decreased by k , new average = Old average $\pm k$

3 PERCENTAGE

- % = $\left(\frac{\text{Part}}{\text{Whole}}\right) \times 100$
- Increase % = $\left(\frac{\text{Increase}}{\text{Original}}\right) \times 100$
- Decrease % = $\left(\frac{\text{Decrease}}{\text{Original}}\right) \times 100$
- Successive % change:
New Value = Old Value $\times \left(1 + \frac{P}{100}\right) \times \left(1 + \frac{Q}{100}\right) \dots$

4 RATIO & PROPORTION

- If $a:b = c:d \Rightarrow ad = bc$
- If $a:b = m:n$, then
$$\frac{a}{b} = \frac{m}{n} = \frac{a+b}{m+n}$$
- Third proportional to a and $b = \frac{b^2}{a}$
- Fourth proportional to a, b and $c = \frac{bc}{a}$

5 PROFIT & LOSS

- Profit = SP - CP
Loss = CP - SP
- Profit % = $\left(\frac{\text{Profit}}{\text{CP}}\right) \times 100$
Loss % = $\left(\frac{\text{Loss}}{\text{CP}}\right) \times 100$
- SP = CP $\times \frac{(100 + \text{Profit}\%)}{100}$
SP = CP $\times \frac{(100 - \text{Loss}\%)}{100}$
- Discount = MP - SP
Discount % = $\left(\frac{\text{Discount}}{\text{MP}}\right) \times 100$

6 SIMPLE INTEREST

- SI = $\frac{P \times R \times T}{100}$
Where,
P = Principle, R = Rate %, T = Time (years)
- Amount = P + SI

7 COMPOUND INTEREST

- Amount = $P \left(1 + \frac{R}{100}\right)^T$
- CI = Amount - P
- Half yearly: $A = P \left(1 + \frac{R}{200}\right)^{2T}$
Quarterly: $A = P \left(1 + \frac{R}{400}\right)^{4T}$

8 TIME, SPEED & DISTANCE

- Speed = $\frac{\text{Distance}}{\text{Time}}$
- Time = $\frac{\text{Distance}}{\text{Speed}}$
- Distance = Speed $\times$ Time
- If speed is increased by $p\%$,
New time = Old time $\times \frac{100}{100+p}$
- If speed is decreased by $p\%$,
New time = Old time $\times \frac{100}{100-p}$

9 TIME & WORK

- Work = Time $\times$ Rate
- Time = $\frac{\text{Work}}{\text{Rate}}$
- If A alone takes x days, B alone takes y days,
 $A + B$ together = $\frac{xy}{x+y}$ days
- If A's 1 day work = $\frac{1}{x}$, B's 1 day work = $\frac{1}{y}$,
 $(A + B)$'s 1 day work = $\frac{1}{x} + \frac{1}{y}$

10 BOATS & STREAMS

- Downstream Speed = Speed in still water + Speed of stream
- Upstream Speed = Speed in still water - Speed of stream
- Speed in still water = $\frac{\text{Down} + \text{Up}}{2}$
- Speed of stream = $\frac{\text{Down} - \text{Up}}{2}$

11 PIPE & CISTERN

- Work = Part of tank filled
- If pipe A can fill tank in x hours,
A's 1 hour work = $\frac{1}{x}$
- If A and B together can fill in T hours,
 $(A + B)$'s 1 hour work = $\frac{1}{T}$

12 ALGEBRA

- $(a+b)^2 = a^2 + b^2 + 2ab$
- $(a-b)^2 = a^2 + b^2 - 2ab$
- $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$
- $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
- $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

13 GEOMETRY

2D FIGURES

- Square
Area = a^2
Perimeter = $4a$
- Rectangle
Area = $l \times b$
Perimeter = $2(l+b)$
- Circle
Area = πr^2
Circumference = $2\pi r$

3D FIGURES

- Cube
Volume = a^3
Surface Area = $6a^2$
- Cuboid
Volume = $l \times b \times h$
Surface Area = $2(lb + bh + hl)$
- Cylinder
Volume = $\pi r^2 h$
Curved Surface Area = $2\pi rh$
Total Surface Area = $2\pi r(r+h)$
- Sphere
Volume = $\frac{4}{3}\pi r^3$
Surface Area = $4\pi r^2$

14 MENSURATION

- Cone
Volume = $\frac{1}{3}\pi r^2 h$
Curved Surface Area = πrl
Total Surface Area = $\pi r(l+r)$
- Hemisphere
Volume = $\frac{2}{3}\pi r^3$
Curved Surface Area = $2\pi r^2$
Total Surface Area = $3\pi r^2$
- Pyramid
Volume = $\frac{1}{3} \times \text{Base Area} \times h$

IMPORTANT CONSTANTS

- π (pi) = $\frac{22}{7}$ or 3.1416
- $\sqrt{2} = 1.414$
 $\sqrt{3} = 1.732$
 $\sqrt{5} = 2.236$
 $\sqrt{10} = 3.162$
- 1 km = 1000 m
1 m = 100 cm
1 m = 1000 mm
1 L = 1000 ml

SELF CHECK

- ☐ Revise all formulas once
- ☐ Solve 2 full length mocks
- ☐ Focus on weak topics
- ☐ Stay positive & confident

“The formula for success is simple -
Practice + Patience +
Persistence = Placement
Success! ★

ANSWER KEY (Page 1)



1. b 2. c 3. a 4. d 5. c 6. b 7. d 8. a 19. c
11. c 12. a 13. c 14. b 15. a 16. d 17. b 19. a 20. d



THIS IS YOUR FINAL DAY - GIVE YOUR BEST & FINISH STRONG! ALL THE BEST! 🍀





QUICK REVISION TIPS

- ✓ Practice more, think less.
- ✓ Identify pattern & apply shortcut.
- ✓ Eliminate options smartly.
- ✓ Accuracy + Speed = Success
- ✓ Stay calm, stay focused.

SUPER SHORTCUTS

- Even number of letters - Same mirror image
- Opposite faces of dice - (1-6), (2-5), (3-4)
- In seating - Fix one position, others relative
- In puzzles - Make table/diagram
- Always check all conditions before final answer.

TYPES TO FOCUS

- Syllogism
- Seating Arrangement
- Blood Relation
- Coding - Decoding
- Direction Sense
- Order & Ranking
- Non-Verbal
- Puzzle & Miscellaneous

1. SERIES COMPLETION

Find the next number/letter.

- 2, 6, 12, 20, 30, ?
a) 40 b) 42 c) 44 d) 48
- 5, 10, 20, 40, 80, ?
a) 100 b) 120 c) 140 d) 160
- A, C, F, J, O, ?
a) U b) V c) W d) Y
- 1, 1, 2, 3, 5, 8, ?
a) 11 b) 12 c) 13 d) 15

2. CODING - DECODING

If 'APPLE' is coded as '61516125'.

How is 'GRAPES' coded?

- a) 71216151919 b) 71216151915
c) 71216191519 d) 71216151918

If 'BANK' is coded as '31916'.

How is 'KNAB' coded?

- a) 16319 b) 16391
c) 19316 d) 91631

3. DIRECTION SENSE

- If a man walks 5 km to the North, then 3 km to the East, then 5 km to the South, then 3 km to the West. In which direction is he from the starting point?

- a) North b) South c) East d) West

- Pointing towards the North, Rahul turns right, walks 10 m, turns left, walks 5 m, turns left again and walks 10 m. In which direction is he from the starting point?

- a) North b) South c) East d) West

4. SEATING ARRANGEMENT



Eight friends P, Q, R, S, T, U, V and W are sitting around a circular table facing the center.

- P is between Q and R.
- S is not the neighbor of P.
- T is second to the right of S.
- U is between T and V.

Who is between Q and V?

- a) R b) T c) S d) U

5. BLOOD RELATIONS



- A is the brother of B. C is the mother of A. D is the father of C. How is D related to B?

- a) Father b) Grandfather
c) Uncle d) Brother

- P is Q's mother. R is P's father. S is R's only daughter. How is S related to Q?

- a) Sister b) Daughter
c) Niece d) Cousin

6. ORDER & RANKING



- In a class of 40 students, Rahul ranks 12th from the top and 15th from the bottom. How many students are between Rahul and the top ranker?

- a) 11 b) 12 c) 13 d) 14

- Among 6 friends, A is older than B but younger than C. D is older than A but younger than E. F is the youngest. Who is the eldest?

- a) A b) C c) D d) E

7. SYLLOGISM



Statements: All pens are books.
Some books are copies.

Conclusions:

- Some pens are copies.
 - Some copies are pens.
- a) Only conclusion I follows
b) Only conclusion II follows
c) Both I and II follow
d) Neither I nor II follows

8. MIRROR IMAGES

Select the correct mirror image of the given figure. (Mirror is shown by line MN).

R E K

- a) K E R
b) R E K
c) K E R
d) R E K

9. CUBES & DICE



Two positions of a cube are given. Which number will be on the face opposite to face having '2'?



- a) 3 b) 4 c) 5 d) 6

10. PUZZLE & MISCELLANEOUS



Eight people A, B, C, D, E, F, G and H are sitting in a straight row facing North.

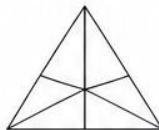
- Only three people sit between A and B.
- C sits immediate left of A.
- D is not an immediate neighbor of A.
- G sits second to the right of D.
- F sits at one of the ends.
- H sits to the immediate left of F.

Who sits fourth to the left of B?

- a) C b) D c) E d) G

11. FIGURE COUNTING

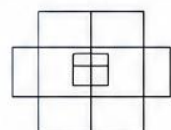
Count the number of triangles in the figure.



- a) 16 b) 18 c) 20 d) 22

12. EMBEDDED FIGURE

How many squares are there in the figure?



- a) 14 b) 16 c) 18 d) 20

LOGICAL REASONING TRICKS

- Series → Find pattern (Diff/Multiple/Alternate)
- Coding → Letter position / Reverse / Pattern
- Seating → Draw rough circle / row
- Direction → Draw diagram
- Syllogism → Draw Venn diagram
- Puzzle → Make table
- Cube → Opposite faces sum = 7 (1-6, 2-5, 3-4)

ANSWER KEY (Page 2)

1. d 2. a 3. b 4. c 5. b 6. a
7. a 8. d 9. b 10. b 11. b 12. b

SCORE BOOSTER CHECKLIST

- ✓ Attempt easy questions first.
- ✓ Use elimination technique.
- ✓ Don't get stuck on one question.
- ✓ Review if time left.
- ✓ Stay positive till the end!





FINAL REVISION - STRONG GRAMMAR, STRONG SCORE !



QUICK TIPS

- ✓ Read the sentence fully.
- ✓ Identify the error or best option.
- ✓ Use rules, not guesses.
- ✓ Eliminate & choose smartly.
- ✓ Practice daily, improve surely !



PARTS OF SPEECH

Noun	: Naming word (boy, city)
Pronoun	: Replaces a noun (he, it)
Verb	: Action / State (run, is)
Adjective	: Describes noun (good, big)
Adverb	: Describes verb/adj/adv (quickly)
Preposition	: Relation (in, on, at)
Conjunction	: Connects words (and, but)
Interjection	: Sudden feeling (oh!, wow!)

COMMON GRAMMAR RULES

- ♦ Subject - Verb Agreement
- ♦ Tense Consistency
- ♦ Articles (a, an, the)
- ♦ Prepositions
- ♦ Active - Passive Voice
- ♦ Direct - Indirect Speech
- ♦ Modifiers
- ♦ Conjunctions
- ♦ Punctuation

1 SUBJECT - VERB AGREEMENT

- Singular subject → Singular verb
- Plural subject → Plural verb

Examples:

- ✓ He goes to school.
- ✓ He go to school.
- ✓ They are playing.
- ✗ They is playing.

Tip: Check the subject, then choose the verb.

2 TENSE CONSISTENCY

- Keep the tense same throughout.
- Match with the time clue.

Examples:

- ✓ She finished her work and went home. (Past - Past)
- ✗ She finish her work and went home.
- ✓ I will call you tomorrow.
- ✗ I will called you tomorrow.

Tip: Identify the main verb tense first.

3 ARTICLES

- a → before consonant sound
- an → before vowel sound
- the → specific / known
- No article → general / plural

Examples:

- ✓ She is an honest girl.
- ✗ She is a honest girl.
- ✓ Please close the door.
- ✗ Please close a door.

Tip: Use 'the' for unique things.

4 PREPOSITIONS

- Preposition shows relation.
- Common: in, on, at, by, with, for, to

Examples:

- ✓ The book is on the table.
- ✗ The book is on table.
- ✓ He is good at maths.
- ✗ He is good in maths.

Tip: Prepositions are followed by noun/pronoun.

5 ACTIVE - PASSIVE VOICE

- Active: Subject does action
- Passive: Subject receives action

Examples:

- ✓ Active: She writes a letter.
Passive: A letter is written by her.
- ✓ Active: They will solve the problem.
Passive: The problem will be solved by them.

Tip: In passive, use 'by' + object.

6 DIRECT - INDIRECT SPEECH

- Change the reporting style.
- Change tense, pronouns, time & place.

Examples:

- ✓ Direct: He said, "I am happy."
Indirect: He said that he was happy.
- ✓ Direct: She said, "I will help you."
Indirect: She said that she would help me.

Tip: No quotes in indirect speech.

7 MODIFIERS

- Use correct word to describe.
- Place adjective/adverb properly.

Examples:

- ✓ He is a very intelligent boy.
- ✗ He is a intelligent very boy.
- ✓ She sings beautifully.
- ✗ She beautifully sings.

Tip: Adverb usually comes after verb.

8 CONJUNCTIONS

- And, but, or, so, because, although, etc.
- Join words, phrases or clauses.

Examples:

- ✓ I wanted to go, but it started raining.
- ✓ He is smart and hardworking.
- ✗ I wanted to go but and it rained.

Tip: Use comma before 'but'.

9 PUNCTUATION

- Capital letter at start.
- Full stop at the end.
- Comma (,), Apostrophe ('), etc.

Examples:

- ✓ Wow! You did it.
- ✓ Yes, I will go.
- ✓ It's raining heavily.
- ✗ its raining heavily

Tip: Punctuation shows meaning clearly.

PRACTICE QUESTIONS - CHOOSE THE CORRECT OPTION



- He _____ playing cricket every evening.
a) is b) are c) am d) were
- She is _____ honest girl.
a) a b) an c) the d) - (no article)
- I have finished my work, _____ I will go out.
a) and b) but c) or d) so
- The letter _____ by my friend yesterday.
a) write b) wrote c) is written d) was written

- He said, "I have done my task."
(Indirect Speech)
a) He said that he had done his task.
b) He said that I have done my task.
c) He said that he did his task.
d) He said that he has done his task.
- She sings _____.
a) beautiful b) beautifully
c) beauties d) more beautiful

- The keys are _____ the table.
a) in b) on c) at d) by
- He is good _____ solving puzzles.
a) at b) in c) on d) for
- I couldn't go out _____ it was raining.
a) because b) so c) but d) and
- Wow! _____ amazing performance.
a) what b) what a c) how d) how a

COMMONLY CONFUSED WORDS

- Affect - Verb (to influence)
- Effect - Noun (result)
- Accept - To receive
- Except - Other than
- Compliment - Praise
- Complement - Complete



ANSWER KEY (Page 3)



- a
- b
- b
- d
- a
- b
- b
- a
- a
- b

FINAL REMINDER

- ✓ Revise all grammar rules.
- ✓ Solve more practice sets.
- ✓ Avoid silly mistakes.
- ✓ Stay positive & confident !



SMART REVISION TODAY, TOP PERFORMANCE TOMORROW !





FINAL REVISION - WORDS ARE POWER, USE THEM WISELY !



QUICK TIPS

- ✓ Learn a few words daily.
- ✓ Use in sentences.
- ✓ Revise regularly.
- ✓ Write > Speak > Remember!
- ✓ Consistency = Success



HOW TO REMEMBER ?

1. Read the word
2. Understand meaning
3. Make a sentence
4. Use it in daily life
5. Revise after 24 hrs



WORD POWER BOOSTERS

- ★ Synonyms help in GD & Interview
- ★ Good Vocabulary = Good Impression
- ★ Use Specific words, not Simple words
- ★ Avoid Repetition of words
- ★ Confidence comes from Preparation



A. CONFIDENCE BUILDERS (1-50)

1. Abundant (Adj.) - Plenty
2. Achieve (V.) - Accomplish
3. Adapt (V.) - Adjust
4. Adept (Adj.) - Skilled
5. Ambitious (Adj.) - Driven
6. Analyze (V.) - Examine
7. Approve (V.) - Sanction
8. Assert (V.) - State firmly
9. Assume (V.) - Take for granted
10. Authentic (Adj.) - Genuine
11. Benevolent (Adj.) - Kind
12. Capable (Adj.) - Competent
13. Careful (Adj.) - Cautious
14. Clarify (V.) - Explain
15. Commit (V.) - Dedicate
16. Communicate (V.) - Convey
17. Competent (Adj.) - Efficient
18. Confident (Adj.) - Sure
19. Connect (V.) - Relate
20. Contribute (V.) - Give
21. Creative (Adj.) - Innovative
22. Decisive (Adj.) - Firm
23. Dedicate (V.) - Devote
24. Define (V.) - Describe
25. Deliver (V.) - Provide
26. Dependable (Adj.) - Reliable
27. Detail (N.) - Particular
28. Develop (V.) - Progress
29. Diligent (Adj.) - Hardworking
30. Determine (V.) - Decide
31. Effective (Adj.) - Productive
32. Efficient (Adj.) - Capable
33. Enhance (V.) - Improve
34. Enthusiastic (Adj.) - Eager
35. Establish (V.) - Set up
36. Evaluate (V.) - Assess
37. Excel (V.) - Perform greatly
38. Execute (V.) - Carry out
39. Expand (V.) - Increase
40. Flexible (Adj.) - Adaptable
41. Focus (V./N.) - Concentrate
42. Goal (N.) - Target
43. Identify (V.) - Recognize
44. Implement (V.) - Put into use
45. Improve (V.) - Make better
46. Initiative (N.) - First step
47. Inspire (V.) - Motivate
48. Interpret (V.) - Explain
49. Motivate (V.) - Encourage
50. Neutral (Adj.) - Impartial

B. POSITIVE IMPACT (51-100)

51. Optimize (V.) - Make best use
52. Organize (V.) - Arrange
53. Original (Adj.) - Unique
54. Overcome (V.) - Defeat
55. Participate (V.) - Take part
56. Perform (V.) - Act
57. Positive (Adj.) - Optimistic
58. Practical (Adj.) - Realistic
59. Predict (V.) - Foresee
60. Prepare (V.) - Get ready
61. Prioritize (V.) - Give priority
62. Proactive (Adj.) - Active
63. Process (N.) - Procedure
64. Progress (N.) - Advancement
65. Promote (V.) - Encourage
66. Provide (V.) - Supply
67. Qualify (V.) - Meet criteria
68. Question (V./N.) - Enquire
69. Quality (N.) - Standard
70. Quantify (V.) - Measure
71. Realize (V.) - Understand
72. Recommend (V.) - Suggest
73. Recognize (V.) - Acknowledge
74. Record (V./N.) - Document
75. Reduce (V.) - Decrease
76. Reflect (V.) - Think deeply
77. Relevant (Adj.) - Pertinent
78. Reliable (Adj.) - Trustworthy
79. Research (N./V.) - Study
80. Resolve (V.) - Solve
81. Resource (N.) - Asset
82. Respect (V./N.) - Admire
83. Respond (V.) - React
84. Responsible (Adj.) - Answerable
85. Result (N.) - Outcome
86. Review (V./N.) - Recheck
87. Support (V./N.) - Back
88. Success (N.) - Achievement
89. Suggest (V.) - Propose
90. Suitable (Adj.) - Appropriate
91. Summarize (V.) - Brief
92. Superior (Adj.) - Better
93. Systematic (Adj.) - Organized
94. Transform (V.) - Change
95. Trust (V./N.) - Believe
96. Understand (V.) - Comprehend
97. Utilize (V.) - Use
98. Value (N./V.) - Worth
99. Verify (V.) - Confirm
100. Vision (N.) - Outlook

C. SMART CHOICE (101-150)

101. Accomplish (V.) - Achieve
102. Advocate (V.) - Support
103. Alert (Adj.) - Aware
104. Allocate (V.) - Distribute
105. Alternative (N.) - Option
106. Evidence (N.) - Proof
107. Estimate (V.) - Approximate
108. Evaluate (V.) - Judge
109. Excellent (Adj.) - Outstanding
110. Expand (V.) - Extend
111. Factor (N.) - Element
112. Generate (V.) - Produce
113. Identify (V.) - Detect
114. Impact (N.) - Effect
115. Indicate (V.) - Show
116. Influence (V./N.) - Affect
117. Insight (N.) - Understanding
118. Integrate (V.) - Combine
119. Issue (N.) - Problem
120. Justify (V.) - Prove
121. Logical (Adj.) - Reasonable
122. Maintain (V.) - Keep up
123. Manage (V.) - Handle
124. Minimize (V.) - Reduce
125. Monitor (V.) - Observe
126. Objective (Adj.) - Goal-oriented
127. Operate (V.) - Function
128. Oppose (V.) - Disagree
129. Organize (V.) - Systemize
130. Perspective (N.) - Viewpoint
131. Plan (V./N.) - Strategy
132. Precise (Adj.) - Exact
133. Predictable (Adj.) - Expected
134. Prevent (V.) - Stop
135. Process (N.) - Method
136. Propose (V.) - Put forward
137. React (V.) - Respond
138. Recommend (V.) - Advise
139. Reflect (V.) - Contemplate
140. Relevant (Adj.) - Important
141. Report (N./V.) - Inform
142. Research (V.) - Investigate
143. Significant (Adj.) - Important
144. Solve (V.) - Find solution
145. Strength (N.) - Power
146. Structure (N.) - Framework
147. Technical (Adj.) - Practical
148. Test (V./N.) - Examine
149. Validate (V.) - Confirm
150. Worthwhile (Adj.) - Valuable

D. COMMUNICATION BOOST (151-200)

151. Accurate (Adj.) - Correct
152. Appropriate (Adj.) - Suitable
153. Attractive (Adj.) - Appealing
154. Balanced (Adj.) - Fair
155. Benefit (N./V.) - Advantage
156. Brief (Adj./V.) - Short
157. Clear (Adj.) - Evident
158. Concept (N.) - Idea
159. Consider (V.) - Think of
160. Correct (Adj.) - Right
161. Data (N.) - Facts
162. Describe (V.) - Illustrate
163. Distinct (Adj.) - Different
164. Efficiently (Adv.) - Effectively
165. Ensure (V.) - Guarantee
166. Essential (Adj.) - Necessary
167. Exhibit (V.) - Display
168. Express (V.) - Convey
169. Facilitate (V.) - Help
170. Feedback (N.) - Response
171. Formulate (V.) - Frame
172. Highlight (V.) - Emphasize
173. Ignore (V.) - Disregard
174. Inform (V.) - Notify
175. Involve (V.) - Include
176. Learn (V.) - Gain knowledge
177. Listen (V.) - Hear
178. Message (N.) - Information
179. Notice (V.) - Observe
180. Notify (V.) - Inform
181. Observe (V.) - Watch
182. Opinion (N.) - View
183. Participate (V.) - Engage
184. Present (V.) - Introduce
185. Prove (V.) - Demonstrate
186. Question (V.) - Ask
187. Reply (V.) - Answer
188. Request (V./N.) - Ask
189. Respond (V.) - Answer
190. Review (V.) - Revisit
191. Share (V.) - Distribute
192. Speak (V.) - Talk
193. Statement (N.) - Remark
194. Term (N.) - Word
195. Understand (V.) - Grasp
196. Update (V.) - Modify
197. Use (V.) - Utilize
198. View (N.) - Opinion
199. Write (V.) - Compose
200. Yield (V./N.) - Produce

E. ADVANCED WORDS (201-250)

201. Abate (V.) - Reduce
202. Accelerate (V.) - Speed up
203. Acquire (V.) - Obtain
204. Alleviate (V.) - Relieve
205. Analyze (V.) - Examine
206. Apparent (Adj.) - Obvious
207. Ascertain (V.) - Find out
208. Authentic (Adj.) - Real
209. Bolster (V.) - Support
210. Commence (V.) - Begin
211. Compensate (V.) - Reward
212. Comprehend (V.) - Understand
213. Conclude (V.) - End
214. Constrain (V.) - Restrict
215. Contradict (V.) - Oppose
216. Convince (V.) - Persuade
217. Critical (Adj.) - Crucial
218. Delegate (V.) - Assign
219. Demonstrate (V.) - Prove
220. Derive (V.) - Obtain
221. Devise (V.) - Plan
222. Distinguish (V.) - Differentiate
223. Eliminate (V.) - Remove
224. Emphasize (V.) - Stress
225. Enhance (V.) - Augment
226. Enrich (V.) - Improve
227. Escalate (V.) - Increase
228. Evaluate (V.) - Appraise
229. Exemplify (V.) - Illustrate
230. Formulate (V.) - Devise
231. Illustrate (V.) - Explain
232. Impart (V.) - Teach
233. Imply (V.) - Suggest
234. Infer (V.) - Deduce
235. Initiate (V.) - Start
236. Inspect (V.) - Examine
237. Intervene (V.) - Interfere
238. Legitimate (Adj.) - Valid
239. Persist (V.) - Continue
240. Prior (Adj.) - Earlier
241. Reiterate (V.) - Repeat
242. Suppress (V.) - Restrain
243. Substitute (V.) - Replace
244. Sustain (V.) - Maintain
245. Validate (V.) - Authenticate
246. Verify (V.) - Check
247. Vigilant (Adj.) - Watchful
248. Withdraw (V.) - Take back
249. Yield (V.) - Surrender
250. Zeal (N.) - Enthusiasm

DAILY PRACTICE IDEA

- ✓ Pick 10 words daily
- ✓ Make 1 sentence each
- ✓ Use in speaking/writing
- ✓ Revise weekly



ANSWER KEY (Page 4)

1. a 2. d 3. b 4. c 5. a 6. d 7. b 8. c 9. a 10. b
11. c 12. a 13. d 14. b 15. c 16. a 17. d 18. b 19. a 20. c

REMEMBER



- ✓ Good Vocabulary
- ✓ Better Communication
- ✓ Great Confidence
- ✓ Bright Future !



A GREAT VOCABULARY CAN OPEN ANY DOOR - KEEP LEARNING, KEEP GROWING !



5. TECHNICAL

100 MIXED MCQs + INTERVIEW QUICK REVISION



FINAL REVISION - PRACTICE WELL, PERFORM WELL !



QUICK TIPS

- ✓ Revise core concepts daily.
- ✓ Practice MCQs regularly.
- ✓ Understand, don't mug up.
- ✓ Stay calm in the exam.
- ✓ Concepts + Practice = Success !



IMPORTANT AREAS COVERED



Programming (C / Python)



OOPs Concepts



Data Structures



Database (SQL)



Operating Systems



Computer Networks



Computer Fundamentals



Software Engineering



Web Basics



Cloud Computing (AWS)



Cyber Security Basics



Miscellaneous

100 MIXED MCQs (PRACTICE)

1. PROGRAMMING (C / PYTHON)



What is the output of the following code?

int a = 5;

printf("%d", ++a * 2);

- a) 10 b) 11
c) 12 d) 14

4. DATABASE (SQL)



Which SQL command is used to remove the structure of a table?

- a) DROP TABLE b) DELETE
c) TRUNCATE d) REMOVE

7. COMPUTER FUNDAMENTALS



Which of the following is volatile memory?

- a) ROM b) Cache
c) Hard Disk d) RAM

10. CLOUD COMPUTING (AWS)



Which AWS service provides scalable virtual servers?

- a) S3 b) EC2
c) RDS d) IAM

2. OOPS CONCEPTS



Which of the following is NOT a feature of Object Oriented Programming?

- a) Inheritance b) Polymorphism
c) Compilation d) Encapsulation

5. OPERATING SYSTEMS



Which scheduling algorithm is used in Round Robin?

- a) SJF b) Priority
c) FCFS d) Time Quantum

8. SOFTWARE ENGINEERING



What is the first phase of SDLC?

- a) Testing b) Deployment
c) Requirement Analysis d) Design

11. CYBER SECURITY BASICS



Which of the following is an example of malware?

- a) Firewall b) Antivirus
c) Trojan Horse d) VPN

3. DATA STRUCTURES



Which data structure uses FIFO principle?

- a) Stack b) Queue
c) Tree d) Graph

6. COMPUTER NETWORKS



Which layer of OSI model is responsible for end-to-end communication?

- a) Transport Layer b) Session Layer
c) Network Layer d) Data Link Layer

9. WEB BASICS



Which tag is used to create a hyperlink in HTML?

- a) <link> b) <a>
c) <href> d) <hyper>

12. MISCELLANEOUS



Which of the following is an example of Artificial Intelligence?

- a) Email b) Chess Game
c) Calculator d) Word Processor

★ INTERVIEW QUICK REVISION ★

PROGRAMMING (C / PYTHON)



- C is case-sensitive.
- "0" is the null character in C.
- Python is an interpreted language.
- Use "==" to compare values.
- Indentation is important in Python.

OOPS CONCEPTS



- Class is a blueprint of objects.
- Object is an instance of class.
- Inheritance promotes reusability.
- Polymorphism allows one interface, multiple implementations.
- Encapsulation binds data & methods.

DATA STRUCTURES



- Array - Fixed size, contiguous memory.
- Linked List - Dynamic size.
- Stack - LIFO (Last In First Out)
- Queue - FIFO (First In First Out)
- Tree - Hierarchical structure.

DATABASE / SQL



- PRIMARY KEY - Unique & not null.
- FOREIGN KEY - References another table's primary key.
- SELECT - Fetches data.
- WHERE - Filters records.
- GROUP BY - Groups rows.

OPERATING SYSTEMS



- OS manages hardware & software.
- Process - A program in execution.
- Thread - Smallest unit of process.
- Deadlock - Occurs when processes wait indefinitely.
- Virtual Memory extends RAM.

COMPUTER NETWORKS



- IP Address - Logical address.
- DNS - Domain Name System.
- HTTP - Hyper Text Transfer Protocol.
- TCP - Reliable, connection-oriented.
- UDP - Faster, connectionless.

COMPUTER FUNDAMENTALS



- CPU - Central Processing Unit.
- ALU - Arithmetic Logic Unit.
- RAM - Read/Write memory.
- ROM - Read Only Memory.
- Cache memory is faster than RAM.

SOFTWARE ENGINEERING



- SDLC - Plan, Analyse, Design, Code, Test, Deploy, Maintain.
- Waterfall - Linear model.
- Agile - Iterative model.
- Testing ensures quality.
- Documentation is important.

WEB BASICS



- HTML - Structure
- CSS - Styling
- JavaScript - Behavior
- <a> tag - Link
- tag - Image

CLOUD COMPUTING (AWS)



- EC2 - Virtual machines
- S3 - Object storage
- RDS - Managed Database
- IAM - User access management
- AWS is pay-as-you-go.

CYBER SECURITY BASICS



- Phishing - Fake emails/websites
- Malware - Malicious software
- Antivirus - Detects malware
- Firewall - Monitors traffic
- Strong passwords keep safe.

MISCELLANEOUS



- AI - Machines mimicking human intelligence.
- ML - Subset of AI.
- DL - Subset of ML.
- Data is the new oil.



FORMULA & SHORTCUT REMINDER

- Binary to Decimal → Multiply by 2ⁿ
- Time Complexity (Linear Search) → O(n)
- Time Complexity (Binary Search) → O(log n)
- Queue Operations → Enqueue / Dequeue
- Stack Operations → Push / Pop



ANSWER KEY (Page 5)



1. c 2. c 3. b 4. a 5. d 6. a
7. d 8. c 9. b 10. b 11. c 12. b

FINAL REMINDER

- ✓ Revise all important concepts.
- ✓ Solve more mock tests.
- ✓ Manage time in exam.
- ✓ Believe in yourself. You can do it !

